

第九章 空间解析几何

习题 9.2 向量代数

1. 已知两点 $A(4, \sqrt{2}, 1)$ 和 $B(3, 0, 2)$. (1) 求 $\overrightarrow{AB}$ 的模; (2) 求与 $\overrightarrow{AB}$ 平行的单位向量;

(3) 求 $\overrightarrow{AB}$ 的方向角.

【解析】本题考查向量的坐标形式, 向量的模计算, 单位向量的计算以及方向角计算公式的应用.

$$(1) \overrightarrow{AB} = (-1, -\sqrt{2}, 1), \quad |\overrightarrow{AB}| = \sqrt{(-1)^2 + (-\sqrt{2})^2 + 1^2} = 2;$$

$$(2) \overrightarrow{AB}^\circ = \frac{1}{2}\overrightarrow{AB} = \frac{1}{2}(-1, -\sqrt{2}, 1); \text{ 与其平行的单位向量为 } \pm \frac{1}{2}(-1, -\sqrt{2}, 1).$$

$$(3) \cos \alpha = -\frac{1}{2}, \cos \beta = -\frac{\sqrt{2}}{2}, \cos \gamma = \frac{1}{2}, \text{ 则 } \alpha = \frac{2\pi}{3}, \beta = \frac{3\pi}{4}, \gamma = \frac{\pi}{3}.$$

2. 已知 $\vec{\alpha} = (a, 5, 1)$ 与 $\vec{\beta} = (3, 1, b)$ 共线, 求 a 与 b 的值.

【解析】本题考查两个向量平行的坐标关系.

$\vec{\alpha} = (a, 5, 1)$ 与 $\vec{\beta} = (3, 1, b)$ 共线, 则对应分量成比例, 即 $\frac{a}{3} = \frac{5}{1} = \frac{1}{b}$, 则 $a = 15, b = \frac{1}{5}$.

3. 设 $\vec{\alpha} = (3, -1, -2), \vec{\beta} = (1, 2, -1)$, 求 (1) $\vec{\alpha} \cdot \vec{\beta}$ 及 $\vec{\alpha} \times \vec{\beta}$; (2) $(-2\vec{\alpha}) \cdot (3\vec{\beta})$ 及 $\vec{\alpha} \times 2\vec{\beta}$;

(3) $\vec{\alpha}$ 与 $\vec{\beta}$ 的夹角余弦; (4) 以 $\vec{\alpha}, \vec{\beta}$ 为邻边的平行四边形面积; (5) 既垂直于 $\vec{\alpha}$ 又垂直于 $\vec{\beta}$ 的一个向量; (6) $\vec{\alpha} \cdot (\vec{\beta} \times \vec{\alpha})$.

【解析】本题考查向量的数量积、向量积、混合积在坐标形式下的计算公式; 向量夹角的计算; 向量积的概念和向量积模的几何意义.

$$(1) \vec{\alpha} \cdot \vec{\beta} = 3 \times 1 + (-1) \times 2 + (-2) \times (-1) = 3, \quad \vec{\alpha} \times \vec{\beta} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & -1 & -2 \\ 1 & 2 & -1 \end{vmatrix} = (5, 1, 7);$$

$$(2) (-2\vec{\alpha}) \cdot 3\vec{\beta} = -6\vec{\alpha} \cdot \vec{\beta} = -18, \quad \vec{\alpha} \times 2\vec{\beta} = 2\vec{\alpha} \times \vec{\beta} = (10, 2, 14);$$

$$(3) \cos \angle(\vec{\alpha}, \vec{\beta}) = \frac{\vec{\alpha} \cdot \vec{\beta}}{|\vec{\alpha}| |\vec{\beta}|} = \frac{3}{\sqrt{14} \cdot \sqrt{6}} = \frac{\sqrt{21}}{14};$$

$$(4) S = |\vec{\alpha} \times \vec{\beta}| = \left| \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & -1 & -2 \\ 1 & 2 & -1 \end{vmatrix} \right| = |(5, 1, 7)| = 5\sqrt{3};$$

(5) 既垂直于 $\vec{\alpha}$ 又垂直于 $\vec{\beta}$ 的一个向量是 $\vec{\alpha} \times \vec{\beta} = (5, 1, 7)$;

$$(6) \vec{\alpha} \cdot (\vec{\beta} \times \vec{\alpha}) = \begin{vmatrix} 3 & -1 & -2 \\ 1 & 2 & -1 \\ 3 & -1 & -2 \end{vmatrix} = 0.$$

4. 已知 $\vec{\alpha}$ 与 $\vec{\beta}$ 垂直, 且 $|\vec{\alpha}| = 3, |\vec{\beta}| = 4$, 求 $|(3\vec{\alpha} - \vec{\beta}) \times (\vec{\alpha} - 2\vec{\beta})|$.

【解析】考查抽象的向量的性质、运算了的计算.

$$\textcircled{1} (3\vec{\alpha} - \vec{\beta}) \times (\vec{\alpha} - 2\vec{\beta}) = 3\vec{\alpha} \times \vec{\alpha} - 6\vec{\alpha} \times \vec{\beta} - \vec{\beta} \times \vec{\alpha} + 2\vec{\beta} \times \vec{\beta} = -5\vec{\alpha} \times \vec{\beta};$$

$$\textcircled{2} |(3\vec{\alpha} - \vec{\beta}) \times (\vec{\alpha} - 2\vec{\beta})| = |-5\vec{\alpha} \times \vec{\beta}| = 5|\vec{\alpha} \times \vec{\beta}| = 5|\vec{\alpha}||\vec{\beta}|\sin \angle(\vec{\alpha}, \vec{\beta});$$

$$\textcircled{3} \text{ 因为 } \vec{\alpha} \text{ 与 } \vec{\beta} \text{ 垂直, 所以 } \angle(\vec{\alpha}, \vec{\beta}) = \frac{\pi}{2}, \text{ 原式} = 5 \times 3 \times 4 \times \sin \frac{\pi}{2} = 60.$$

5. 已知 $|\vec{\alpha}| = 10$, $|\vec{\beta}| = 2$. (1) 若 $\vec{\alpha} \cdot \vec{\beta} = 12$, 求 $|\vec{\alpha} \times \vec{\beta}|$; (2) 若 $|\vec{\alpha} \times \vec{\beta}| = 16$, 求 $\vec{\alpha} \cdot \vec{\beta}$.

【解析】考查抽象向量的数量积, 向量积的运算性质计算.

$$(1) \textcircled{1} \vec{\alpha} \cdot \vec{\beta} = |\vec{\alpha}||\vec{\beta}|\cos \theta = 20\cos \theta = 12 \Rightarrow \cos \theta = \frac{3}{5} \Rightarrow \sin \theta = \frac{4}{5},$$

$$\textcircled{2} |\vec{\alpha} \times \vec{\beta}| = |\vec{\alpha}||\vec{\beta}|\sin \theta = 20 \times \frac{4}{5} = 16;$$

$$(2) \textcircled{1} |\vec{\alpha} \times \vec{\beta}| = |\vec{\alpha}||\vec{\beta}|\sin \theta = 16 \Rightarrow \sin \theta = \frac{4}{5} \Rightarrow \cos \theta = \pm \frac{3}{5},$$

$$\textcircled{2} \vec{\alpha} \cdot \vec{\beta} = |\vec{\alpha}||\vec{\beta}|\cos \theta = 20 \times \left(\pm \frac{3}{5}\right) = \pm 12.$$

6. 设 $\vec{a}$, $\vec{b}$, $\vec{c}$ 均为单位向量, 且满足 $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, 求 $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$.

【解析】考查单位向量概念, 数量积的应用.

$$\textcircled{1} \vec{a}, \vec{b}, \vec{c} \text{ 均为单位向量, 则 } |\vec{a}| = |\vec{b}| = |\vec{c}| = 1;$$

$$\textcircled{2} \vec{a} + \vec{b} + \vec{c} = \vec{0} \text{ 两边右乘 } \vec{b} \text{ 得, } \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b} + \vec{c} \cdot \vec{b} = 0 \Rightarrow \vec{a} \cdot \vec{b} + \vec{c} \cdot \vec{b} = -1,$$

$$\vec{a} + \vec{b} + \vec{c} = \vec{0} \text{ 两边右乘 } \vec{c} \text{ 得, } \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{c} = 0 \Rightarrow \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c} = -1,$$

$$\vec{a} + \vec{b} + \vec{c} = \vec{0} \text{ 两边右乘 } \vec{a} \text{ 得, } \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{a} + \vec{c} \cdot \vec{a} = 0 \Rightarrow \vec{b} \cdot \vec{a} + \vec{c} \cdot \vec{a} = -1,$$

$$\text{上面三式相加, 得 } \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{3}{2}$$

习题 9.3 空间的平面与直线

1. 求平面 $2x - 2y + z + 5 = 0$ 与各坐标面间夹角的余弦.

【解析】考查平面夹角的计算

① 平面法向量 $\vec{n} = (2, -2, 1)$;

② xoy 面 $\vec{n}_1 = (0, 0, 1)$, 则 $\cos \theta_1 = \frac{|\vec{n} \cdot \vec{n}_1|}{|\vec{n}| |\vec{n}_1|} = \frac{1}{3}$;

xoz 面 $\vec{n}_2 = (0, 1, 0)$, 则 $\cos \theta_2 = \frac{|\vec{n} \cdot \vec{n}_2|}{|\vec{n}| |\vec{n}_2|} = \frac{2}{3}$;

$yo z$ 面 $\vec{n}_3 = (1, 0, 0)$, 则 $\cos \theta_3 = \frac{|\vec{n} \cdot \vec{n}_3|}{|\vec{n}| |\vec{n}_3|} = \frac{2}{3}$.

2. 求过点 $M_1(4, 1, 2)$ 和 $M_2(-3, 5, -1)$ 且垂直于平面 $\pi: 6x - 2y + 3z + 7 = 0$ 的平面方程.

【解析】考查点法式方法构建平面方程

① $\overrightarrow{M_1M_2} = (-7, 4, -3)$, $\vec{n} = (6, -2, 3)$;

② 所求平面法向量 $\vec{n}_1 = \overrightarrow{M_1M_2} \times \vec{n} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -7 & 4 & -3 \\ 6 & -2 & 3 \end{vmatrix} = (6, 3, -10)$;

③ 所求平面方程为: $6(x-4) + 3(y-1) - 10(z-2) = 0$, 即 $6x + 3y - 10z - 7 = 0$.

3. 求通过点 $(2, 1, 1)$ 且垂直于直线 $\begin{cases} x + 2y - z + 1 = 0 \\ 2x + y - z = 0 \end{cases}$ 的平面方程.

【解析】考查点法式方法构建平面方程

① $\vec{n}_1 = (1, 2, -1)$, $\vec{n}_2 = (2, 1, -1)$;

② 所求平面法向量 $\vec{n} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & -1 \\ 2 & 1 & -1 \end{vmatrix} = (-1, -1, -3) = -(1, 1, 3)$;

③ 所求平面方程为: $(x-2) + (y-1) + 3(z-1) = 0$, 即 $x + y + 3z - 6 = 0$.

4. 一直线过点 $(-1, 2, 1)$ 且平行于直线 $\begin{cases} x + y - 2z - 1 = 0 \\ x + 2y - z + 1 = 0 \end{cases}$, 求该直线的方程.

【解析】考查点向式方法构建直线方程

① $\vec{n}_1 = (1, 1, -2)$, $\vec{n}_2 = (1, 2, -1)$;

② $\vec{s} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & -2 \\ 1 & 2 & -1 \end{vmatrix} = (3, -1, 1)$;

③ 所求直线方程为 $\frac{x+1}{3} = \frac{y-2}{-1} = \frac{z-1}{1}$.

5. 一直线过点 $(1, 2, 1)$, 又与直线 $\frac{x}{2} = \frac{y}{1} = -z$ 相交且垂直于直线 $\frac{x-1}{3} = \frac{y}{2} = \frac{z+1}{1}$, 求该直

线
方程.

【解析】直线方程的计算

① $\frac{x}{2} = \frac{y}{1} = -z$, $P_1(0,0,0)$, $\vec{s}_1 = (2,1,-1)$;

② $\frac{x-1}{3} = \frac{y}{2} = \frac{z+1}{1}$, $P_2(1,0,-1)$, $\vec{s}_2 = (3,2,1)$;

③ 设所求直线 $\vec{s} = (a,b,c)$, 因为 $\vec{s} \perp \vec{s}_2 \Rightarrow \vec{s} \cdot \vec{s}_2 = 0$, 即 $3a+2b+c=0$ (*)

④ 记 $P(1,2,1)$, 所求直线与 $\frac{x}{2} = \frac{y}{1} = -z$ 相交 $\Leftrightarrow (\overrightarrow{PP_1}, \vec{s}, \vec{s}_1) = 0$, 而 $\overrightarrow{PP_1} = (-1,-2,-1)$, 即

$$\begin{vmatrix} -1 & -2 & -1 \\ a & b & c \\ 2 & 1 & -1 \end{vmatrix} = 0, \text{ 即} \quad a-b+c=0 \quad (*)$$

⑤ (*) 与 (2) 联立, 得 $a = -\frac{3}{2}b, c = \frac{5}{2}b$, 则 $a:b:c = -3:2:5$;

⑥ $\frac{x-1}{-3} = \frac{y-2}{2} = \frac{z-1}{5}$ 为所求直线方程.

6. 求直线 $\begin{cases} 2x-4y+z=0 \\ 3x-y-2z-9=0 \end{cases}$ 在平面 $4x-y+z=1$ 上的投影直线的方程.

【解析】考查平面束方程的应用

① 设过直线的平面束方程为 $\Pi_1: (2x-4y+z) + \lambda(3x-y-2z-9) = 0$, 化简得 $(2+3\lambda)x - (4+\lambda)y + (1-2\lambda)z - 9\lambda = 0$, 则 $\vec{n}_1 = (2+3\lambda, -4-\lambda, 1-2\lambda)$;

② 设已知平面方程 $4x-y+z=1$ 为 Π , 因为 $\Pi \perp \Pi_1$, 则 $\vec{n} \perp \vec{n}_1$, 即

$$4(2+3\lambda) + 4 + \lambda + (1-2\lambda) = 0 \Rightarrow \lambda = -\frac{13}{11}, \text{ 所以 } \Pi_1: 17x+31y-37z-117=0;$$

③ 投影方程为: $\begin{cases} 17x+31y-37z-117=0 \\ 4x-y+z=1 \end{cases}$

习题 9.4 几种常见的二次曲面

1. 求以点 $A(3, 2, 1)$ 为球心, 且与平面 $x + 2y - 3z = 18$ 相切的球面方程.

【解析】考查点到平面的距离公式及球面方程.

$$\textcircled{1} R = \frac{|3 + 2 \times 2 - 3 \times 1 - 18|}{\sqrt{1 + 2^2 + (-3)^2}} = \sqrt{14};$$

$$\textcircled{2} \text{ 所求球面方程为: } (x - 3)^2 + (y - 2)^2 + (z - 1)^2 = 14.$$

2. 求下列旋转面的方程, 并指出它的名称.

$$(1) \text{ 曲线 } \begin{cases} y = 2x \\ z = 0 \end{cases} \text{ 绕 } y \text{ 轴旋转一周; } \quad \text{【解】 } y^2 = 4(x^2 + z^2) \text{ 锥面.}$$

$$(2) \text{ 曲线 } \begin{cases} z^2 = 5x \\ y = 0 \end{cases} \text{ 绕 } x \text{ 轴旋转一周; } \quad \text{【解】 } 5x = y^2 + z^2 \text{ 抛物面.}$$

$$(3) \text{ 曲线 } \begin{cases} x^2 + z^2 = 9 \\ y = 0 \end{cases} \text{ 绕 } z \text{ 轴旋转一周. } \quad \text{【解】 } x^2 + y^2 + z^2 = 9 \text{ 球面.}$$

4. 求两曲面 $x^2 + y^2 + z^2 = 2$ 和 $z^2 = x^2 + y^2$ 的交线在 xOy 坐标面的投影曲线的方程, 并作图.

【解析】投影曲面的计算

$$\textcircled{1} \begin{cases} x^2 + y^2 + z^2 = 2 \\ z^2 = x^2 + y^2 \end{cases} \Rightarrow 2(x^2 + y^2) = 2 \Rightarrow x^2 + y^2 = 1,$$

$$\textcircled{2} \text{ 投影曲线方程为 } \begin{cases} x^2 + y^2 = 1 \\ z = 0 \end{cases}.$$

自 测 题

一、填空题（每小题 4 分，共 20 分）.

1. 设 $\overrightarrow{AB} = \{-3, 0, 4\}$, $\overrightarrow{AC} = \{5, -2, -14\}$, 则 $\angle BAC$ 的平分线上的单位向量是_____.

【解析】本题思路不可直接用 $\overrightarrow{AB} + \overrightarrow{AC}$ 作为角平分线向量，因为平行四边形对角线未必为角平分线，所以正确方法是将 $\overrightarrow{AB}, \overrightarrow{AC}$ 单位化后构造平行四边形，即菱形，而菱形的对角线为角平分线.

$$\textcircled{1} \quad \overrightarrow{AB}^o = \frac{1}{5}(-3, 0, 4), \quad \overrightarrow{AC}^o = \frac{1}{15}(5, -2, -14);$$

$$\textcircled{2} \quad \vec{a} = \overrightarrow{AB}^o + \overrightarrow{AC}^o = \left(-\frac{4}{15}, -\frac{2}{15}, -\frac{2}{15}\right);$$

$$\textcircled{3} \quad \vec{a}^o = \frac{1}{|\vec{a}|} \vec{a} = \frac{15}{\sqrt{24}} \left(-\frac{4}{15}, -\frac{2}{15}, -\frac{2}{15}\right) = \left(-\frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right).$$

2. 已知 $|\vec{a}| = 2, |\vec{b}| = 5, \angle(\vec{a}, \vec{b}) = \frac{2\pi}{3}, \vec{c} = 3\vec{a} - \vec{b}, \vec{d} = \lambda\vec{a} + 17\vec{b}$, 若 $\vec{c}$ 与 $\vec{d}$ 垂直, 则 $\lambda =$ _____.

【解析】 $\vec{c}$ 与 $\vec{d}$ 垂直, 则 $\vec{c} \cdot \vec{d} = 0$, 即

$$\begin{aligned} 0 &= \vec{c} \cdot \vec{d} = (3\vec{a} - \vec{b}) \cdot (\lambda\vec{a} + 17\vec{b}) = 3\lambda\vec{a} \cdot \vec{a} + (51 - \lambda)\vec{a} \cdot \vec{b} - 17\vec{b} \cdot \vec{b} \\ &= 3\lambda|\vec{a}|^2 + (51 - \lambda)|\vec{a}| \cdot |\vec{b}| \cos \frac{2\pi}{3} - 17|\vec{b}|^2 \\ &= 12\lambda - 5(51 - \lambda) - 425 = 17\lambda - 780 \Rightarrow \lambda = 40. \end{aligned}$$

3. 直线 $\frac{x}{-1} = \frac{y-1}{1} = \frac{z-1}{2}$ 与平面 $2x + y - z - 3 = 0$ 的夹角是_____.

【解析】 $\vec{s} = (-1, 1, 2), \vec{n} = (2, 1, -1), \sin \theta = \frac{|-1 \times 2 + 1 \times 1 + 2 \times (-1)|}{\sqrt{6} \cdot \sqrt{6}} = \frac{1}{2}$, 得 $\theta = \frac{\pi}{6}$.

4. 过直线 $L_1: \frac{x-1}{1} = \frac{y-2}{0} = \frac{z-3}{-1}$ 且平行于直线 $L_2: \frac{x+2}{2} = \frac{y-1}{1} = \frac{z}{1}$ 的平面方程为_____.

【解析】 $\vec{s}_1 = (1, 0, -1), \vec{s}_2 = (2, 1, 1), \vec{n} = \vec{s}_1 \times \vec{s}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -1 \\ 2 & 1 & 1 \end{vmatrix} = (1, -3, 1)$, 所求平面方程为

$$x - 3y + z + 2 = 0.$$

5. 曲线 $\begin{cases} x^2 + y^2 + z^2 = 1 \\ x + y + z = 0 \end{cases}$ 在 xoy 坐标面上的投影曲线为_____.

【解析】方程组联立消 z , 得 $\begin{cases} 2x^2 + 2y^2 + 2xy = 1 \\ z = 0 \end{cases}$.

二、选择题（每小题 4 分，共 20 分）.

6. 下列方程表示的直线中与直线 $L: \begin{cases} x + y + z = 1 \\ x - y - 2z = 1 \end{cases}$ 平行的是 ().

$$(A) \frac{x-1}{1} = \frac{y-2}{-3} = \frac{z}{-2}$$

$$(B) \frac{x-1}{1} = \frac{y-2}{3} = \frac{z}{-2}$$

$$(C) \frac{x-1}{1} = \frac{y-2}{3} = \frac{z}{2}$$

$$(D) \frac{x-1}{1} = \frac{y-2}{-3} = \frac{z}{2}$$

【解析】 $\vec{s} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 1 \\ 1 & -1 & -2 \end{vmatrix} = (-1, 3, -2),$

7. 两直线 $L_1: \frac{x-1}{1} = \frac{y-5}{-2} = \frac{z+8}{1}$ 与 $L_2: \begin{cases} x-y=6 \\ 2y+z=3 \end{cases}$ 的夹角为 ().

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

【解析】 L_1 的方向向量 $\vec{s}_1 = (1, -2, 1)$, L_2 的方向向量 $\vec{s}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 0 \\ 0 & 2 & 1 \end{vmatrix} = (-1, -1, 2),$

夹角余弦 $\cos \theta = \frac{|1 \times (-1) + (-2) \times (-1) + 1 \times 2|}{\sqrt{6} \cdot \sqrt{6}} = \frac{1}{2}$, 则 $\theta = \frac{\pi}{3}$, 所以答案选 C.

8. 设有直线 $L: \begin{cases} x+3y+2z+1=0 \\ 2x-y-10z+3=0 \end{cases}$ 及平面 $\pi: 4x-2y+z-2=0$, 则直线 L ().

- (A) 平行于 π (B) 在 π 上 (C) 垂直于 π (D) 与 π 斜交

【解析】直线 L 的方向向量 $\vec{s} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 3 & 2 \\ 2 & -1 & -10 \end{vmatrix} = -7(4, -2, 1)$, 平面法向量 $\vec{n} = (4, -2, 1)$, 所以答

案选 C

9. 直线 $\frac{x-1}{4} = \frac{y-3}{-2} = \frac{z}{1}$ 与直线 $\frac{x}{0} = \frac{y}{2} = \frac{z+2}{1}$ 的位置关系是 ().

- (A) 平行 (B) 相交于一点 (C) 异面 (D) 重合

【解析】 $P_1(1, 3, 0)$, $P_2(0, 0, -2)$, $\overrightarrow{P_2P_1} = (1, 3, 2)$, $\vec{s}_1 = (4, -2, 1)$, $\vec{s}_2 = (0, 2, 1)$, 则 $\begin{vmatrix} 1 & 3 & 2 \\ 4 & -2 & 1 \\ 0 & 2 & 1 \end{vmatrix} = 0,$

由定理可知两直线相交, 所以答案选 B.

10. xOz 坐标面上曲线 $z = e^x (x > 0)$ 绕 z 轴旋转所得的旋转面方程为 ().

- (A) $\sqrt{y^2 + z^2} = e^x$ (B) $y^2 + z^2 = e^x$ (C) $z = e^{x^2+y^2}$
(D) $z = e^{\sqrt{x^2+y^2}}$

【解析】由选择曲面方程构造公式, 得答案选 D.

三、解答题 (每小题 10 分, 共 60 分).

11. 设 $\vec{a} = \{3, 0, 4\}$, $\vec{b} = \{-1, 2, -2\}$, 求与向量 $\vec{a}$ 和 $\vec{b}$ 均垂直的单位向量.

【解析】① $\vec{a}$ 与 $\vec{b}$ 垂直向量记为 $\vec{c} = \vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 0 & 4 \\ -1 & 2 & 2 \end{vmatrix} = (-8, 2, 6) = 2(-4, 1, 3);$

② $\vec{c}^o = \pm \frac{1}{\sqrt{26}}(-4, 1, 3)$ 即为所求.

12. 设一平面经过原点及点 $(6, -3, 2)$, 且与平面 $4x - y + 2z = 8$ 垂直, 求此平面方程.

【解析】① $O(0, 0, 0)$, $P(6, -3, 2)$, 则 $\overrightarrow{OP} = (6, -3, 2)$, $\vec{n}_1 = (4, -1, 2);$

② 所求平面法向量 $\vec{n} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 6 & -3 & 2 \\ 4 & -1 & 2 \end{vmatrix} = (-4, -4, 6) = -2(2, 2, -3)$;

③ $2x + 2y - 3z = 0$ 为所求平面方程.

13. 过平面 $\pi_1: x + 28y - 2z + 17 = 0$ 和 $\pi_2: 5x + 8y - z + 1 = 0$ 的交线, 作球面 $x^2 + y^2 + z^2 = 1$ 的切平面, 求该切平面方程.

【解析】① 过平面 Π_1 与 Π_2 交线的平面束方程为 $(x + 28y - 2z + 17) + \lambda(5x + 8y - z + 1) = 0$, 即

$$(1 + 5\lambda)x + (28 + 8\lambda)y - (2 + \lambda)z + 17 + \lambda = 0;$$

② 由题意, 球面的球心到切平面距离为半径 1, 即

$$\frac{|17 + \lambda|}{\sqrt{(1 + 5\lambda)^2 + (28 + 8\lambda)^2 + [-(2 + \lambda)]^2}} = 1,$$

化简得 $89\lambda^2 + 428\lambda + 500 = 0$, 解得: $\lambda = -\frac{250}{89}$ 或 -2 ;

③ 切平面方程为: $387x - 164y - 24z - 421 = 0$ 或 $3x - 4 - 5 = 0$.

14. 求过点 $M_0(2, 1, 3)$ 且与直线 $l: \begin{cases} 2x + y + 2z = 0 \\ x + y - 3 = 0 \end{cases}$ 垂直相交的直线方程.

【解析】① 设所求直线方向向量 $\vec{s} = (a, b, c)$;

② l 的方向向量 $\vec{s}_1 = \vec{n}_1 \times \vec{n}_2 = (-2, 2, 1)$;

③ $\begin{cases} 2x + y + 2z = 0 \\ x + y - 3 = 0 \end{cases} \Rightarrow x + 2z + 3 = 0$, 令 $z = 1$, 则 $x = -5, y = 8$, 交点记为 $M_1(-5, 8, 1)$;

④ 两直线垂直得: $-2a + 2b + c = 0$;

$$\text{两直线相交得: } \begin{vmatrix} \overrightarrow{M_0M_1} & \vec{s} & \vec{s}_1 \end{vmatrix} = \begin{vmatrix} -7 & 7 & -2 \\ a & b & c \\ -2 & 2 & 1 \end{vmatrix} = 0, \text{ 即 } a + b = 0;$$

两式整理得: $a = -b, c = -4b$;

⑤ 则 $a : b : c = -1 : 1 : (-4)$;

⑥ 所求直线方程为: $\frac{x-2}{-1} = \frac{y-1}{1} = \frac{z-3}{-4}$.

★★15. 设 l_1, l_2 为两条共面直线, l_1 的方程为 $\frac{x-7}{1} = \frac{y-3}{2} = \frac{z-5}{2}$; l_2 通过点 $(2, -3, -1)$,

且与 x 轴正向夹

角为 $\frac{\pi}{3}$, 与 z 轴正向夹角为锐角, 求 l_2 的方程.

【解析】① 若 $l_1 // l_2$, 则 l_1 的方向向量 $(1, 2, 2)$ 也为 l_2 的方向向量, 则 l_2 与 x 轴夹角余弦为 $\frac{(1, 2, 2) \cdot (1, 0, 0)}{\sqrt{1+2^2+2^2}} = \frac{1}{3}$, 不可能为 $\frac{\pi}{3}$, 所以 l_1 与 l_2 不平行;

② l_1 与 l_2 只能相交. 不妨令交点为 P , 所以 P 在 l_1 上, 由 l_1 的参数方程可设 P 的坐标为 $(t+7, 2t+3, 2t+5)$, 又 l_2 过点 $Q(2, -3, -1)$, 则 $\overrightarrow{QP} = (t+5, 2t+6, 2t+6)$ 平行于 l_2 , 取 $(\lambda c, c, c)$ 为 l_2 的方向向量, 其中 $c = 2t+6$, 则 $\lambda = \frac{t+5}{2t+6}$; 又 l_2 与 x 轴夹角为锐角, 取 $c = 1$, 又夹角为 $\frac{\pi}{3}$,

$$\text{即 } \cos \frac{\pi}{3} = \frac{(\lambda, 1, 1) \cdot (1, 0, 0)}{\sqrt{\lambda^2 + 1 + 1}} = \frac{\lambda}{\sqrt{\lambda^2 + 2}} \Rightarrow \lambda = \frac{\sqrt{6}}{3};$$

所以 l_2 的方向向量为 $\left(\frac{\sqrt{6}}{3}, 1, 1\right)$ ，其对应的方程为 $\frac{x-2}{\frac{\sqrt{6}}{3}} = \frac{y+3}{1} = \frac{z+1}{1}$ ，化简得

$$\frac{x-2}{2} = \frac{y+3}{\sqrt{6}} = \frac{z+1}{\sqrt{6}}.$$

16. 求旋转抛物面 $z = x^2 + y^2$ 与平面 $y + z = 1$ 的交线在 xoy 坐标面上的投影方程，并确定交线类型.

【解析】① $\begin{cases} z = x^2 + y^2 \\ y + z = 1 \end{cases}$ 消 z 得 $1 - y = x^2 + y^2$ ，即 $x^2 + y^2 + y = 1 \Rightarrow x^2 + \left(y + \frac{1}{2}\right)^2 = \frac{5}{4}$;

② 投影方程为 $\begin{cases} x^2 + \left(y + \frac{1}{2}\right)^2 = \frac{5}{4} \\ z = 0 \end{cases}$ ，为 xoy 面上的一个圆.

第十章 多元函数微分学

习题 10.1 多元函数的基本概念

1. 已知 $f(x+y, \frac{y}{x}) = x^2 - y^2$, 求 $f(x, y)$.

【解析】 $\begin{cases} x+y=u \\ \frac{y}{x}=v \end{cases} \Rightarrow \begin{cases} x=\frac{u}{1+v} \\ y=\frac{uv}{1+v} \end{cases} \Rightarrow f(u, v) = \left(\frac{u}{1+v}\right)^2 - \left(\frac{uv}{1+v}\right)^2 = \frac{u^2(1-v)}{(1+v)^2} = \frac{u^2(1-v)}{1+v}$, 则

$$f(x, y) = \frac{x^2(1-y)}{1+y}.$$

2. 求下列函数的极限.

$$(1) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow a}} \frac{\sin xy}{x} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow a}} \frac{\sin xy}{xy} \cdot y = 1 \cdot a = a;$$

$$(2) \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{xy+4}-2} = \lim_{(x,y) \rightarrow (0,0)} \frac{xy(\sqrt{xy+4}+2)}{xy} = \lim_{(x,y) \rightarrow (0,0)} (\sqrt{xy+4}+2) = 4;$$

$$(3) \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2+y^2}}$$

【解析】 因为 $0 \leq \left| \frac{xy}{x^2+y^2} \right| \leq \left| \frac{xy}{\sqrt{2xy}} \right| = \frac{1}{2} \sqrt{xy}$, 由夹边定理可知 $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2+y^2}} = 0$.

3. 证明下列极限不存在.

$$(1) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y^2}{x^2 y^2 + (x-y)^2}$$

【解析】 ①沿 $y=x$ 的路径趋于 $(0,0)$, $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y^2}{x^2 y^2 + (x-y)^2} = \lim_{x \rightarrow 0} \frac{x^4}{x^4} = 1;$

②沿 y 轴趋于 $(0,0)$, $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y^2}{x^2 y^2 + (x-y)^2} = 0;$

③沿两条不同路径趋于 $(0,0)$ 点, 所得极限存在但不相等, 所以原式极限不存在.

$$(2) \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y}{x^4 + y^2}$$

【解析】 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y}{x^4 + y^2} = \lim_{\substack{x \rightarrow 0 \\ y=kx^2}} \frac{x^2 y}{x^4 + y^2} = \lim_{x \rightarrow 0} \frac{kx^4}{(1+k^2)x^4} = \frac{k}{1+k^2}$, 沿不同路径极限存在不唯一,

所以原式极限不存在.

习题 10.2 偏导数与全微分

1. 求下列函数的一阶偏导数.

$$(1) z = \frac{x^2 + y^2}{xy};$$

【解析】① $z = \frac{x^2 + y^2}{xy} = \frac{x}{y} + \frac{y}{x};$

$$② \frac{\partial z}{\partial x} = \frac{1}{y} - \frac{y}{x^2} = \frac{x^2 - y^2}{x^2 y}, \quad \frac{\partial z}{\partial y} = -\frac{x}{y^2} + \frac{1}{x} = \frac{y^2 - x^2}{xy^2}.$$

$$(2) z = \ln(x + \sqrt{x^2 + y^2});$$

【解析】 $\frac{\partial z}{\partial x} = \frac{1}{x + \sqrt{x^2 + y^2}} \cdot \left[1 + \frac{1}{2\sqrt{x^2 + y^2}} \cdot 2x \right] = \frac{1}{\sqrt{x^2 + y^2}},$

$$\frac{\partial z}{\partial y} = \frac{1}{x + \sqrt{x^2 + y^2}} \cdot \frac{1}{2\sqrt{x^2 + y^2}} \cdot 2y = \frac{y}{(x + \sqrt{x^2 + y^2}) \cdot \sqrt{x^2 + y^2}}.$$

$$(3) u = x^{\frac{y}{z}};$$

【解析】 $\frac{\partial u}{\partial x} = \frac{y}{z} x^{\frac{y}{z}-1}, \quad \frac{\partial u}{\partial y} = \left(x^{\frac{y}{z}} \cdot \ln x \right) \cdot \frac{1}{z} = \frac{\ln x}{z} \cdot x^{\frac{y}{z}}, \quad \frac{\partial u}{\partial z} = \left(x^{\frac{y}{z}} \cdot \ln x \right) \cdot \left(-\frac{y}{z^2} \right) = -\frac{y \ln x}{z^2} \cdot x^{\frac{y}{z}}.$

$$(4) u = \arctan(x - y)^2.$$

【解析】 $\frac{\partial u}{\partial x} = \frac{1}{1 + (x - y)^4} \cdot 2(x - y) \cdot 1 = \frac{2(x - y)}{1 + (x - y)^4}, \quad \frac{\partial u}{\partial y} = \frac{1}{1 + (x - y)^4} \cdot 2(x - y) \cdot (-1) = -\frac{2(x - y)}{1 + (x - y)^4}.$

2. 设 $f(x, y, z) = (z - a^{xy}) \sin \ln x$, 求 $f(x, y, z)$ 在点 $(1, 0, 2)$ 处的 3 个一阶偏导数.

【解析】

方法一: 先分别对 x, y, z 求出偏导函数, 然后在代值;

方法二: 亦可以用偏导数定义的方法求之;

方法三: $f'_x(1, 0, 2) = \left[\frac{d}{dx} f(x, 0, 2) \right]_{x=1} = \left[\frac{d}{dx} \sin(\ln x) \right]_{x=1} = \left[\frac{1}{x} \cos(\ln x) \right]_{x=1} = 1;$

$$f'_y(1, 0, 2) = \left[\frac{d}{dy} f(1, y, 2) \right]_{y=1} = \left[\frac{d}{dy} (0) \right]_{y=1} = 0;$$

$$f'_z(1, 0, 2) = \left[\frac{d}{dz} f(1, 0, z) \right]_{z=1} = \left[\frac{d}{dz} (0) \right]_{z=1} = 0.$$

3. 设 $u = e^{\frac{x}{y^2}}$, 证明 $2x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$.

【解析】① $\frac{\partial u}{\partial x} = e^{\frac{x}{y^2}} \cdot \frac{1}{y^2}, \quad \frac{\partial u}{\partial y} = e^{\frac{x}{y^2}} \cdot x \cdot (-2) \cdot \frac{1}{y^3} = -\frac{2x}{y^3} e^{\frac{x}{y^2}};$

$$② \quad 2x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2x \cdot e^{\frac{x}{y^2}} \cdot \frac{1}{y^2} - y \cdot \frac{2x}{y^3} e^{\frac{x}{y^2}} = 0.$$

4. 设 $z = x \ln(xy)$, 求 $\frac{\partial^3 z}{\partial x^2 \partial y}$ 与 $\frac{\partial^3 z}{\partial x \partial y^2}$.

【解析】① $\frac{\partial z}{\partial x} = \ln(xy) + x \cdot \frac{1}{xy} \cdot y = 1 + \ln(xy)$, $\frac{\partial^2 z}{\partial x^2} = \frac{1}{xy} \cdot y = \frac{1}{x}$, $\frac{\partial^3 z}{\partial x^2 \partial y} = 0$;

$$\textcircled{2} \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{1}{y}, \quad \frac{\partial^3 z}{\partial x \partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial^2 z}{\partial x \partial y} \right) = -\frac{1}{y^2}.$$

5. 设 $z = \arctan \frac{x+y}{1-xy}$, 求 $\frac{\partial^2 z}{\partial x^2}$, $\frac{\partial^2 z}{\partial x \partial y}$, $\frac{\partial^2 z}{\partial y \partial x}$, $\frac{\partial^2 z}{\partial y^2}$.

【解析】

$$\textcircled{1} \quad \frac{\partial z}{\partial x} = \frac{1}{1 + \left(\frac{x+y}{1-xy} \right)^2} \cdot \frac{1 \cdot (1-xy) - (x+y)(-y)}{(1-xy)^2} = \frac{1}{1+x^2}, \quad \frac{\partial z}{\partial y} = \frac{1}{1 + \left(\frac{x+y}{1-xy} \right)^2} \cdot \frac{(1-xy) - (x+y)(-x)}{(1-xy)^2} = \frac{1}{1+y^2};$$

$$\textcircled{2} \quad \frac{\partial^2 z}{\partial x^2} = -\frac{2x}{(1+x^2)^2}, \quad \frac{\partial^2 z}{\partial y^2} = -\frac{2y}{(1+y^2)^2};$$

$$\textcircled{3} \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = 0.$$

6. 求下列函数的全微分.

$$(1) z = \arctan \frac{x+y}{x-y}$$

$$\text{【解析】} \textcircled{1} \quad \frac{\partial z}{\partial x} = \frac{1}{1 + \left(\frac{x+y}{x-y} \right)^2} \cdot \frac{(x-y) - (x+y)}{(x-y)^2} = \frac{-y}{x^2 + y^2}, \quad \frac{\partial z}{\partial y} = \frac{x}{x^2 + y^2};$$

$$\textcircled{2} \quad dz = \frac{-ydx + xdy}{x^2 + y^2}.$$

$$(2) u = \ln(x^2 - y^2 + e^z)$$

$$\text{【解析】} \textcircled{1} \quad \frac{\partial u}{\partial x} = \frac{2x}{x^2 - y^2 + e^z}, \quad \frac{\partial u}{\partial y} = \frac{-2y}{x^2 - y^2 + e^z}, \quad \frac{\partial u}{\partial z} = \frac{e^z}{x^2 - y^2 + e^z};$$

$$\textcircled{2} \quad dz = \frac{2xdx - 2ydy + e^z dz}{x^2 - y^2 + e^z}.$$

$$7. \text{ 设 } u = \left(\frac{y}{z} \right)^{\frac{1}{x}}, \text{ 求 } du(1,1,1).$$

$$\text{【解析】} \textcircled{1} \quad \frac{\partial u}{\partial x} = \left(\frac{y}{z} \right)^{\frac{1}{x}} \cdot \ln \frac{y}{z} \cdot \left(-\frac{1}{x^2} \right), \quad \left. \frac{\partial u}{\partial x} \right|_{(1,1,1)} = 0;$$

$$\frac{\partial u}{\partial y} = \frac{1}{x} \cdot \left(\frac{y}{z} \right)^{\frac{1}{x}-1} \cdot \frac{1}{z}, \quad \left. \frac{\partial u}{\partial y} \right|_{(1,1,1)} = 1;$$

$$\frac{\partial u}{\partial z} = \frac{1}{x} \cdot \left(\frac{y}{z} \right)^{\frac{1}{x}-1} \cdot \left(-\frac{y}{z^2} \right), \quad \left. \frac{\partial u}{\partial z} \right|_{(1,1,1)} = -1;$$

$$\textcircled{2} \quad du(1,1,1) = dy - dz.$$

安徽大学 ACM/ICPC 实验室

习题 10.3 多元复合函数微分法

1. 求下列复合函数的偏导数.

$$(1) z = \sin(2u + 3v), u = xy, v = x^2 + y^2, \text{ 求 } \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}.$$

【解析】

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = \cos(2u + 3v) \cdot 2 \cdot y + \cos(2u + 3v) \cdot 3 \cdot 2x = \cos(2xy + 3x^2 + 3y^2) \cdot (2y + 6x),$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} = \cos(2u + 3v) \cdot 2 \cdot x + \cos(2u + 3v) \cdot 3 \cdot 2y = \cos(2xy + 3x^2 + 3y^2) \cdot (2x + 6y).$$

$$(2) z = u^2 \ln v, \text{ 其中 } u = \frac{x}{y}, v = 3x - 2y, \text{ 求 } \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}.$$

【解析】

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = 2u \ln v \cdot \frac{1}{y} + \frac{u^2}{v} \cdot 3 = \frac{2x}{y^2} \ln(3x - 2y) + \frac{3x}{(3x - 2y)y^2},$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} = 2u \ln v \cdot \left(-\frac{x}{y}\right) + \frac{u^2}{v} \cdot (-2) = -\frac{2x}{y^2} \ln(3x - 2y) - \frac{2x^2}{(3x - 2y)y^2}.$$

$$2. \text{ 设 } z = f(xy, \frac{x}{y}) + g(\frac{y}{x}), \text{ 其中 } f \text{ 具有二阶连续偏导数, } g \text{ 具有二阶连续导数, 求 } \frac{\partial^2 z}{\partial x \partial y}.$$

$$\text{【解析】 } \textcircled{1} \frac{\partial z}{\partial x} = f'_1 \cdot y + f'_2 \cdot \frac{1}{y} + g' \cdot \left(-\frac{y}{x^2}\right),$$

$$\textcircled{2} \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial z}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial z}{\partial y} \left[f'_1 \cdot y + f'_2 \cdot \frac{1}{y} + g' \cdot \left(-\frac{y}{x^2}\right) \right]$$

$$= y \left[f''_{11} \cdot x + f''_{12} \cdot \left(-\frac{x}{y^2}\right) \right] + f'_1 + \left[f''_{21} \cdot x + f''_{22} \cdot \left(-\frac{x}{y^2}\right) \right] \cdot \frac{1}{y} + f'_2 \cdot \left(-\frac{1}{y^2}\right) + g'' \cdot \frac{1}{x} \cdot \left(-\frac{y}{x^2}\right) + g' \cdot \left(-\frac{1}{x^2}\right)$$

$$= xyf''_{11} - \frac{x}{y^3} f''_{22} + f'_1 - \frac{1}{y^2} f'_2 - \frac{y}{x^3} g'' - \frac{1}{x^2} g'.$$

$$3. \text{ 设 } z = \frac{y}{f(u)}, \text{ 其中 } u = x^2 - y^2, f(u) \text{ 为可导函数, 求 } \frac{1}{x} \frac{\partial z}{\partial x} + \frac{1}{y} \frac{\partial z}{\partial y}.$$

$$\text{【解析】 } \textcircled{1} \frac{\partial z}{\partial x} = -\frac{yf'(u)}{f^2(u)} \cdot \frac{\partial u}{\partial x} = -\frac{2xyf'(u)}{f^2(u)}, \quad \frac{\partial z}{\partial y} = \frac{1}{f(u)} - \frac{yf'(u)}{f^2(u)} \cdot \frac{\partial u}{\partial y} = \frac{1}{f(u)} + \frac{2y^2 f'(u)}{f^2(u)},$$

$$\textcircled{2} \quad \frac{1}{x} \frac{\partial z}{\partial x} + \frac{1}{y} \frac{\partial z}{\partial y} = -\frac{2yf'(u)}{f^2(u)} + \frac{1}{yf(u)} + \frac{2yf'(u)}{f^2(u)} = \frac{1}{yf(u)} = \frac{z}{y^2}.$$

4. 设 $f(u, v, w)$ 具有二阶连续偏导数, 求函数 $z = f(\sin x, \cos y, e^{x+y})$ 的二阶连续偏导数

$$\frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial x \partial y}, \frac{\partial^2 z}{\partial y^2} \text{ 及 } dz.$$

【解析】① $\frac{\partial z}{\partial x} = f'_1 \cdot \cos x + f'_3 \cdot e^{x+y}, \quad \frac{\partial z}{\partial y} = f'_1 \cdot (-\sin y) + f'_3 \cdot e^{x+y},$

$$\textcircled{2} \quad \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial x} (f'_1 \cdot \cos x + f'_3 \cdot e^{x+y})$$

$$= -\sin x \cdot f'_1 + \cos x \cdot (f''_{11} \cdot \cos x + f''_{13} \cdot e^{x+y}) + e^{x+y} \cdot f'_3 + e^{x+y} \cdot (f''_{31} \cdot \cos x + f''_{33} \cdot e^{x+y})$$

$$= -\sin x \cdot f'_1 + e^{x+y} \cdot f'_3 + \cos^2 x \cdot f''_{11} + 2 \cos x f''_{13} \cdot e^{x+y} + e^{2(x+y)} \cdot f''_{33}.$$

$$\textcircled{3} \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial y} (f'_1 \cdot \cos x + f'_3 \cdot e^{x+y})$$

$$= \cos x \cdot [f''_{12} \cdot (-\sin y) + f''_{13} \cdot e^{x+y}] + e^{x+y} \cdot f'_3 + e^{x+y} \cdot [f''_{32} \cdot (-\sin y) + f''_{33} \cdot e^{x+y}]$$

$$= e^{x+y} \cdot f'_3 - \cos x \cdot \sin y \cdot f''_{12} + \cos x \cdot e^{x+y} \cdot f''_{13} - \sin y \cdot e^{x+y} \cdot f''_{23} + f''_{33} \cdot e^{2(x+y)}$$

$$\textcircled{4} \quad \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial y} [f'_1 \cdot (-\sin y) + f'_3 \cdot e^{x+y}]$$

$$= -\cos y \cdot f'_1 - \sin y [f''_{22} \cdot (-\sin y) + f''_{23} \cdot e^{x+y}] + e^{x+y} \cdot f'_3 + e^{x+y} \cdot [f''_{32} \cdot (-\sin y) + f''_{33} \cdot e^{x+y}]$$

$$= -\cos y \cdot f'_1 + e^{x+y} \cdot f'_3 + \sin^2 y \cdot f''_{22} - 2e^{x+y} \cdot \sin y \cdot f''_{23} + f''_{33} \cdot e^{2(x+y)}$$

$$\textcircled{5} \quad dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = (f'_1 \cdot \cos x + f'_3 \cdot e^{x+y}) dx + (f'_1 \cdot (-\sin y) + f'_3 \cdot e^{x+y}) dy$$

5. 设 $u = f(r), r = \sqrt{x^2 + y^2 + z^2}$, 若 u 满足调和方程 $\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$, 试求函数 u .

【解析】① $\frac{\partial u}{\partial x} = f'(r) \cdot \frac{2x}{2\sqrt{x^2 + y^2 + z^2}} = \frac{x}{r} f'(r), \quad \frac{\partial u}{\partial y} = \frac{y}{r} f'(r), \quad \frac{\partial u}{\partial z} = \frac{z}{r} f'(r),$

$$\textcircled{2} \quad \frac{\partial^2 u}{\partial x^2} = \left[\frac{x}{r} f'(r) \right]'_x = \frac{1}{r} f'(r) + x \left(-\frac{1}{r^2} \right) \cdot \frac{x}{r} f'(r) + \frac{x}{r} f''(r) \cdot \frac{x}{r} = \frac{1}{r} f'(r) - \frac{x^2}{r^3} f'(r) + \frac{x^2}{r^2} f''(r);$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{1}{r} f'(r) - \frac{y^2}{r^3} f'(r) + \frac{y^2}{r^2} f''(r); \quad \frac{\partial^2 u}{\partial z^2} = \frac{1}{r} f'(r) - \frac{z^2}{r^3} f'(r) + \frac{z^2}{r^2} f''(r);$$

$$\textcircled{3} \quad \Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{2}{r} f'(r) + f''(r) = 0,$$

解微分方程得: $\frac{f''(r)}{f'(r)} = -\frac{2}{r} \Rightarrow \ln f'(r) = -2 \ln r + \ln C_1 \Rightarrow f'(r) = \frac{C_1}{r^2} \Rightarrow f(r) = -\frac{C_1}{r} + C_2$;

即 $u = -\frac{C_1}{\sqrt{x^2 + y^2 + z^2}} + C_2$, C_1, C_2 为任意实数.

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习题 10.4 隐函数求导法则

1. 设 $\frac{x}{z} = \ln \frac{z}{y}$, 求 $\frac{\partial z}{\partial x}$ 及 $\frac{\partial z}{\partial y}$.

【解析】① $F(x, y, z) = \frac{x}{z} - \ln \frac{z}{y} = \frac{x}{z} - \ln z + \ln y$,

② $F'_x = \frac{1}{z}$, $F'_y = \frac{1}{y}$, $F'_z = -\frac{x}{z^2} - \frac{1}{z}$,

③ $\frac{\partial z}{\partial x} = -\frac{F'_x}{F'_z} = -\frac{\frac{1}{z}}{-\frac{x}{z^2} - \frac{1}{z}} = \frac{x}{x+z}$, $\frac{\partial z}{\partial y} = -\frac{F'_y}{F'_z} = -\frac{\frac{1}{y}}{-\frac{x}{z^2} - \frac{1}{z}} = \frac{z^2}{y(x+z)}$.

2. 设 $e^z - xyz = 0$, 求 $\frac{\partial^2 z}{\partial x^2}$.

【解析】① $F(x, y, z) = e^z - xyz$,

② $F'_x = -yz$, $F'_z = e^z - xy$, $\frac{\partial z}{\partial x} = -\frac{F'_x}{F'_z} = \frac{yz}{e^z - xy}$,

③

$$\begin{aligned} \frac{\partial^2 z}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \frac{y \cdot \frac{\partial z}{\partial x} \cdot (e^z - xy) - yz \left(e^z \cdot \frac{\partial z}{\partial x} - y \right)}{(e^z - xy)^2} = \frac{y \cdot \frac{yz}{e^z - xy} \cdot (e^z - xy) - yz \left(e^z \cdot \frac{yz}{e^z - xy} - y \right)}{(e^z - xy)^2} \\ &= \frac{2y^2 z (e^z - xy) - y^2 z^2 e^z}{(e^z - xy)^3}. \end{aligned}$$

3. 设 $u = u(x, y)$, $v = v(x, y)$ 是由方程组 $\begin{cases} u^2 - v + x = 0 \\ u + v^2 - y = 0 \end{cases}$ 确定的 x, y 的隐函数, 求 $\frac{\partial u}{\partial x}$, $\frac{\partial v}{\partial y}$.

【解析】方程组两边对 x, y 分别求微分, 得

$$\begin{cases} 2u du - dv + dx = 0 \\ du + 2v dv - dy = 0 \end{cases}$$

消去 du : $(4uv + 1)dv = -dx + 2udy \Rightarrow dv = \frac{-dx + 2udy}{4uv + 1} \Rightarrow \frac{\partial v}{\partial y} = \frac{2u}{4uv + 1}$;

消去 dv : $(4uv + 1)du = -2vdx + dy \Rightarrow du = \frac{-2vdx + dy}{4uv + 1} \Rightarrow \frac{\partial u}{\partial x} = \frac{-2v}{4uv + 1}$.

4. 设 $\begin{cases} x = e^u + u \sin v \\ y = e^u - u \cos v \end{cases}$, 求 $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$, $\frac{\partial v}{\partial y}$.

【解析】方程组两边对 x, y 分别求偏导，得

$$1 = e^u \cdot \frac{\partial u}{\partial x} + \frac{\partial u}{\partial x} \cdot \sin v + u \cdot \cos v \cdot \frac{\partial v}{\partial x} \quad (1)$$

$$0 = e^u \cdot \frac{\partial u}{\partial y} + \frac{\partial u}{\partial y} \cdot \sin v + u \cdot \cos v \cdot \frac{\partial v}{\partial y} \quad (2)$$

$$0 = e^u \cdot \frac{\partial u}{\partial x} - \frac{\partial u}{\partial x} \cdot \cos v + u \cdot \sin v \cdot \frac{\partial v}{\partial x} \quad (3)$$

$$1 = e^u \cdot \frac{\partial u}{\partial y} - \frac{\partial u}{\partial y} \cdot \cos v + u \cdot \sin v \cdot \frac{\partial v}{\partial y} \quad (4)$$

(1)(3) 联立，解得

$$\frac{\partial u}{\partial x} = \frac{\sin v}{e^u (\sin v - \cos v) + 1}, \quad \frac{\partial v}{\partial x} = \frac{\cos v - e^u}{u [e^u (\sin v - \cos v) + 1]};$$

(2)(4) 联立，解得

$$\frac{\partial u}{\partial y} = \frac{-\cos v}{e^u (\sin v + \cos v) + 1}, \quad \frac{\partial v}{\partial y} = \frac{\sin v + e^u}{u [e^u (\sin v + \cos v) + 1]}.$$

5. 设 $u = f(x, y, z)$ 有连续偏导数， $y = y(x), z = z(x)$ 分别由方程 $e^{xy} - y = 0$ 和 $e^z - xz = 0$

所确定，求 $\frac{du}{dx}$.

【解析】① $\frac{du}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} + \frac{\partial f}{\partial z} \cdot \frac{dz}{dx}$,

$$② \quad e^{xy} - y = 0 \Rightarrow e^{xy}(y + xy') - y' = 0 \Rightarrow y' = \frac{ye^{xy}}{1 - xe^{xy}},$$

$$③ \quad e^z - xz = 0 \Rightarrow e^z \cdot z' - z - xz' = 0 \Rightarrow z' = \frac{z}{e^z - x},$$

$$④ \quad \frac{du}{dx} = f'_x + f'_y \cdot \frac{ye^{xy}}{1 - xe^{xy}} + f'_z \cdot \frac{z}{e^z - x}.$$

6. 设 $y = g(x, z)$ ，而 z 是由方程 $f(x - z, xy) = 0$ 所确定的 x, y 的函数，其中 g, f 具有一阶

偏导连续， $f'_1 - xf'_2 g'_2 \neq 0$ ，求 $\frac{dz}{dx}$.

【解析】 $f(x - z, xy) = 0$ 两边对 x 求导，即

$$f'_1 \cdot \left(1 - \frac{dz}{dx}\right) + f'_2 \cdot \left[y + x \left(g'_1 + g'_2 \cdot \frac{dz}{dx}\right)\right] = 0,$$

化简得

$$\frac{dz}{dx} = \frac{f'_1 + yf'_2 + xf'_2 \cdot g'_2}{f'_1 - xf'_2 \cdot g'_2}.$$

习题 10.5 偏导数在几何上的应用

1. 在曲线 $x=t, y=-t^2, z=t^3$ 的所有切线中, 求与平面 $x+2y+z+4=0$ 平行的切线方程.

【解析】切点 P 对应参数为 t_0 , 切向量 $\vec{T} = (1, -2t, 3t^2)|_{t=t_0} = (1, -2t_0, 3t_0^2)$ 与 $\vec{n} = (1, 2, 1)$ 垂直, 即

$$1 - 4t_0 + 3t_0^2 = 0 \Rightarrow t_0 = 1 \text{ 或 } \frac{1}{3},$$

$\vec{T}_1 = (1, -2, 3)$ 或 $\vec{T}_2 = \left(1, -\frac{2}{3}, \frac{1}{3}\right)$, 则切线方程为:

$$\frac{x-1}{1} = \frac{y+1}{-2} = \frac{z-1}{3} \text{ 或 } \frac{x-\frac{1}{3}}{1} = \frac{y+\frac{1}{9}}{-\frac{2}{3}} = \frac{z-\frac{1}{27}}{\frac{1}{3}}.$$

2. 求曲线 $\begin{cases} x^2 + y^2 + z^2 - 3x = 0 \\ 2x - 3y + 5z - 4 = 0 \end{cases}$ 在 $M(1, 1, 1)$ 处的切线与法平面方程.

【解析】方程组两边对 x 求导, 得 $\begin{cases} 2x + 2y \cdot \frac{dy}{dx} + 2z \cdot \frac{dz}{dx} - 3 = 0 \\ 2 - 3 \cdot \frac{dy}{dx} + 5 \cdot \frac{dz}{dx} = 0 \end{cases} \Rightarrow \begin{cases} 2y \cdot \frac{dy}{dx} + 2z \cdot \frac{dz}{dx} = 3 - 2x \\ 3 \cdot \frac{dy}{dx} - 5 \cdot \frac{dz}{dx} = 2 \end{cases}$

两式联立解得 $\begin{cases} \frac{dy}{dx} = \frac{15 - 10x + 4z}{6z + 10y} \\ \frac{dz}{dx} = \frac{9 - 6x - 4y}{6z + 10y} \end{cases}$

则 $\vec{T}|_{(1,1,1)} = \left(1, \frac{dy}{dx}, \frac{dz}{dx}\right)|_{(1,1,1)} = \left(1, \frac{9}{16}, -\frac{1}{16}\right) = \frac{1}{16}(16, 9, -1),$

即切线方程为: $\frac{x-1}{16} = \frac{y-1}{9} = \frac{z-1}{-1};$

法平面方程为: $16(x-1) + 9(y-1) - (z-1) = 0$, 化简得 $16x + 9y - z - 24 = 0.$

3. 在曲面 $z = xy$ 上求一点, 使得曲面在该点的法线垂直于平面 $x + 3y + z = 0$, 并求法线方程.

【解析】设该点坐标 $P_0(x_0, y_0, z_0)$, 该点的法向量为 $\vec{n}|_{P_0} = (-y_0, -x_0, 1),$

记平面 $x + 3y + z = 0$ 的法向量为 $\vec{n}_1 = (1, 3, 1)$, 由题意可知 $\vec{n}|_{P_0} \perp \vec{n}_1$, 则 $\frac{-y_0}{1} = \frac{-x_0}{3} = 1$, 解

$$x_0 = -3, y_0 = -1, z_0 = x_0 y_0 = 3,$$

所以 $P_0(-3, -1, 3)$, $\vec{n}|_{P_0} = (1, 3, 1),$

则法线方程为: $\frac{x+3}{1} = \frac{y+1}{3} = \frac{z-3}{1}$.

4. 设直线 $l_1: \begin{cases} x+y+b=0 \\ x+ay-z-3=0 \end{cases}$ 在平面 π 上, 而平面 π 与曲面 $z=x^2+y^2$ 相切于点

$(1, -2, 5)$, 求 a, b 之值.

【解析】设 $F(x, y, z) = x^2 + y^2 - z$, 则曲面 $S: z = x^2 + y^2$ 在 $(1, -2, 5)$ 处的法向量为

$$(F'_x, F'_y, F'_z)|_{(1, -2, 5)} = (2x, 2y, -1)|_{(1, -2, 5)} = (2, -4, -1),$$

由题意可知平面 π 的方程为: $2(x-1) - 4(y+2) - (z-5) = 0$, 化简得 $2x - 4y - z - 5 = 0$;

由 l_1 的方程可知 $y = -b - x$, 所以 $z = x + ay - 3 = x + a(-b - x) - 3 = (1-a)x - ab - 3$ 代入平面 π

方程, 得

$$2x - 4(-b - x) - (1-a)x + ab + 3 - 5 = 0,$$

化简得 $(5+a)x + 4b + ab - 2 = 0$, 即 $\begin{cases} 5+a=0 \\ 4b+ab-2=0 \end{cases} \Rightarrow \begin{cases} a=-5 \\ b=-2 \end{cases}$.

习题 10.7 多元函数的极值

1. 求函数 $z = e^{2x}(x + y^2 + 2y)$ 的极值.

【解析】(1) $\begin{cases} z'_x = 2e^{2x}(x + y^2 + 2y) + e^{2x} = e^{2x}[2x + 2y^2 + 4y + 1] = 0 \\ z'_y = e^{2x}(2y + 2) = 0 \Rightarrow y = -1 \end{cases}$, 得驻点 $P\left(\frac{1}{2}, -1\right)$;

(2) $A = z''_{xx} = e^{2x}(2x + 2y^2 + 4y + 2)$, $B = z''_{xy} = 4e^{2x}(y + 1)$, $C = z''_{yy} = 2e^{2x}$;

(3) $A|_P = 2e$, $B|_P = 0$, $C|_P = 2e$;

(4) $B^2 - AC = -4e^2 < 0$, $A|_P = 2e > 0$, 则 $f_{\text{极小值}}(P) = -\frac{e}{2}$.

2. 求函数 $z = x^2y(4 - x - y)$ 在 $x = 0, y = 0$ 及 $x + y = 6$ 围成的区域上的最大值及最小值.

【解析】(1) 先求出函数在 D 内的所有驻点和偏导数不存在的点, 解方程得:

$$\begin{cases} f'_x(x, y) = 2xy(4 - x - y) - x^2y = xy(8 - 3x - 2y) = 0 \\ f'_y(x, y) = x^2(4 - x - y) - x^2y = x^2(4 - x - 2y) = 0 \end{cases}$$

得到区域 D 内的唯一驻点 $(2, 1)$, 且 $f(2, 1) = 4$;

(2) 再求 $f(x, y)$ 在 D 的边界上的极值.

① 在边界 $x=0$ 和 $y=0$ 上 $f(x, y) = 0$;

② 在边界 $x+y=6$ 上, 即 $y=6-x (0 \leq x \leq 6)$, 于是

$$f(x, y) = -2x^2(6 - x) \quad (0 \leq x \leq 6)$$

$f'_x = 4x(x - 6) + 2x^2 = 0 \Rightarrow x_1 = 0, x_2 = 4$, 则驻点为 $(0, 6), (4, 2)$, 则 $f(0, 6) = 0$, $f(4, 2) = -64$; 又 $f(6, 0) = 0$

(3) 综上所述得到 $f(2, 1) = 4$ 为最大值, $f(4, 2) = -64$ 为最小值.

3. 求内接于半径为 R 的球且有最大体积的长方体.

【解析】设长方体的长、宽、高分别为 $2x, 2y, 2z$, 则长方体体积 $V = 8xyz$ ($x, y, z > 0$), 而 x, y, z 应满

$$x^2 + y^2 + z^2 = R^2;$$

构建拉格朗日辅助函数: $L(x, y, z, \lambda) = 8xyz + \lambda(x^2 + y^2 + z^2 - R^2)$

$$\begin{cases} L'_x = 8yz + 2\lambda x = 0 \\ L'_y = 8xz + 2\lambda y = 0 \\ L'_z = 8xy + 2\lambda z = 0 \\ L'_\lambda = x^2 + y^2 + z^2 - R^2 = 0 \end{cases}$$

上述方程满足轮换对称式, 则可知 $x = y = z$ 代入最后一个方程, 解得 $x = y = z = \frac{R}{\sqrt{3}}$, 且 $V_{\text{max}} = \frac{8}{3\sqrt{3}}R^3$.

4. 抛物面 $z = x^2 + y^2$ 与平面 $x + y + z = 1$ 的交线为一椭圆, 求原点到这椭圆的最长与最短距离.

【解析】设从原点到椭圆上任一点 (x, y, z) 的距离为 $d = \sqrt{x^2 + y^2 + z^2}$.

构建拉格朗日辅助函数: $L(x, y, z, \lambda) = x^2 + y^2 + z^2 + \lambda(x^2 + y^2 - z) + \mu(x + y + z - 1)$

$$\begin{cases} L'_x = 2x + 2\lambda x + \mu = 0 & (1) \\ L'_y = 2y + 2\lambda y + \mu = 0 & (2) \\ L'_z = 2z - \lambda + \mu = 0 & (3) \\ L'_\lambda = x^2 + y^2 - z = 0 & (4) \\ L'_\mu = x + y + z - 1 = 0 & (5) \end{cases}$$

由 (1) (2) 推出 $x = y$ 代入 (4) (5) 得 $2y^2 - z = 0, 2y + z - 1 = 0$ ，联立解得

$$y = \frac{-1 \pm \sqrt{3}}{2}, x = \frac{-1 \pm \sqrt{3}}{2}, z = 2 \mp \sqrt{3},$$

$$\text{则 } d_{\min} \left(\frac{-1 + \sqrt{3}}{2}, \frac{-1 + \sqrt{3}}{2}, 2 - \sqrt{3} \right) = \sqrt{9 - 5\sqrt{3}}, \quad d_{\max} \left(\frac{-1 - \sqrt{3}}{2}, \frac{-1 - \sqrt{3}}{2}, 2 + \sqrt{3} \right) = \sqrt{9 + 5\sqrt{3}}.$$

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自测题

一、填空题（每题 4 分，共 20 分）.

1. 极限 $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \sin \frac{1}{x^2 + y^2} = \underline{\hspace{2cm}}.$

【解析】答案是 0. 利用有界量乘无穷小量仍为无穷小量.

2. 设 $z = e^{\sin xy}$, 则 $dz = \underline{\hspace{2cm}}.$

【解】 $dz = e^{\sin xy} \cdot \cos xy \cdot (ydx + xdy).$

3. 设 $z = z(x, y)$ 可微, 且满足 $\frac{\partial z}{\partial y} = x^2 + 2y$, 且 $z(x, x^2) = 1$, 则 $z(x, y) = \underline{\hspace{2cm}}.$

【解析】 $\frac{\partial z}{\partial y} = x^2 + 2y \Rightarrow z = x^2 y + y^2 + \varphi(x)$, 又 $z(x, x^2) = 1$, 得 $\varphi(x) = 1 - 2x^4$, 则

$$z(x, y) = x^2 y + y^2 + 1 - 2x^4.$$

4. 设 $f(x, y, z) = e^x y z^2$, 其中 $z = z(x, y)$ 是由 $x + y + z + xyz = 0$ 确定的隐函数, 则 $f'_x(0, 1, -1) = \underline{\hspace{2cm}}.$

【解析】 $f'_x = e^x y z^2 + e^x y \cdot 2z \cdot \frac{\partial z}{\partial x}$; 下只要求 $\frac{\partial z}{\partial x}$ 即可;

方程 $x + y + z + xyz = 0$ 两边对 x 求偏导, 得 $1 + z'_x + yz + xyz'_x = 0$, 代值得 $z'_x|_{(0,1,-1)} = 0$, 进而得 $f'_x(0, 1, -1) = 1.$

5. 函数 $z = x^3 - 4x^2 + 2xy - y^2$ 的极值是 $\underline{\hspace{2cm}}.$

【解析】按照无条件极值的计算方法, 计算得极值为 0.

二、选择题（每小题 4 分，共 20 分）.

6. 设 $z = \begin{cases} \frac{x^2 y}{x^2 + y^2}, & x^2 + y^2 \neq 0 \\ 0, & x^2 + y^2 = 0 \end{cases}$, 则函数 z 在点 $(0, 0)$ 处 ().

(A) 不连续

(B) 连续, 但偏导数 $z'_x(0, 0)$ 和 $z'_y(0, 0)$ 不存在

(C) 连续且偏导数 $z'_x(0, 0)$ 和 $z'_y(0, 0)$ 都存在, 但不可微 (D) 可微

【解析】答案选 C. 上课作为例题详细讲解过.

7. 考虑二元函数 $f(x, y)$ 下面 4 条性质:

① $f(x, y)$ 在点 (x_0, y_0) 处连续 ② $f(x, y)$ 在点 (x_0, y_0) 处的两个偏导数连续

③ $f(x, y)$ 在点 (x_0, y_0) 处可微 ④ $f(x, y)$ 在点 (x_0, y_0) 处的两个偏导数存在

若用“ $P \Rightarrow Q$ ”表示可由性质 P 推出性质 Q , 则有 ().

(A) ② $\Rightarrow$ ③ $\Rightarrow$ ①

(B) ③ $\Rightarrow$ ② $\Rightarrow$ ①

$$(C) ③ \Rightarrow ④ \Rightarrow ①$$

$$(D) ③ \Rightarrow ① \Rightarrow ④$$

【解析】答案选择 A. 上课讲解过之间关系图

8. 已知函数 $f(x+y, x-y) = x^2 - y^2$ 对任何 x 与 y 成立, 则 $\frac{\partial f(x, y)}{\partial x} + \frac{\partial f(x, y)}{\partial y}$ 等于 ().

$$(A) 2x - 2y$$

$$(B) 2x + 2y$$

$$(C) x + y$$

$$(D) x - y$$

【解析】由题意可知 $f(x, y) = xy$, 得答案选 C.

9. 曲线 $\begin{cases} z = \frac{1}{4}(x^2 + y^2) \\ y = 4 \end{cases}$ 在 $P_0(2, 4, 5)$ 点的法平面方程为 ().

$$(A) x + y - 7 = 0$$

$$(B) x + z - 7 = 0$$

$$(C) x - y + 7 = 0$$

$$(D) x - z - 7 = 0$$

【解析】方程组两边分别对 x 求导, 得 $\begin{cases} z'_x = \frac{1}{4}(2x + 2y \cdot y'_x) \\ y'_x = 0 \end{cases} \Rightarrow \begin{cases} z'_x = \frac{1}{2}x \\ y'_x = 0 \end{cases}$, 则切向量 $\vec{T} = (1, 0, \frac{1}{2}x)$,

切向量坐标为 $\vec{T}|_{(2, 4, 5)} = (1, 0, 1)$, 进而法平面方程为 $(x-2) + (z-5) = 0$, 化简得 $x + z - 7 = 0$, 答案选 B.

10. 函数 $f(x, y) = x^2 - ay^2$ ($a > 0$) 在 $(0, 0)$ 处 ().

$$(A) \text{不取极值}$$

$$(B) \text{取极小值}$$

$$(C) \text{取极大值}$$

$$(D) \text{是否取极值依赖于 } a$$

【解析】由极值的定义可知正确答案选 A.

三、解答题 (每小题 10 分, 共 60 分).

11. 设 $z = (x^2 + y^2)e^{-\arctan \frac{y}{x}}$, 求 $dz, \frac{\partial^2 z}{\partial x \partial y}$.

【解析】(1) $\frac{\partial z}{\partial x} = 2xe^{-\arctan \frac{y}{x}} + (x^2 + y^2) \cdot e^{-\arctan \frac{y}{x}} \cdot \left[-\frac{1}{1 + \left(\frac{y}{x}\right)^2} \right] \cdot \left(-\frac{y}{x^2}\right) = (2x + y)e^{-\arctan \frac{y}{x}}$, $\frac{\partial z}{\partial x} = (2y - x)e^{-\arctan \frac{y}{x}}$;

$$(2) dz = (2x + y)e^{-\arctan \frac{y}{x}} dx + (2y - x)e^{-\arctan \frac{y}{x}} dy;$$

$$(3) \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left[(2x + y)e^{-\arctan \frac{y}{x}} \right] = e^{-\arctan \frac{y}{x}} + (2x + y)e^{-\arctan \frac{y}{x}} \cdot \left[-\frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \frac{1}{x} \right] = \frac{y^2 - x^2 - xy}{x^2 + y^2} e^{-\arctan \frac{y}{x}}.$$

12. 设 $u = xy, v = \frac{x}{y}, z = z(u, v)$ 对每个变量有二阶连续偏导数, 计算 $x^2 \frac{\partial^2 z}{\partial x^2} - y^2 \frac{\partial^2 z}{\partial y^2}$.

【解析】(1) $\frac{\partial z}{\partial x} = z'_1 \cdot y + z'_2 \cdot \frac{1}{y}$, $\frac{\partial z}{\partial y} = z'_1 \cdot x + z'_2 \cdot \left(-\frac{x}{y^2}\right) = z'_1 \cdot x - \frac{x}{y^2} z'_2$;

$$(2) \frac{\partial^2 z}{\partial x^2} = y \cdot \left(z''_{11} \cdot y + z''_{12} \cdot \frac{1}{y} \right) + \frac{1}{y} \cdot \left(z''_{21} \cdot y + z''_{22} \cdot \frac{1}{y} \right) = y^2 \cdot z''_{11} + 2z''_{12} + \frac{1}{y^2} z''_{22},$$

$$\frac{\partial^2 z}{\partial y^2} = x \cdot \left[z''_{11} \cdot x + z''_{12} \cdot \left(-\frac{x}{y^2} \right) \right] + \frac{2x}{y^3} \cdot z'_2 - \frac{x}{y^2} \left[z''_{21} \cdot x + z''_{22} \cdot \left(-\frac{x}{y^2} \right) \right] = x^2 \cdot z''_{11} - \frac{2x^2}{y^2} z''_{12} + \frac{2x}{y^3} z'_2 + \frac{x^2}{y^4} z''_{22} ;$$

$$(3) \quad x^2 \frac{\partial^2 z}{\partial x^2} - y^2 \frac{\partial^2 z}{\partial y^2} = 4x^2 \cdot z''_{12} - \frac{2x}{y} \cdot z'_2 .$$

13. 设 $z = f(2x-y) + g(x, xy)$, 其中 $f(t)$ 二阶可导, $g(u, v)$ 具有连续二阶偏导数, 求 $\frac{\partial^2 z}{\partial x \partial y}$.

【解析】(1) $\frac{\partial z}{\partial x} = f' \cdot 2 + g'_1 \cdot 1 + g'_2 \cdot y = 2f' + g'_1 + yg'_2$;

$$(2) \quad \frac{\partial^2 z}{\partial x \partial y} = (2f' + g'_1 + yg'_2)'_y = 2f'' \cdot (-1) + g''_{12} \cdot x + g'_2 + yg''_{22} \cdot x = -2f'' + xg''_{12} + g'_2 + xyg''_{22} .$$

14. 设 $u = u(x, y), v = v(x, y)$, 由方程组 $\begin{cases} u^2 - v = 3x + y \\ u^2 - 2v^2 = x - 2y \end{cases}$ 确定, 求 $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$.

【解析】方程组两边同时取微分, 得 $\begin{cases} 2udu - dv = 3dx + dy & (1) \\ 2udu - 4vdv = dx - 2dy & (2) \end{cases}$,

$$(1) \times 4v - (2) \text{ 得 } (8uv - 2u)du = (12v - 1)dx + (4v + 2)dy \Rightarrow du = \frac{12v - 1}{8uv - 2u}dx + \frac{4v + 2}{8uv - 2u}dy, \text{ 则}$$

$$\frac{\partial u}{\partial x} = \frac{12v - 1}{8uv - 2u}, \quad \frac{\partial u}{\partial y} = \frac{4v + 2}{8uv - 2u}.$$

$$(1) - (2) \text{ 得 } (4v - 1)dv = 2dx + 3dy \Rightarrow dv = \frac{2}{4v - 1}dx + \frac{3}{4v - 1}dy, \text{ 则}$$

$$\frac{\partial v}{\partial x} = \frac{2}{4v - 1}, \quad \frac{\partial v}{\partial y} = \frac{3}{4v - 1}.$$

15. 设曲面 $F(x, y, z) = 0$ 在点 $P(1, 1, 1)$ 处法向量为 $\vec{n} = \{1, 2, 3\}$, 求曲面 $F(x, y^2, z^3) = 0$ 在点 $P(1, 1, 1)$ 处的法线与切平面方程.

【解析】(1) $F'_x(1, 1, 1) = 1, F'_y(1, 1, 1) = 2, F'_z(1, 1, 1) = 3$,

(2) $\vec{n} = (F'_x \cdot 1, F'_y \cdot 2y, F'_z \cdot 3z^2)$, 则 $\vec{n}|_P = (1, 4, 9)$;

(3) 法线方程为: $\frac{x-1}{1} = \frac{y-1}{4} = \frac{z-1}{9}$;

(4) 切平面方程为: $(x-1) + 4(y-1) + 9(z-1) = 0$, 化简得 $x + 4y + 9z - 14 = 0$.

16. 在第一卦限内作椭球面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ 的切平面, 使该切平面与三个坐标平面围成四面体体积最小, 求切点坐标.

【解析】(1) 设 $P_0(x_0, y_0, z_0)$ 为椭球面上任一点, x_0, y_0, z_0 均大于零.

在 P_0 处法向量为 $(F'_x, F'_y, F'_z)|_{P_0} = \left(\frac{2x_0}{a^2}, \frac{2y_0}{b^2}, \frac{2z_0}{c^2} \right)$, 则切平面方程为

$$\frac{x_0}{a^2}(x - x_0) + \frac{y_0}{b^2}(y - y_0) + \frac{z_0}{c^2}(z - z_0) = 0,$$

化简为: $\frac{x_0}{a^2}x + \frac{y_0}{b^2}y + \frac{z_0}{c^2}z = 1$, 所以切平面在三个坐标轴上的截距分别为 $\frac{a^2}{x_0}, \frac{b^2}{y_0}, \frac{c^2}{z_0}$, 于是由该切平面与三坐标轴围

成四面体体积为 $V = \frac{a^2 b^2 c^2}{6x_0 y_0 z_0}$.

(2) 要使得 $V = \frac{a^2 b^2 c^2}{6xyz}$ 取得最小值, 只要 $u = xyz$ 取得最大值即可, 故原问题转化为求 $u = xyz$ 在 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ 下最大值.

构建拉格朗日辅助函数 $L = xyz + \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 \right)$, 则

$$\begin{cases} L'_x = yz + \frac{2\lambda x}{a^2} = 0 & (1) \end{cases}$$

$$\begin{cases} L'_y = zx + \frac{2\lambda y}{b^2} = 0 & (2) \end{cases}$$

$$\begin{cases} L'_z = xy + \frac{2\lambda z}{c^2} = 0 & (3) \end{cases}$$

$$\begin{cases} L'_\lambda = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0 & (4) \end{cases}$$

由 (1) (2) (3) 联立, 得 $-xyz = \frac{2\lambda x^2}{a^2} = \frac{2\lambda y^2}{b^2} = \frac{2\lambda z^2}{c^2}$, 则 $\frac{x^2}{a^2} = \frac{y^2}{b^2} = \frac{z^2}{c^2}$, 代入 (4) 中, 得

$$x = \frac{a}{\sqrt{3}}, y = \frac{b}{\sqrt{3}}, z = \frac{c}{\sqrt{3}},$$

所以 $u = xyz$ 在 $\left(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}}, \frac{c}{\sqrt{3}} \right)$ 处取最大值, 故切点坐标为 $\left(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}}, \frac{c}{\sqrt{3}} \right)$, $V_{\min} = \frac{\sqrt{3}}{3} abc$.

第十一章 重积分

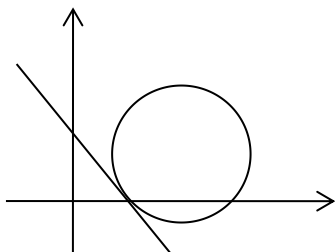
习题 11.1 二重积分的概念与性质

1. 利用二重积分的性质, 比较下列二重积分的大小.

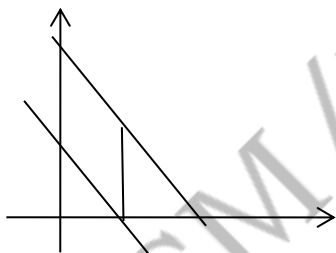
(1) $\iint_D (x+y)^2 d\sigma$ 与 $\iint_D (x+y)^3 d\sigma$, 其中 D 是由圆周 $(x-2)^2 + (y-1)^2 = 2$ 所围成.

【解析】由题意及图像可知 $x+y=1$ 为切线, $x+y \geq 1$, 则 $(x+y)^2 \leq (x+y)^3$, 由保号性可知

$$\iint_D (x+y)^2 d\sigma \leq \iint_D (x+y)^3 d\sigma$$



(2) $\iint_D \ln(x+y) d\sigma$ 与 $\iint_D [\ln(x+y)]^2 d\sigma$, 其中 D 是由三角形闭区域, 三顶点分别为 $(1,0)$, $(1,1)$, $(2,0)$.



【解析】

由图像可知 $1 \leq x+y \leq 2 < e$, 则 $0 \leq \ln(x+y) < 1$, 所以 $\ln(x+y) \geq [\ln(x+y)]^2$,

则

$$\iint_D \ln(x+y) d\sigma \geq \iint_D [\ln(x+y)]^2 d\sigma$$

2. 利用二重积分的性质估计下列积分的值.

(1) $I = \iint_D \sin^2 x \sin^2 y d\sigma$, 其中 $D = \{(x,y) | 0 \leq x \leq \pi, 0 \leq y \leq \pi\}$.

【解析】(1) $S_D = \pi^2$;

(2) $f(x,y) = \sin^2 x \sin^2 y$, $0 \leq f(x,y) \leq 1$, 则 $I \in [0, \pi^2]$.

(2) $I = \iint_D (x^2 + 4y^2 + 9) d\sigma$, 其中 $D = \{(x,y) | x^2 + y^2 \leq 4\}$.

【解析】(1) $S_D = 4\pi$;

(2) $f(x,y) = x^2 + 4y^2 + 9$ 在 D 上最值问题, 计算最大值为 25, 最小值为 9;

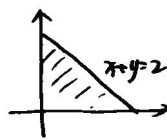
(3) $I \in [36\pi, 100\pi]$.

习题 11.2 二重积分的计算

1. 计算下列二重积分.

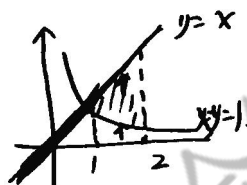
(1) $\iint_D (3x+2y)d\sigma$, 其中 D 是由 $x=0, y=0$ 及直线 $x+y=2$ 所围成的区域.

【解析】原式 $= \int_0^2 dx \int_0^{2-x} (3x+2y) dy = \frac{20}{3}$



(2) $\iint_D \frac{y^2}{x^2} dx dy$, 其中 D 是由直线 $x=2, y=x$ 及双曲线 $xy=1$ 所围成的区域.

【解析】原式 $= \int_1^2 dx \int_{\frac{1}{x}}^x \frac{y^2}{x^2} dy$
 $= \int_1^2 \frac{1}{x^2} \cdot \left[\frac{1}{3} y^3 \right]_{\frac{1}{x}}^x dx = \int_1^2 \frac{1}{x^2} \cdot \frac{1}{3} \left(x^3 - \frac{1}{x^3} \right) dx = \frac{27}{64}$

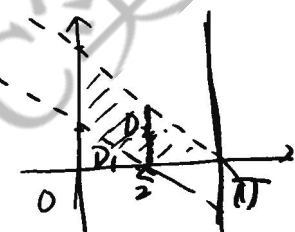


(3) $\iint_D |\cos(x+y)| dx dy$, 其中 D 是由 $0 \leq x \leq \pi, 0 \leq y \leq \pi-x$ 确定的区域.

【解析】 D 被 $x+y=\frac{\pi}{2}$ 划分为 D_1, D_2 , 如图所示

原式 $= \iint_{D_1} \cos(x+y) dx dy + \iint_{D_2} [-\cos(x+y)] dx dy$

$= \int_0^{\frac{\pi}{2}} dx \int_0^{\frac{\pi}{2}-x} \cos(x+y) dy + \int_0^{\frac{\pi}{2}} dx \int_{\frac{\pi}{2}-x}^{\pi-x} [-\cos(x+y)] dy + \int_{\frac{\pi}{2}}^{\pi} dx \int_0^{\pi-x} [-\cos(x+y)] dy = \pi$



2. 画出下列二次积分所表示的二重积分的积分区域, 并交换积分次序.

(1) $\int_1^e dx \int_0^{\ln x} f(x, y) dy$;

【解析】原式 $= \int_0^1 dy \int_{e^y}^e f(x, y) dx$

(2) $\int_0^1 dx \int_0^{x^2} f(x, y) dy + \int_1^2 dx \int_0^{\sqrt{1-(x-1)^2}} f(x, y) dy$.

【解析】原式 $= \int_0^1 dy \int_{\sqrt{y}}^{1+\sqrt{1-y^2}} f(x, y) dx$

3. 计算下列二重积分, 必要时交换积分次序.

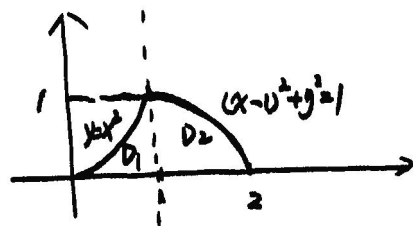
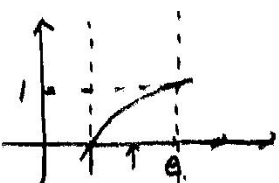
(1) $\int_0^a dx \int_x^a e^{y^2} dy$;

【解析】 $\int_0^a dx \int_x^a e^{y^2} dy = \int_0^a dy \int_0^y e^{y^2} dx = \int_0^a y e^{y^2} dy$
 $= \frac{1}{2} \int_0^a e^{y^2} dy^2 = \frac{1}{2} e^{y^2} \Big|_0^a = \frac{1}{2} (e^{a^2} - 1)$

(2) $\int_0^1 dx \int_x^{\sqrt{x}} \frac{\sin y}{y} dy$.

【解析】 $\int_0^1 dx \int_x^{\sqrt{x}} \frac{\sin y}{y} dy = \int_0^1 dy \int_{y^2}^y \frac{\sin y}{y} dy$

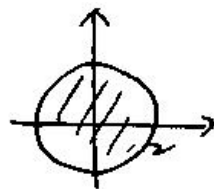
$= \int_0^1 \frac{\sin y}{y} (y - y^2) dy = \int_0^1 (\sin y - y \sin y) dy$



$$= \int_0^1 \sin y dy - \int_0^1 y \sin y dy = \sin 1 - 1$$

4. 选择适当的坐标系计算下列积分:

(1) $\iint_D e^{x^2+y^2} dx dy$, 其中 D 是由圆周 $x^2 + y^2 = 4$ 所围成的区域.



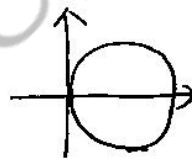
【解析】 $\iint_D e^{x^2+y^2} dx dy = \int_0^{2\pi} d\theta \int_0^2 e^{r^2} \cdot r dr = 2\pi \cdot \frac{1}{2} e^{r^2} \Big|_0^2 = \pi(e^4 - 1)$

(2) $\iint_D (x+y) dx dy$, 其中 $D = \{(x, y) | x^2 + y^2 \leq x + y\}$.



【解析】 $\iint_D (x+y) dx dy = \int_{-\pi/4}^{3\pi/4} d\theta \int_0^{\sin\theta+\cos\theta} (r \cos \theta + r \sin \theta) \cdot r dr$
 $= \int_{-\pi/4}^{3\pi/4} (\cos \theta + \sin \theta) d\theta \int_0^{\sin\theta+\cos\theta} r^2 dr = \frac{1}{3} \int_{-\pi/4}^{3\pi/4} (\cos \theta + \sin \theta)^4 d\theta = \frac{\pi}{2}$

(3) $\iint_D \frac{x^2}{x^2 + y^2} dx dy$, 其中 $D = \{(x, y) | x^2 + y^2 \leq x\}$.



【解析】 $\iint_D \frac{x^2}{x^2 + y^2} dx dy = \int_{-\pi/2}^{\pi/2} d\theta \int_0^{\cos\theta} \frac{r^2 \cos^2 \theta}{r^2} \cdot r dr$
 $= \int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta \int_0^{\cos\theta} r dr = \frac{1}{2} \int_{-\pi/2}^{\pi/2} \cos^4 \theta d\theta = \int_0^{\pi/2} \cos^4 \theta d\theta = \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} = \frac{3\pi}{16}$

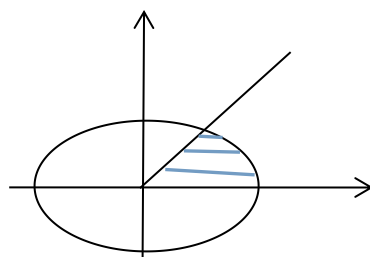
(4) $\iint_D \sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2}} dx dy$, 其中 D 是由椭圆 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 4$ 和直线 $y=0, y=x$ 所围成的第一象限部分.

【解析】 利用广义极坐标: $x = ar \cos \theta, y = br \sin \theta$, 其中 $0 \leq \theta \leq \arctan \frac{a}{b}, 0 \leq r \leq 2$;

则 $\frac{\partial(x, y)}{\partial(r, \theta)} = abr, dx dy = r dr d\theta$;

$$\iint_D \sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2}} dx dy = \int_0^{\arctan \frac{a}{b}} d\theta \int_0^2 \sqrt{\frac{(ar \cos \theta)^2}{a^2} + \frac{(br \sin \theta)^2}{b^2}} \cdot abr dr$$

$$= \arctan \frac{a}{b} \int_0^2 abr^2 dr = \frac{8}{3} ab \cdot \arctan \frac{a}{b}$$



(5) $\iint_D xy dx dy$, 其中 D 由 $xy=a, xy=b, y^2=cx, y^2=dx$ 所围成的第一象限部分 ($0 < a < b, 0 < c < d$).

【解析】 利用任意坐标变换: 令 $xy = u, \frac{y^2}{x} = v \Rightarrow x = \left(\frac{u^2}{v}\right)^{\frac{1}{3}}, y = (uv)^{\frac{1}{3}}$, 则

$$D_1: a \leq u \leq b, c \leq v \leq d \text{ 且 } \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \frac{1}{3v},$$

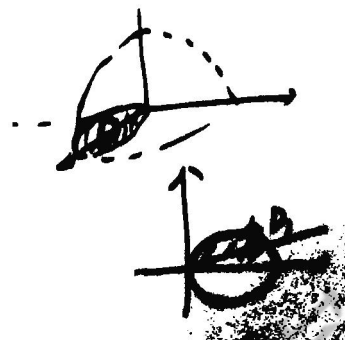
$$\iint_D xy dx dy = \iint_{D_1} u \cdot \frac{1}{3v} du dv = \frac{1}{3} \int_a^b u du \int_c^d \frac{1}{v} dv = \frac{1}{6} (b^2 - a^2) \ln \frac{d}{c}$$

5. 求由柱面 $x^2 + y^2 = 2ax$ 围成的柱体被球面 $x^2 + y^2 + z^2 = 4a^2$ 所截得部的体积.

【解析】所求体积以 $D: (x-a)^2 + y^2 \leq a^2$ 为底，以 $z = \sqrt{4a^2 - x^2 - y^2}$ 围成体积的 2 倍，则

$$V = 2 \iint_D \sqrt{4a^2 - x^2 - y^2} dx dy = 4 \iint_{D_1} \sqrt{4a^2 - x^2 - y^2} dx dy$$

$$= 4 \int_0^{\frac{\pi}{2}} d\theta \int_0^{2a \cos \theta} \sqrt{4a^2 - r^2} \cdot r dr = -\frac{8a^3}{3} \int_0^{\frac{\pi}{2}} (\sin^3 \theta - 1) d\theta = \frac{32a^3}{3} \left(\frac{\pi}{2} - \frac{2}{3} \right)$$



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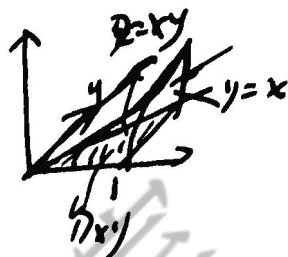
习题 11.3 三重积分

1. 计算下列三重积分:

(1) $\iiint_V xy^2 z^3 dx dy dz$, 其中 V 由 $z = xy, y = x, x = 1$ 和 $z = 0$ 所围成.

【解析】 V 投影到 xoy 面的投影区域 $D_{xy} : \begin{cases} 0 \leq x \leq 1 \\ 0 \leq y \leq x \end{cases}, 0 \leq z \leq xy$

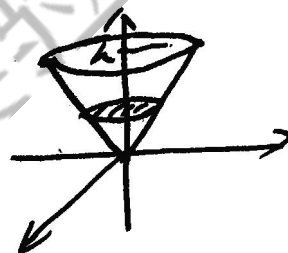
$$\text{原式} = \iint_{D_{xy}} dx dy \int_0^{xy} xy^2 z^3 dz = \frac{1}{4} \iint_{D_{xy}} xy^2 \cdot (xy)^4 dx dy = \frac{1}{4} \int_0^1 dx \int_0^x x^5 y^6 dy = \frac{1}{364}.$$



(2) $\iiint_V z dx dy dz$, 其中 V 由 $z = \frac{h}{R} \sqrt{x^2 + y^2}$ 与平面 $z = h$ ($R > 0, h > 0$) 所围成.

【解析】 $D_z : x^2 + y^2 \leq \left(\frac{R}{h} z\right)^2, 0 \leq z \leq h$

$$\text{原式} = \int_0^h z dz \iint_{D_z} dx dy = \int_0^h z \cdot \pi \left(\frac{R}{h} z\right)^2 dz = \pi \cdot \frac{R^2}{h^2} \int_0^h z^3 dz = \frac{\pi}{4} R^2 h^2$$

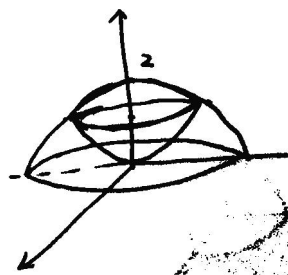


(3) $\iiint_V z dx dy dz$, 其中 V 由球面 $z = \sqrt{4 - x^2 - y^2}$ 与抛物面 $z = \frac{1}{3}(x^2 + y^2)$ 所围成.

【解析】 $D_{xy} : x^2 + y^2 \leq 3$, 利用柱坐标方程计算; $x = r \cos \theta, y = r \sin \theta, z = z$

则 $V' : 0 \leq \theta \leq 2\pi, 0 \leq r \leq \sqrt{3}, \frac{1}{3}r^2 \leq z \leq \sqrt{4 - r^2}$;

$$\text{原式} = \int_0^{2\pi} d\theta \int_0^{\sqrt{3}} r dr \int_{\frac{1}{3}r^2}^{\sqrt{4-r^2}} z dz = 2\pi \int_0^{\sqrt{3}} r \cdot \frac{1}{2} \left[4 - r^2 - \frac{4}{9}r^4 \right] dr = \frac{13}{4}\pi$$



(4) $\iiint_V z dx dy dz$, 其中 V 由 $x^2 + y^2 + (z - a)^2 \leq a^2$ 及 $x^2 + y^2 \leq z^2$ 所围成.

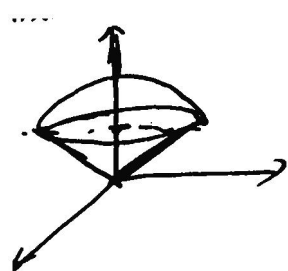
【解析】 $\begin{cases} x^2 + y^2 + (z - a)^2 = a^2 \\ x^2 + y^2 = z^2 \end{cases} \Rightarrow z = 0 \text{ 或 } z = a, D_{xy} : x^2 + y^2 \leq a^2$;

选择球坐标方程: $x = r \sin \varphi \cos \theta, y = r \sin \varphi \sin \theta, z = r \cos \varphi$,

得 $0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \frac{\pi}{4}, 0 \leq r \leq 2a \cos \varphi$,

$$\text{原式} = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^{2a \cos \varphi} (r \cos \varphi)(r^2 \sin \varphi) dr$$

$$= 2\pi \int_0^{\frac{\pi}{4}} \cos \varphi \sin \varphi d\varphi \int_0^{2a \cos \varphi} r^3 dr = 2\pi \int_0^{\frac{\pi}{4}} 4a^4 \cos^5 \varphi \sin \varphi d\varphi = \frac{7}{6}\pi a^4$$



$$(5) \iiint_V \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2}} dx dy dz, \text{ 其中 } V \text{ 为椭球体 } \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1.$$

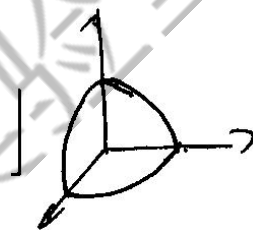
【解析】利用广义球坐标方程: $x = ra \sin \varphi \cos \theta, y = rb \sin \varphi \sin \theta, z = rc \cos \varphi$, 得

$$0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \pi, 0 \leq r \leq 1, dx dy dz = abcr^2 \sin \varphi,$$

$$\text{原式} = \int_0^{2\pi} d\theta \int_0^\pi d\varphi \int_0^1 \sqrt{1-r^2} \cdot ab r^2 \sin \varphi dr = \frac{\pi^2}{4} abc$$

2. 设物体占有的空间区域为球面 $x^2 + y^2 + z^2 = 1$ 及三个坐标面在第一卦限内的部分, 点 (x, y, z) 处的体密度为 $\rho(x, y, z) = xyz$, 求物体的质量.

【解析】 $M = \iiint_V xyz dV$, 其中 $V: x^2 + y^2 + z^2 \leq 1, x \geq 0, y \geq 0, z \geq 0$;



选择球坐标方程: $x = r \sin \varphi \cos \theta, y = r \sin \varphi \sin \theta, z = r \cos \varphi$,

得 $0 \leq \theta \leq \frac{\pi}{2}, 0 \leq \varphi \leq \frac{\pi}{2}, 0 \leq r \leq 1$, 即

$$\begin{aligned} \text{原式} &= \int_0^{\frac{\pi}{2}} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^1 r \sin \varphi \cos \theta \cdot r \sin \varphi \sin \theta \cdot r^2 \sin \varphi dr \\ &= \int_0^{\frac{\pi}{2}} \cos \theta \sin \theta d\theta \int_0^{\frac{\pi}{2}} \sin^3 \varphi d\varphi \int_0^1 r^5 dr = \frac{1}{2} \times \frac{1}{4} \times \frac{1}{6} = \frac{1}{48} \end{aligned}$$

习题 11.4 重积分的应用

1. 求锥面 $z = \sqrt{x^2 + y^2}$ 被柱面 $z^2 = 2x$ 所割下部分的曲面面积.

【解析】(1) 所求面积的曲面方程为 $z = \sqrt{x^2 + y^2}$;

$$(2) \quad \sqrt{1 + z'_x{}^2 + z'_y{}^2} = \sqrt{1 + \frac{x^2}{x^2 + y^2} + \frac{y^2}{x^2 + y^2}} = \sqrt{2};$$

$$(3) \quad \begin{cases} z = \sqrt{x^2 + y^2} \\ z^2 = 2x \end{cases} \Rightarrow x^2 + y^2 = 2x \text{ 为在 } xoy \text{ 面投影曲线边界方程};$$

(4)

$$A = \iint_{D_{xy}} \sqrt{2} dx dy = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{2\cos\theta} \sqrt{2} r dr = \sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1}{2} r^2 \right) \Big|_0^{2\cos\theta} d\theta = \sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2\cos^2\theta d\theta = \sqrt{2}\pi$$



2. 求由曲面 $z = \sqrt{2 - x^2 - y^2}$, $z = x^2 + y^2$ 所围立体的表面积.

【解析】(1) 消 z , 得

$$(x^2 + y^2)^2 = 2 - (x^2 + y^2) \Rightarrow (x^2 + y^2)^2 + (x^2 + y^2) - 2 = 0, \text{ 即}$$

$(x^2 + y^2 - 1)(x^2 + y^2 + 2) = 0$, 则 $x^2 + y^2 = 1$ 为积分区域 D 的边界;

$$(2) \quad S_1 = \iint_D \sqrt{1 + z'_x{}^2 + z'_y{}^2} dx dy = \iint_D \sqrt{1 + 4x^2 + 4y^2} dx dy \\ = \int_0^{2\pi} d\theta \int_0^1 \sqrt{1 + 4r^2} \cdot r dr = \frac{\pi}{6} (5\sqrt{5} - 1);$$

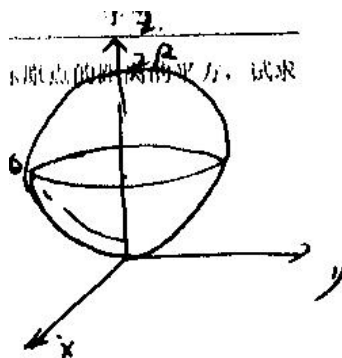
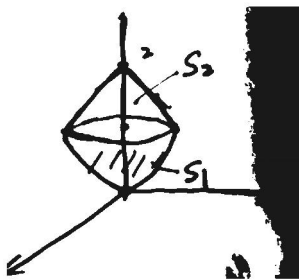
$$(3) \quad S_2 = \iint_D \sqrt{1 + z'_x{}^2 + z'_y{}^2} dx dy = \iint_D \sqrt{1 + \frac{x^2}{2 - x^2 - y^2} + \frac{y^2}{2 - x^2 - y^2}} dx dy \\ = \sqrt{2} \iint_D \frac{1}{\sqrt{2 - x^2 - y^2}} dx dy = \sqrt{2} \int_0^{2\pi} d\theta \int_0^1 \frac{1}{\sqrt{2 - r^2}} \cdot r dr = (4 - 2\sqrt{2})\pi;$$

$$(4) \quad S = S_1 + S_2 = \frac{\pi}{6} (5\sqrt{5} - 1) + (4 - 2\sqrt{2})\pi$$

3. 球体 $x^2 + y^2 + z^2 \leq 2Rz$ 内, 各点处的密度大小等于该点到坐标原点的距离的平方, 试求该球体的质心.

【解析】(1) V 为球体空间区域, 所给球体质量分布对称于 z 轴, 质点位于 z 轴上, 由对称性可知, $\bar{x} = 0, \bar{y} = 0$, 所以只要求 $\bar{z}$;

$$(2) \quad \text{密度 } \rho = x^2 + y^2 + z^2;$$



$$(3) \quad M = \iiint_V (x^2 + y^2 + z^2) dV \quad \text{利用球坐标}$$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^{2R\cos\varphi} r^2 \cdot r^2 \sin\varphi dr$$

$$= 2\pi \int_0^{\frac{\pi}{2}} \sin\varphi \cdot \frac{1}{5} (2R\cos\varphi)^5 d\varphi = \frac{32}{15} \pi R^5;$$

$$(4) \quad \bar{z} = \frac{1}{M} \iiint_V z \rho(x, y, z) dV = \frac{1}{M} \iiint_V z(x^2 + y^2 + z^2) dV$$

$$= \frac{1}{M} \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^{2R\cos\varphi} r \cos\varphi \cdot r^2 \cdot r^2 \sin\varphi dr = \frac{2\pi}{M} \int_0^{\frac{\pi}{2}} \cos\varphi \cdot \sin\varphi \cdot \frac{1}{6} (2R\cos\varphi)^6 d\varphi$$

$$= \frac{1}{M} \cdot \frac{8}{3} \pi R^5 = \frac{5}{4} R$$

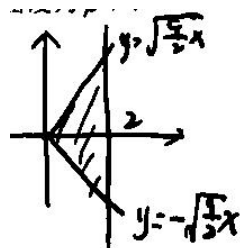
所以质心为 $\left(0, 0, \frac{5}{4}R\right)$.

4. 求由 $y^2 = \frac{9}{2}x$ 和 $x=2$ 围成的均匀薄板对 x 轴及 y 轴的转动惯量 (设面密度为 ρ).

【解析】 $I_x = \iint_D y^2 \rho d\sigma = \rho \int_0^2 dx \int_{-\sqrt{\frac{9}{2}x}}^{\sqrt{\frac{9}{2}x}} y^2 dy = \rho \int_0^2 \left(\frac{1}{3} y^3 \right) \bigg|_{-\sqrt{\frac{9}{2}x}}^{\sqrt{\frac{9}{2}x}} dx$

$$= \frac{2}{3} \rho \int_0^2 \left(\sqrt{\frac{9}{2}x} \right)^3 dx = \frac{2}{3} \rho \int_0^2 \frac{9}{2} \cdot \frac{3}{\sqrt{2}} \cdot x^{\frac{3}{2}} dx = \frac{9}{\sqrt{2}} \rho \int_0^2 x^{\frac{3}{2}} dx = \frac{72}{5} \rho.$$

$$I_y = \iint_D x^2 \rho d\sigma = \rho \int_0^2 dx \int_{-\sqrt{\frac{9}{2}x}}^{\sqrt{\frac{9}{2}x}} x^2 dy = 2\rho \int_0^2 x^2 \cdot \sqrt{\frac{9}{2}x} dx = 2\rho \cdot \frac{3}{\sqrt{2}} \int_0^2 x^{\frac{5}{2}} dx = \frac{96}{7} \rho.$$



自 测 题

一、填空题（每题 4 分，共 20 分）.

1. 交换二次积分次序:

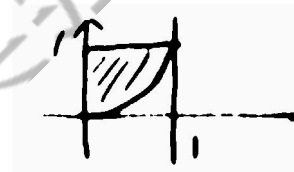
$$\int_0^1 dx \int_{1-x^2}^1 f(x, y) dy + \int_1^e dx \int_{\ln x}^1 f(x, y) dy = \underline{\hspace{2cm}}.$$

【答案】 $\int_0^1 dy \int_{\sqrt{1-y^2}}^{e^y} f(x, y) dx$



2. 计算 $I = \int_0^1 dx \int_{x^2}^1 \frac{xy}{\sqrt{1+y^3}} dy = \underline{\hspace{2cm}}.$

【解析】 交换积分顺序，计算得 $\frac{1}{3}(\sqrt{2}-1)$



3. 设 $f(x)$ 连续, $f(1)=1, F(t) = \iint_{x^2+y^2 \leq t^2} f(x^2+y^2) dx dy, (t \geq 0)$, 则 $F'(1) = \underline{\hspace{2cm}}.$

【解析】 $F(t) = \iint_{x^2+y^2 \leq t^2} f(x^2+y^2) dx dy = \int_0^{2\pi} d\theta \int_0^t f(r^2) \cdot r dr = 2\pi \int_0^t f(r^2) \cdot r dr,$

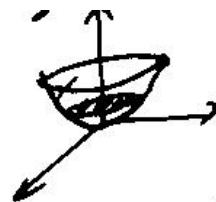
则 $F'(t) = 2\pi t f(t^2)$, 进而 $F'(1) = 2\pi$.

4. 计算 $\iiint_{\Omega} \frac{z \ln(x^2+y^2+z^2+1)}{x^2+y^2+z^2+1} dx dy dz = \underline{\hspace{2cm}},$ 其中 Ω 是球面 $x^2+y^2+z^2=1$ 所围成的闭区域.

【解析】 利用三重积分的对称性计算可知积分值为 0.

5. 设立体 Ω 由曲面 $z = x^2 + y^2$ 及平面 $z = 1$ 围成, 则其体积为 $\underline{\hspace{2cm}}.$

【解析】 $V = \iiint_{\Omega} 1 dV = \int_0^1 dz \iint_{D_z} 1 dV = \int_0^1 \pi z dz = \frac{\pi}{2}$



二、选择题（每小题 4 分，共 20 分）.

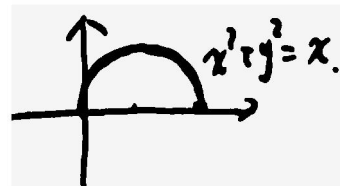
6. 设 $f(u, v)$ 连续, 则 $\int_0^{\frac{\pi}{2}} d\theta \int_0^{\cos \theta} f(r \cos \theta, r \sin \theta) r dr$ 等于 ().

(A) $\int_0^{\frac{1}{2}} dy \int_0^1 f(x, y) dx$

(B) $\int_0^1 dx \int_0^{\sqrt{x-x^2}} f(x, y) dy$

(C) $\int_0^1 dx \int_0^{\sqrt{2x-x^2}} f(x, y) dy$

(D) $\int_0^{\frac{1}{2}} dy \int_{\frac{1}{2}-\sqrt{\frac{1}{4}-y^2}}^{\frac{1}{2}+\sqrt{\frac{1}{4}-y^2}} f(x, y) dx$



【解析】 由图形可知正确答案选择 B

7. 设平面区域 $D = \{(x, y) | -a \leq x \leq a, x \leq y \leq a\}$, D_1 表示 D 在第一象限的部分, 则

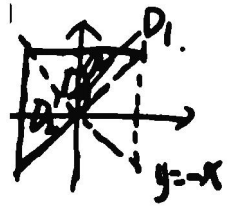
$$\iint_D (xy + \cos x \sin y) dx dy = ()$$

$$(A) 2 \iint_{D_1} \cos x \sin y dx dy$$

$$(B) 2 \iint_{D_1} xy dx dy$$

$$(C) 4 \iint_{D_1} (xy + \cos x \sin y) dx dy$$

$$(D) 0$$



【解析】添加辅助曲线 $y = -x$, 则 $\iint_D = \iint_{D_1} + \iint_{D_2}$, 如图所示, 利用对称性可知正确答案为 A

8. 设 $f(x, y)$ 为连续函数, 且 $D = \{(x, y) | x^2 + y^2 \leq t^2\}$, 则 $\lim_{t \rightarrow 0^+} \frac{1}{\pi t^2} \iint_D f(x, y) dx dy =$

()

$$(A) f(0, 0)$$

$$(B) -f(0, 0)$$

$$(C) f'(0, 0)$$

$$(D) \text{不存在}$$

【解析】

$$\lim_{t \rightarrow 0^+} \frac{1}{\pi t^2} \iint_D f(x, y) dx dy = \lim_{t \rightarrow 0^+} \frac{1}{\pi t^2} f(\xi, \eta) \cdot \pi t^2 = \lim_{t \rightarrow 0^+} f(\xi, \eta) = \lim_{\substack{\xi \rightarrow 0^+ \\ \eta \rightarrow 0^+}} f(\xi, \eta) = f(0, 0)$$

9. 设有空间区

$$\Omega_1 = \{(x, y, z) | x^2 + y^2 + z^2 \leq R^2, z \geq 0\}, \Omega_2 = \{(x, y, z) | x^2 + y^2 + z^2 \leq R^2$$

则有 ()

$$(A) \iiint_{\Omega_1} x dx dy dz = 4 \iiint_{\Omega_2} x dx dy dz$$

$$(B) \iiint_{\Omega_1} y dx dy dz = 4 \iiint_{\Omega_2} y dx dy dz$$

$$(C) \iiint_{\Omega_1} z dx dy dz = 4 \iiint_{\Omega_2} z dx dy dz$$

$$(D) \iiint_{\Omega_1} xyz dx dy dz = 4 \iiint_{\Omega_2} xyz dx dy dz$$



【解析】利用对称性和保号性可知正确答案为 C

10. 已知空间区域 Ω 由 $x^2 + y^2 \leq z, 1 \leq z \leq 2$ 确定, $f(z)$ 连续, 则 $\iiint_{\Omega} f(z) dv = ()$

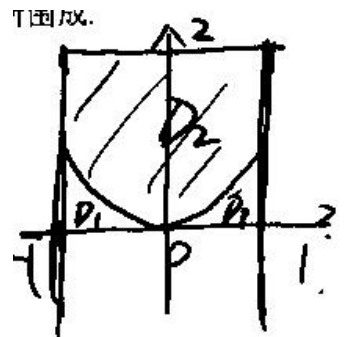
$$(A) \pi \int_1^2 z^2 f(z) dz \quad (B) 2\pi \int_1^2 f(z) dz \quad (C) 2\pi \int_1^2 z f(z) dz \quad (D) \pi \int_1^2 z f(z) dz$$

【解析】利用三重积分的截面法得正确答案为 D

三、解答题 (每小题 10 分, 共 60 分) .

11. 计算二重积分 $\iint_D |y - x^2| dx dy$, 其中 D 由 $|x| \leq 1, 0 \leq y \leq 2$ 所围成.

【解析】(1) 利用 $y = x^2$ 将 D 划分为 D_1, D_2 , 如图所示;

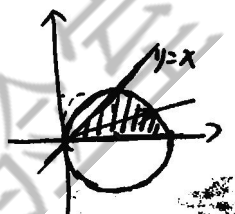


$$\begin{aligned}
 (2) \quad I &= \iint_{D_1} (x^2 - y) dx dy + \iint_{D_1} (y - x^2) dx dy \\
 &= \int_{-1}^1 dx \int_0^{x^2} (x^2 - y) dy + \int_{-1}^1 dx \int_0^{x^2} (y - x^2) dy \\
 &= \int_{-1}^1 \frac{1}{2} x^4 dx + \int_{-1}^1 (2 - 2x^2 + \frac{1}{4} x^4) dx = \frac{46}{15}
 \end{aligned}$$

12. 计算二重积分 $\iint_D \sqrt{x^2 + y^2} dx dy$, 其中 $D = \{(x, y) | 0 \leq y \leq x, x^2 + y^2 \leq 2x\}$.

【解析】 $\iint_D \sqrt{x^2 + y^2} dx dy = \int_0^{\frac{\pi}{4}} d\theta \int_0^{2\cos\theta} r \cdot r dr$

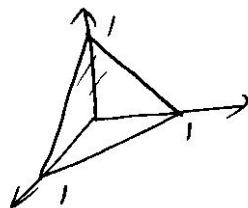
$$= \int_0^{\frac{\pi}{4}} \frac{1}{3} r^3 \Big|_0^{2\cos\theta} d\theta = \frac{8}{3} \int_0^{\frac{\pi}{4}} \cos^3 \theta d\theta = \frac{10}{9} \sqrt{2}$$



13. 计算三重积分 $\iiint_V \frac{dx dy dz}{(1+x+y+z)^3}$, 其中 V 由 $x=0, y=0, z=0$ 和 $x+y+z=1$ 所围成.

【解析】 $\iiint_V \frac{dx dy dz}{(1+x+y+z)^3} = \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} \frac{1}{(1+x+y+z)^3} dz$

$$= \frac{1}{2} \left(\ln 2 - \frac{5}{8} \right)$$



14. 计算三重积分 $\iiint_V (x^2 + y^2 + z) dx dy dz$, 其中 V 是由曲线 $\begin{cases} y^2 = 2z \\ x = 0 \end{cases}$ 绕 z 轴旋转一周而成

的旋转曲面与平面 $z=4$ 所围成的立体.

【解析】(1) 所得旋转曲面方程为: $x^2 + y^2 = 2z$, 如图所示, 其投影区

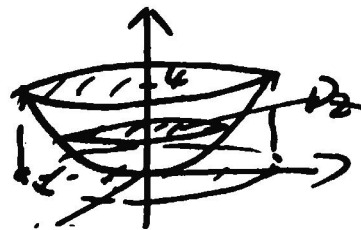
域为 $D_{xy}: x^2 + y^2 \leq 8$;

$$(2) \quad \iiint_V (x^2 + y^2 + z) dx dy dz = \iiint_V (x^2 + y^2) dx dy dz + \iiint_V z dx dy dz;$$

$$(3) \quad \iiint_V (x^2 + y^2) dx dy dz = \int_0^4 dz \iint_{D_x} (x^2 + y^2) dx dy = \int_0^4 dz \int_0^{2\pi} d\theta \int_0^{\sqrt{2z}} r^2 \cdot r dr = 2\pi \int_0^4 z^2 dz = \frac{128\pi}{3};$$

$$(4) \quad \iiint_V z dx dy dz = \int_0^4 dz \iint_{D_x} z dx dy = \int_0^4 z \cdot \pi \cdot 2z dz = 2\pi \int_0^4 z^2 dz = \frac{128\pi}{3};$$

$$(5) \quad \text{原积分} = \frac{256\pi}{3}$$



15. 计算三重积分 $\iiint_V \sqrt{x^2 + y^2 + z^2} dx dy dz$, 其中 V 由曲面 $x^2 + y^2 + z^2 = z$ 所围成.

【解析】原式 = $\int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^{\cos\varphi} r \cdot r^2 \sin\varphi dr$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \sin\varphi d\varphi \int_0^{\cos\varphi} r^3 dr =$$

$$= 2\pi \int_0^{\frac{\pi}{2}} \sin\varphi \cdot \left(\frac{1}{4} r^4 \Big|_0^{\cos\varphi} \right) d\varphi = -\frac{\pi}{2} \int_0^{\frac{\pi}{2}} \cos^4\varphi d(\cos\varphi) = \frac{\pi}{10}$$



16. 求由曲面 $z = \sqrt{2 - x^2 - y^2}$, $z = x^2 + y^2$ 所围成立体的表面积.

【解析】本题同 11.4 节第一题一样, 免做!

安徽大学 ACM/ICPC 实验室

第十二章 曲线积分与曲面积分

习题 12.1 第一类曲线积分

1. 计算 $\int_L (x^2 + y^2 + z^2) ds$, 其中 $L: x = a \cos t, y = a \sin t, z = bt, t \in [0, 2\pi]$.

【解析】(1) $ds = \sqrt{x_t'^2 + y_t'^2 + z_t'^2} = \sqrt{a^2 + b^2} dt$

$$\begin{aligned} (2) \int_L (x^2 + y^2 + z^2) ds &= \int_0^{2\pi} [a^2 \cos^2 t + a^2 \sin^2 t + b^2 t^2] \cdot \sqrt{a^2 + b^2} dt = \int_0^{2\pi} (a^2 + b^2 t^2) \cdot \sqrt{a^2 + b^2} dt \\ &= \sqrt{a^2 + b^2} \cdot \left(a^2 t + \frac{1}{3} b^2 t^3 \right) \Big|_0^{2\pi} = \frac{2\pi}{3} (3a^2 + 4\pi b^2) \sqrt{a^2 + b^2} \end{aligned}$$

2. 计算 $\oint_L \cos \sqrt{x^2 + y^2} ds$, 其中 L 为圆周 $x^2 + y^2 = a^2$, 直线 $y = x$ 与 y 轴在第一象限内围成的图形的边界.

【解析】(1) $I = \int_{\overline{OA}} + \int_{\widehat{AB}} + \int_{\overline{OB}}$

$$(2) \overline{OA}: x = 0, y \in [0, a], \int_{\overline{OA}} = \int_0^a \cos y dx = \sin a;$$

$$\widehat{AB}: \int_{\widehat{AB}} = \int_{\widehat{AB}} \cos a ds = \cos a \cdot \frac{1}{8} \cdot 2\pi a = \frac{\pi}{4} a \cos a;$$

$$\overline{OB}: y = x, x \in [0, \frac{\sqrt{2}}{2} a], \int_{\overline{OB}} = \int_0^{\frac{\sqrt{2}}{2} a} \cos \sqrt{2} x \cdot \sqrt{2} dx = \sin a;$$

$$(3) I = 2 \sin a + \frac{\pi}{4} a \cos a$$

3. 计算 $\int_L \sqrt{x^2 + y^2} ds$, 其中 $L: x^2 + y^2 = ax (a > 0)$.

【解析】解法一: 直角坐标系做

$$(1) \text{ 对上半圆弧微分 } ds = \sqrt{1 + \left(\frac{a-2x}{2y} \right)^2} dx = \frac{a}{2y} dx = \frac{a}{2\sqrt{ax-x^2}} dx \quad (0 \leq x \leq a);$$

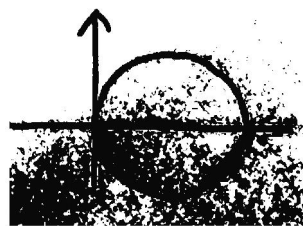
(2) 由对称性可知

$$\int_L \sqrt{x^2 + y^2} ds = a \sqrt{a} \int_0^a \frac{1}{\sqrt{a-x}} dx = 2a^2$$

解法二: 极坐标系做

$$\text{令 } \begin{cases} x = \frac{a}{2} + \frac{a}{2} \cos t \\ y = \frac{a}{2} \sin t \end{cases} \quad (0 \leq t \leq 2\pi)$$

$$\int_L \sqrt{x^2 + y^2} ds = \int_0^{2\pi} \sqrt{\left[\frac{a}{2} + \frac{a}{2} \cos t \right]^2 + \left[\frac{a}{2} \sin t \right]^2} \cdot \sqrt{\left(-\frac{a}{2} \sin t \right)^2 + \left(\frac{a}{2} \cos t \right)^2} dt$$



$$= \frac{a}{\sqrt{2}} \int_0^{2\pi} \sqrt{1 + \cos t} \cdot \frac{a}{2} dt = \frac{a^2}{2\sqrt{2}} \int_0^{2\pi} \sqrt{2 \cos^2 \frac{t}{2}} dt = \frac{a^2}{2\sqrt{2}} \int_0^{2\pi} \left| \cos \frac{t}{2} \right| dt = 2a^2$$

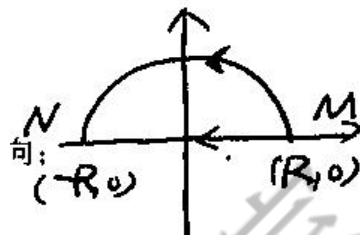
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习题 12.2 第二类曲线积分

1. 计算 $\int_L y^2 dx + x^2 dy$, 其中 L 为

(1) 圆周 $x^2 + y^2 = R^2$ 的上半部分, 方向为逆时针方向;

(2) 从点 $M(R, 0)$ 到点 $N(-R, 0)$ 的直线段.



【解析】(1) 利用极坐标 $x = R \cos \theta, y = R \sin \theta; \theta: 0 \rightarrow \pi$;

$$\begin{aligned} \int_L y^2 dx + x^2 dy &= \int_0^\pi [R^2 \sin^2 \theta \cdot R(-\sin \theta) + R^2 \cos^2 \theta \cdot R \cos \theta] d\theta \\ &= R^3 \int_0^\pi (\cos^3 \theta - \sin^3 \theta) d\theta = R^3 \left[\int_0^\pi \cos^3 \theta d\theta - \int_0^\pi \sin^3 \theta d\theta \right] \end{aligned}$$

$$= R^3 \left[\int_0^\pi \cos^2 \theta d \sin \theta + \int_0^\pi \sin^2 \theta d \cos \theta \right]$$

$$= R^3 \left[\left(\sin \theta - \frac{1}{3} \sin^3 \theta \right) \Big|_0^\pi + \left(\cos \theta - \frac{1}{3} \cos^3 \theta \right) \Big|_0^\pi \right] = -\frac{4}{3} R^3$$

$$(2) \begin{cases} y=0 \\ x=x \end{cases} \quad x: R \rightarrow -R, \quad \text{则} \quad \int_L y^2 dx + x^2 dy = \int_R^{-R} 0 dx = 0$$

2. 计算 $\int_L x dy - y dx, L$: 从 $A(-1, 0)$ 经过 $x^2 + y^2 = 1$ 上半圆到 $B(0, 1)$, 再经过 $y = 1 - x^2$ 到 $C(1, 0)$.

【解析】

$$(1) \int_L = \int_{\overline{AB}} + \int_{\overline{BC}};$$

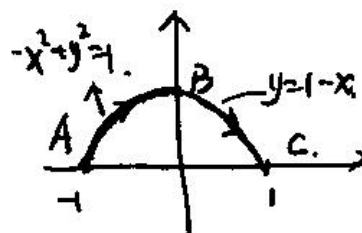
$$(2) \overline{AB}: \begin{cases} x = \cos \theta \\ y = \sin \theta \end{cases} \quad \theta: \pi \rightarrow \frac{\pi}{2}, \quad \text{则}$$

$$\int_{\overline{AB}} = \int_\pi^{\frac{\pi}{2}} (\cos \theta \cdot \sin \theta + \sin \theta \cdot \sin \theta) d\theta = -\frac{\pi}{2};$$

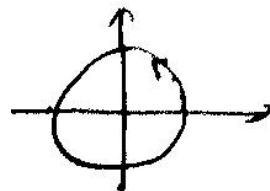
$$(3) \overline{BC}: \begin{cases} y = 1 - x^2 \\ x = x \end{cases} \quad x: 0 \rightarrow 1, \quad \text{则}$$

$$\int_{\overline{BC}} = \int_0^1 [x \cdot (-2x) - (1 - x^2)] dx = \int_0^1 (-x^2 - 1) dx = -\frac{4}{3};$$

$$(4) \int_L = \int_{\overline{AB}} + \int_{\overline{BC}} = -\frac{\pi}{2} - \frac{4}{3}$$



3. 计算第二类曲线积分 $\oint_L \frac{(x+y)dx - (x-y)dy}{x^2 + y^2}$, 其中 L 为圆周 $x^2 + y^2 = a^2$,



方向为逆时针方向.

【解析】(1) $\oint_L \frac{(x+y)dx - (x-y)dy}{x^2 + y^2} = \oint_L \frac{(x+y)dx - (x-y)dy}{a^2};$

(2) $L: x = a \cos \theta, y = a \sin \theta, \quad \theta: 0 \rightarrow 2\pi;$

(3) $\oint_L \frac{(x+y)dx - (x-y)dy}{a^2} = \frac{1}{a^2} \int_0^{2\pi} [a^2(\cos \theta + \sin \theta)(-\sin \theta) - a^2(\cos \theta - \sin \theta) \cdot \cos \theta] d\theta$
 $= -\int_0^{2\pi} d\theta = -2\pi$

4. 计算 $\int_L (x+y)dx + (x-y)dy$, 其中 L 为

(1) 椭圆周 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 的上半部分, 从 $(a, 0)$ 到 $(-a, 0)$;

(2) 从点 $(a, 0)$ 到点 $(-a, 0)$ 的直线段.

【解析】(1) $L: x = a \cos \theta, y = b \sin \theta, \quad \theta: 0 \rightarrow \pi;$

$$\int_L (x+y)dx + (x-y)dy = \int_0^\pi [(a \cos \theta + b \sin \theta) \cdot (-a \sin \theta) + (a \cos \theta - b \sin \theta) \cdot b \cos \theta] d\theta$$
$$= -\int_0^\pi (a^2 + b^2) \sin \theta \cos \theta d\theta + ab \int_0^\pi (\cos^2 \theta - \sin^2 \theta) d\theta = 0$$

(2) $\begin{cases} y=0 \\ x=x \end{cases} \quad x: a \rightarrow -a, \text{ 则 } \int_L = \int_a^{-a} [(x+0) + (x-0) \cdot 0] dx = \int_a^{-a} x dx = 0$

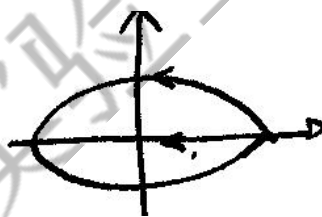
5. 设 $\vec{F} = \{y, z, x\}$, L 为依参数增加方向进行的纽形螺线

$$x = a \cos t, y = a \sin t, z = bt \quad t \in [0, 2\pi]$$

计算 $\int_L \vec{F} \cdot d\vec{r}$.

【解析】

$$\int_L \vec{F} \cdot d\vec{r} = \int_L ydx + zdz + xdz = \int_0^{2\pi} [a \sin t \cdot a(-\cos t) + bt \cdot a \cos t + ab \cos t] dt = -\pi a^2$$



习题 12.3 Green 公式

1. 计算曲线积分 $\oint_{L^+} xy^2 dy - x^2 y dx$, 其中 L 为圆周 $x^2 + y^2 = R^2$.

【解析】(1) $P = -x^2 y$, $Q = xy^2$;

$$(2) \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = y^2 + x^2;$$

$$(3) \oint_{L^+} xy^2 dy - x^2 y dx = \iint_D (x^2 + y^2) dx dy = \int_0^{2\pi} d\theta \int_0^R r^3 dr = \frac{1}{2} \pi R^4$$

2. 计算曲线积分 $\int_L (e^x \sin y - ay) dx + (e^x \cos y - bx) dy$, 其中 L 为从 $A(a, 0)$ 到 $O(0, 0)$ 的上半圆周

$$x^2 + y^2 = ax.$$

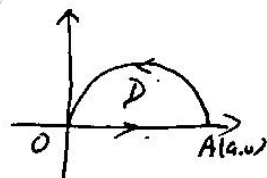
【解析】(1) 添加 $\overline{OA}: y=0$, 从 $O \rightarrow A$;

$$(2) \int_L = \oint_{L+\overline{OA}} - \int_{\overline{OA}};$$

$$(3) \int_L = \oint_{L+\overline{OA}} = \iint_{D_{xy}} \left[\frac{\partial(e^x \cos y - bx)}{\partial x} - \frac{\partial(e^x \sin y - ay)}{\partial y} \right] dx dy = \iint_{D_{xy}} (a-b) dx dy = \frac{\pi a^2}{8} (a-b);$$

$$(4) \overline{OA}: \begin{cases} y=0 \\ x=x \end{cases} \quad x: 0 \rightarrow a, \text{ 则 } \int_{\overline{OA}} = \int_0^a 0 dx = 0;$$

$$(5) \int_L = \frac{\pi a^2}{8} (a-b)$$



3. 计算曲线积分 $\oint_{L^+} \frac{xdy - ydx}{4x^2 + y^2}$, 其中 L 是以点 $(1, 0)$ 为中心, R 为半径的圆周 ($R > 1$), 取

顺时针方向.(提示: 挖去一个小椭圆)

【解析】(1) 添加 $l: 4x^2 + y^2 = \varepsilon^2 (\varepsilon > 0)$, 方向逆时针方向;

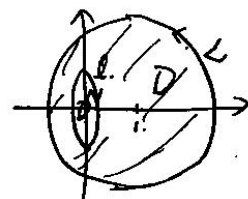
$$(2) P = \frac{-y}{4x^2 + y^2}, Q = \frac{x}{4x^2 + y^2}, \text{ 且 } \frac{\partial P}{\partial y} = \frac{y^2 - 4x^2}{(4x^2 + y^2)^2} = \frac{\partial Q}{\partial x};$$

$$(3) \int_{L^+} = \oint_{L^+ + l} - \oint_l;$$

$$(4) \oint_{L^+ + l} = \iint_D 0 dx dy = 0;$$

$$(5) \oint_l \frac{xdy - ydx}{4x^2 + y^2} = \frac{1}{\varepsilon^2} \oint_l xdy - ydx = \frac{1}{\varepsilon^2} \iint_{D'} 2 dx dy = \frac{1}{\varepsilon^2} \cdot 2\pi \cdot \frac{\varepsilon}{2} \cdot \varepsilon = \pi;$$

$$(6) \int_{L^+} = -\pi$$



4. 曲线积分 $\int_L (e^x + 2f(x))ydx - f(x)dy$ 与路径无关, 且 $f(1)=1$, 求

$$I = \int_{(0,0)}^{(1,1)} (e^x + 2f(x))ydx - f(x)dy.$$

【解析】 $I = \int_{\overline{OB}} + \int_{\overline{BA}} = \int_0^1 (e^x + xf(x)) \cdot 0dx - \int_0^1 f(1)dy = \int_0^1 (-1)dy = -1$



5. 计算 $I = \int_L \frac{xdy - ydx}{x^2 + y^2}$, 其中 L 是从 $A(-1,0)$ 沿抛物线 $y = x^2 - 1$ 到点 $B(2,3)$ 的曲线弧.

【解析】(1) $P = \frac{-y}{x^2 + y^2}, Q = \frac{x}{x^2 + y^2}$, 且 $\frac{\partial P}{\partial y} = \frac{y^2 - x^2}{(x^2 + y^2)^2} = \frac{\partial Q}{\partial x}$, 所以可知积分与路

径无关;

(2) 沿 $A \rightarrow C \rightarrow D \rightarrow B$ 构造新的路径进行计算

$$\overline{AC}: \begin{cases} x = -1 \\ y = y \end{cases}, y: 0 \rightarrow 1, \int_{\overline{AC}} = \int_0^1 \frac{-1}{1+y^2} dy = -\arctan y \Big|_0^1 = -\frac{\pi}{4};$$

$$\overline{CD}: \begin{cases} x = x \\ y = -1 \end{cases}, x: -1 \rightarrow 2, \int_{\overline{CD}} = \int_{-1}^2 \frac{1}{1+x^2} dx = \arctan x \Big|_{-1}^2 = \arctan 2 + \frac{\pi}{4};$$

$$\overline{BD}: \begin{cases} x = 2 \\ y = y \end{cases}, y: -1 \rightarrow 3, \int_{\overline{BD}} = \int_{-1}^3 \frac{2}{4+y^2} dy = \arctan \frac{y}{2} \Big|_{-1}^3 = \arctan \frac{3}{2} + \arctan \frac{1}{2};$$

$$(3) I = \frac{\pi}{2} + \arctan 2 + \arctan \frac{3}{2} + \arctan \frac{1}{2} = \pi + \arctan \frac{3}{2} \quad (\because \arctan 2 + \arctan \frac{1}{2} = \frac{\pi}{2})$$

6. 选择常数 a, b 使得 $(2ax^3y^3 - 3y^2 + 5)dx + (3x^4y^2 - 2bxy - 4)dy$ 是某个二元函数

$U(x, y)$ 在全平面内的

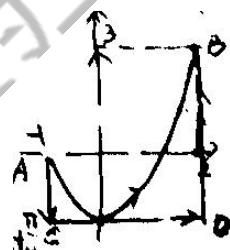
全微分, 并求 $U(x, y)$.

【解析】(1) $P = 2ax^3y^3 - 3y^2 + 5, Q = 3x^4y^2 - 2bxy - 4$, $\frac{\partial P}{\partial y} = 6ax^3y^2 - 6y, \frac{\partial Q}{\partial x} = 12x^3y^2 - 2by$,

$$\text{因为 } \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow \begin{cases} 6a = 12 \\ -2b = -6 \end{cases} \Rightarrow \begin{cases} a = 2 \\ b = 3 \end{cases}$$

$$(2) (4x^3y^3 - 3y^2 + 5)dx + (3x^4y^2 - 6xy - 4)dy$$

$$= 4x^3y^3dx - 3y^2dx + 5dx + 3x^4y^2dy - 6xydy - 4dy$$



$$= (y^3 dx^4 + x^4 dy^3) - 3(y^2 dx + x dy^2) + d(5x - 4y)$$

$$= d(x^4 y^3) - d(3xy^2) + d(5x - 4y) = d(x^4 y^3 - 3xy^2 + 5x - 4y)$$

$$\text{所以 } u(x, y) = x^4 y^3 - 3xy^2 + 5x - 4y + C \quad (C \in \mathbb{R})$$

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习题 12.4 第一类曲面积分

1. 计算曲面积分 $\iint_S \frac{dS}{(1+x+y)^2}$, 其中 S 为四面体 $x+y+z \leq 1, x \geq 0, y \geq 0, z \geq 0$ 的边界.

【解析】(1) S 由四个面组成:

$$S_1: x+y+z=1 \ (x \geq 0, y \geq 0, z \geq 0), \quad S_2: x=0 \ (y+z \leq 1, y \geq 0, z \geq 0),$$

$$S_3: y=0 \ (x+z \leq 1, x \geq 0, z \geq 0), \quad S_4: z=0 \ (x+y \leq 1, x \geq 0, y \geq 0);$$

(2) 对 $S_1: z=1-x-y$, 向 xoy 面做投影, 投影区域 $D_1: x+y \leq 1, x \geq 0, y \geq 0$, 且

$$\sqrt{1+z_x'^2+z_y'^2} = \sqrt{3}, \text{ 则}$$

$$\iint_{S_1} \frac{dS}{(1+x+y)^2} = \iint_{D_1} \frac{\sqrt{3}}{(1+x+y)^2} dx dy = \sqrt{3} \int_0^1 dx \int_0^{1-x} \frac{dy}{(1+x+y)^2} = \sqrt{3} \left(\ln 2 - \frac{1}{2} \right);$$

(3) 对 $S_2: x=0$, 向 $yo z$ 面做投影, 投影区域 $D_2: y+z \leq 1, y \geq 0, z \geq 0$, 且 $\sqrt{1+x_y'^2+x_z'^2} = 1$

$$\iint_{S_2} \frac{dS}{(1+x+y)^2} = \iint_{D_2} \frac{dS}{(1+y)^2} = \iint_{D_2} \frac{dy dz}{(1+y)^2} = \int_0^1 dy \int_0^{1-y} \frac{dz}{(1+y)^2} = 1 - \ln 2;$$

(4) 对 $S_3: y=0$, 向 xoz 面做投影, 投影区域 $D_3: x+z \leq 1, x \geq 0, z \geq 0$, 且 $\sqrt{1+y_x'^2+y_z'^2} = 1$

$$\iint_{S_3} \frac{dS}{(1+x+y)^2} = \iint_{D_3} \frac{dS}{(1+x)^2} = \iint_{D_3} \frac{dx dz}{(1+x)^2} = \int_0^1 dx \int_0^{1-x} \frac{dz}{(1+x)^2} = 1 - \ln 2;$$

(5) 对 $S_4: z=0$, 向 xoy 面做投影, 投影区域 $D_4: x+y \leq 1, x \geq 0, y \geq 0$, 且 $\sqrt{1+z_x'^2+z_y'^2} = 1$

$$\iint_{S_4} \frac{dS}{(1+x+y)^2} = \iint_{D_4} \frac{dx dy}{(1+x+y)^2} = \int_0^1 dx \int_0^{1-x} \frac{dy}{(1+x+y)^2} = \ln 2 - \frac{1}{2};$$

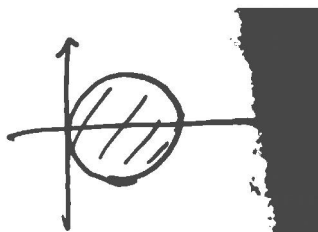
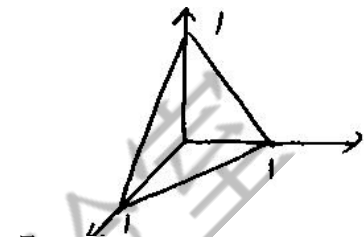
$$(6) \text{ 原式} = \frac{3-\sqrt{3}}{2} + (\sqrt{3}-1)\ln 2$$

2. 计算曲面积分 $\iint_S (xy+yz+zx) dS$, 其中 S 为锥面 $z = \sqrt{x^2+y^2}$ 被曲面 $x^2+y^2 = 2ax$ 所

割下的部分.

【解析】(1) S 向 xoy 面上投影, $D_{xy}: x^2+y^2 \leq 2ax$, 如图所示

$$(2) \sqrt{1+z_x'^2+z_y'^2} = \sqrt{1+\frac{x^2}{x^2+y^2}+\frac{y^2}{x^2+y^2}} = \sqrt{2};$$



$$(3) \text{ 原式} = \iint_{D_{xy}} (xy + y\sqrt{x^2 + y^2} + x\sqrt{x^2 + y^2}) \cdot \sqrt{2} dx dy$$

$$\stackrel{\text{极坐标}}{=} \sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{2a \cos \theta} [r^2 \cos \theta \sin \theta + r^2 \sin \theta + r^2 \cos \theta] \cdot r dr$$

$$= \sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\cos \theta \sin \theta + \sin \theta + \cos \theta) d\theta \int_0^{2a \cos \theta} r^3 dr$$

$$= \sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\cos \theta \sin \theta + \sin \theta + \cos \theta) \cdot \frac{1}{4} (2a \cos \theta)^4 d\theta$$

$$\stackrel{\text{对称性化简}}{=} \sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{4} (2a \cos \theta)^4 \cdot \cos \theta d\theta = 8\sqrt{2} a^4 \int_0^{\frac{\pi}{2}} \cos^5 \theta d\theta = 8\sqrt{2} a^4 \cdot \frac{4}{5} \cdot \frac{2}{3} = \frac{64\sqrt{2}}{15} a^4$$

3. 计算曲面积分 $\oiint_S x^2 dS$, 其中 S 为锥面 $z = \sqrt{x^2 + y^2}$ 与平面 $z=1$ 所围成区域的全部界面.

$$\text{【解析】}(1) \oiint_S x^2 dS = \iint_{S_1} + \iint_{S_2};$$

$$(2) S_1: z=1 \ (x^2 + y^2 \leq 1), \quad \iint_{S_1} x^2 dx dy = \int_0^{2\pi} d\theta \int_0^1 r^2 \cos^2 \theta \cdot r dr = \frac{\pi}{4};$$

$$(3) S_2: z = \sqrt{x^2 + y^2} \ (x^2 + y^2 \leq 1),$$

$$\iint_{S_2} x^2 \cdot \sqrt{1 + z_x'^2 + z_y'^2} dx dy = \sqrt{2} \iint_{D_{xy}} x^2 dx dy = \frac{\pi}{4} \sqrt{2};$$

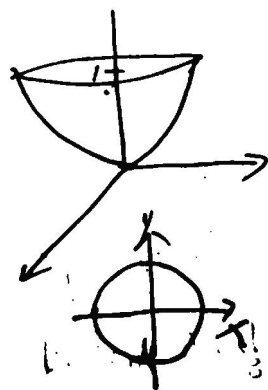
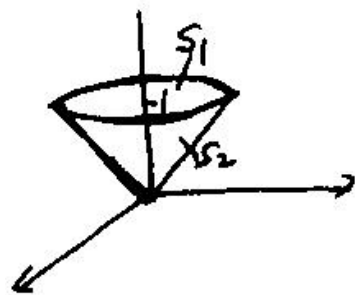
$$(4) \text{ 原式} = \frac{\pi}{4} (\sqrt{2} + 1)$$

4. 求抛物面壳子 $z = \frac{1}{2}(x^2 + y^2) \ (0 \leq z \leq 1)$ 的质量, 此壳的密度按规律 $\rho = z$ 而变更.

$$\text{【解析】}(1) M = \iint_S \rho dS = \iint_S z dS;$$

$$(2) z = \frac{1}{2}(x^2 + y^2), \ D_{xy}: x^2 + y^2 \leq 2;$$

$$\begin{aligned} (3) M &= \iint_{D_{xy}} z \cdot \sqrt{1 + z_x'^2 + z_y'^2} dx dy = \iint_{D_{xy}} z \cdot \sqrt{1 + x^2 + y^2} dx dy \\ &= \frac{1}{2} \iint_{D_{xy}} (x^2 + y^2) \cdot \sqrt{1 + x^2 + y^2} dx dy = \frac{1}{2} \int_0^{2\pi} d\theta \int_0^{\sqrt{2}} r^2 \cdot \sqrt{1 + r^2} \cdot r dr \\ &= \pi \int_0^{\sqrt{2}} r^3 \cdot \sqrt{1 + r^2} dr = \frac{\pi}{2} \int_0^{\sqrt{2}} r^2 \cdot \sqrt{1 + r^2} d(r^2) \end{aligned}$$



$$\begin{aligned}
 &= \frac{\pi}{2} \left[\int_0^{\sqrt{2}} (1+r^2) \cdot \sqrt{1+r^2} d(r^2) - \int_0^{\sqrt{2}} \sqrt{1+r^2} d(r^2) \right] \\
 &= \frac{\pi}{2} \left[\frac{2}{5} (1+r^2)^{\frac{5}{2}} \Big|_0^{\sqrt{2}} - \frac{2}{3} (1+r^2)^{\frac{3}{2}} \Big|_0^{\sqrt{2}} \right] = \frac{2\pi(1+6\sqrt{3})}{15}
 \end{aligned}$$

安徽大学 ACM/ICPC 实验班

习题 12.5 第二类曲面积分

1. 计算下面第二类曲面积分:

(1) $\iint_S x^2 z dy dz + y^2 z dx dz + z dx dy$, 其中 S 为圆柱面 $x^2 + y^2 = 1$ 的前半个柱面界于 $z = 0$ 与

$z = 3$ 之间的部

分, 取前侧.

【解析】(1) 由于曲面 S 垂直于 xoy 面, 所以 $\iint_S z dx dy = 0$;

(2) $S_{\text{右}}: y = \sqrt{1-x^2}$, 方向向右, $S_{\text{左}}: y = -\sqrt{1-x^2}$, 方向向左,

$D_{xz}: 0 \leq x \leq 1, 0 \leq z \leq 3$, 则

$$\iint_S y^2 z dx dz = \iint_{\text{左}} + \iint_{\text{右}} = \iint_{D_{xz}} (\sqrt{1-x^2})^2 dx dz + \left[-\iint_{D_{xz}} (-\sqrt{1-x^2})^2 dx dz \right] = 0;$$

(3) $S_{\text{前}}: x = \sqrt{1-y^2}$, 方向向前, $D_{yz}: -1 \leq y \leq 1, 0 \leq z \leq 3$, 则

$$\iint_S x^2 z dy dz = \iint_{D_{yz}} (1-y^2) z dy dz = \int_0^3 z dz \int_{-1}^1 (1-y^2) dy = 6$$

(2) $\iint_S (x^2 + y^2) dz dx + z dx dy$, 其中 $S: z = \sqrt{x^2 + y^2} (0 \leq z \leq 1)$ 的下侧.

【解析】(1) $D_{xy}: x^2 + y^2 \leq 1$;

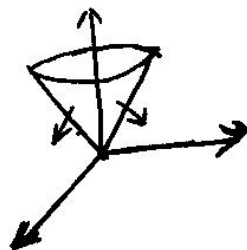
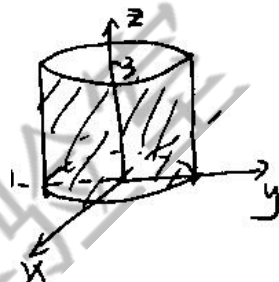
$$\begin{aligned} (2) \quad \iint_S (x^2 + y^2) dz dx + z dx dy &= \iint_S [(x^2 + y^2) \cos \beta + z \cos \gamma] dS \\ &= \iint_S \left[(x^2 + y^2) \frac{\cos \beta}{\cos \gamma} + z \right] \cos \gamma dS \end{aligned}$$

$$\text{又 } \vec{n} = \left(\frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}}, -1 \right), \text{ 则 } \vec{n}^0 = \frac{1}{|\vec{n}|} \left(\frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}}, -1 \right),$$

$$\begin{aligned} \text{则原式} &= \iint_S \left[(x^2 + y^2) \cdot \frac{-y}{\sqrt{x^2 + y^2}} + \sqrt{x^2 + y^2} \right] dx dy \\ &= \iint_S [(1-y)\sqrt{x^2 + y^2}] dx dy = -\iint_{D_{xy}} [(1-y)\sqrt{x^2 + y^2}] dx dy = \iint_{D_{xy}} [(y-1)\sqrt{x^2 + y^2}] dx dy \\ &= \int_0^{2\pi} d\theta \int_0^1 (r \sin \theta - 1) \cdot r \cdot r dr = \int_0^{2\pi} \left(\frac{1}{4} \sin \theta - \frac{1}{3} \right) d\theta = -\frac{2\pi}{3} \end{aligned}$$

(3) $\iint_S x dy dz + y dz dx + z dx dy$, 其中 S 为球面 $x^2 + y^2 + z^2 = R^2$ 的外侧.

【解析】(1) 由轮换对称性可知 $\iint_S x dy dz = \iint_S y dz dx = \iint_S z dx dy$, 所以



$$\iint_S xdydz + ydzdx + zdx dy = 3 \iint_S zdx dy ;$$

(2) 计算 $\iint_S zdx dy$

$\sum_1: z = \sqrt{a^2 - x^2 - y^2}$ 上侧; $\sum_2: z = -\sqrt{a^2 - x^2 - y^2}$, 下侧; $D_{xy}: x^2 + y^2 \leq R^2$;

$$\begin{aligned} \text{则 } \iint_S zdx dy &= \iint_{\sum_1} zdx dy + \iint_{\sum_2} zdx dy \\ &= \iint_{D_{xy}} \sqrt{a^2 - x^2 - y^2} dx dy - \iint_{D_{xy}} \left(-\sqrt{a^2 - x^2 - y^2}\right) dx dy \\ &= 2 \iint_{D_{xy}} \sqrt{a^2 - x^2 - y^2} dx dy = 2 \int_0^{2\pi} d\theta \int_0^R \sqrt{a^2 - r^2} \cdot r dr = \frac{4}{3} \pi R^3 \end{aligned}$$

$$(3) \iint_S xdydz + ydzdx + zdx dy = 3 \cdot \frac{4}{3} \pi R^3 = 4\pi R^3 .$$

2. 已知速度场 $\vec{v}(x, y, z) = \{x, y, z\}$, 求流体在单位时间内通过上半锥面 $z = \sqrt{x^2 + y^2}$ 与平面 $z = 1$ 所围成的锥体表面向外流出的流量. (利用两类曲面积分关系计算)

【解析】(1) $\Phi = \iint_S xdydz + ydzdx + zdx dy$;

(2) $S_1: z = \sqrt{x^2 + y^2}$, $D_{xy}: x^2 + y^2 \leq 1$, 方向向下,

$S_2: z = 1$, $D_{xy}: x^2 + y^2 \leq 1$, 方向向上,

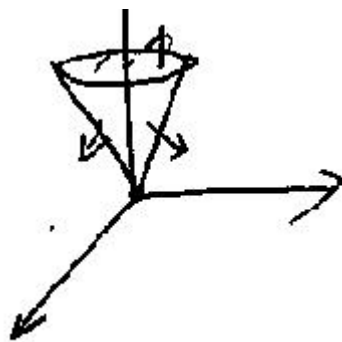
$$\text{则 } \Phi = \iint_S = \iint_{S_1} + \iint_{S_2}$$

$$(3) \iint_{S_1} = \iint_{S_1} (x \cos \alpha + y \cos \beta + z \cos \gamma) dS = \iint_{S_1} \left(x \frac{\cos \alpha}{\cos \gamma} + y \frac{\cos \beta}{\cos \gamma} + z \right) \cos \gamma dS$$

$$\text{又 } \vec{n} = \left(\frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}}, -1 \right), \text{ 则 } \vec{n}^0 = \frac{1}{|\vec{n}|} \left(\frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}}, -1 \right), \text{ 则}$$

$$\begin{aligned} \iint_{S_1} &= \iint_{S_1} \left(x \frac{\cos \alpha}{\cos \gamma} + y \frac{\cos \beta}{\cos \gamma} + z \right) \cos \gamma dS \\ &= \iint_{S_1} \left[x \cdot \left(-\frac{x}{\sqrt{x^2 + y^2}} \right) + y \cdot \left(-\frac{y}{\sqrt{x^2 + y^2}} \right) + \sqrt{x^2 + y^2} \right] dx dy \\ &= - \iint_{D_{xy}} \left[-\frac{x^2}{\sqrt{x^2 + y^2}} - \frac{y^2}{\sqrt{x^2 + y^2}} + \sqrt{x^2 + y^2} \right] dx dy = \iint_{D_{xy}} \left[\frac{x^2}{\sqrt{x^2 + y^2}} + \frac{y^2}{\sqrt{x^2 + y^2}} - \sqrt{x^2 + y^2} \right] dx dy = 0 ; \end{aligned}$$

$$(4) \iint_{S_2} \stackrel{\text{垂直性}}{=} \iint_{S_2} zdx dy = \iint_{D_{xy}} 1 dx dy = \pi ;$$



(5) 原式= π

安徽大学 ACM/ICPC 实验班

习题 12.6 Gauss 公式

1. 计算曲面积分: $I = \oint_S 2xzdydz + yzdzdx - z^2dxdy$, 其中 S 是由曲面

$z = \sqrt{x^2 + y^2}$ 与 $z = \sqrt{2 - x^2 - y^2}$ 所围成立体的表面外侧.

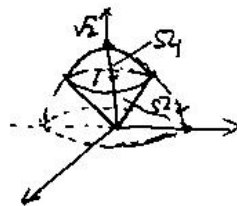
【解析】(1) $I \stackrel{\text{高斯公式}}{=} \iiint_{\Omega} [2z + z + (-2z)]dV = \iiint_{\Omega} z dV$;

(2) $\iiint_{\Omega} z dV = \iiint_{\Omega_1} z dV + \iiint_{\Omega_2} z dV$, 其中 Ω_1, Ω_2 如图所示

$$\begin{aligned} (3) \quad \iiint_{\Omega_1} z dV &= \int_1^{\sqrt{2}} \left[\iint_{D_z} z dxdy \right] dz \quad D_z: x^2 + y^2 \leq 2 - z^2 \\ &= \int_1^{\sqrt{2}} z \cdot \pi(2 - z^2) dz = \pi \int_1^{\sqrt{2}} (2z - z^3) dz = \frac{\pi}{4}; \end{aligned}$$

$$\begin{aligned} (4) \quad \iiint_{\Omega_2} z dV &= \int_0^1 \left[\iint_{D_z} z dxdy \right] dz \quad D_z: x^2 + y^2 \leq z^2 \\ &= \int_0^1 z \cdot \pi z^2 dz = \pi \int_0^1 z^3 dz = \frac{\pi}{4}; \end{aligned}$$

$$(5) \quad I = \frac{\pi}{2}$$



2. 计算 $\iint_S xzdx dz + yzdz dx + x^2dxdy$, 其中 S 是上半球面 $z = \sqrt{a^2 - x^2 - y^2}$ 的内侧.

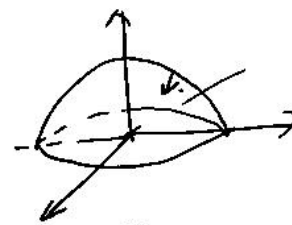
【解析】(1) 添加辅助曲面 $S_1: z = 0$, 方向向上, $D_{xy}: x^2 + y^2 \leq a^2$;

$$(2) \quad \iint_S xzdx dz + yzdz dx + x^2dxdy = \oint_{S+S_1} - \iint_{S_1};$$

$$\begin{aligned} (3) \quad \oint_{S+S_1} &= - \iiint_{\Omega} (z + z + 0) dV = -2 \iiint_{\Omega} z dV \\ &= -2 \int_0^a \left[\iint_{D_z} z dxdy \right] dz = -2 \int_0^a z \cdot \pi(a^2 - z^2) dz = -\frac{\pi}{2} a^4; \end{aligned}$$

$$(4) \quad \iint_{S_1} \stackrel{\text{垂直性}}{=} \iint_{S_1} x^2 dxdy \stackrel{\text{轮换对称}}{=} \iint_{D_{xy}} x^2 dxdy \stackrel{\text{极坐标}}{=} \frac{1}{2} \iint_{D_{xy}} (x^2 + y^2) dxdy = \frac{1}{2} \int_0^{2\pi} d\theta \int_0^a r^2 \cdot r dr = \frac{\pi}{4} a^4$$

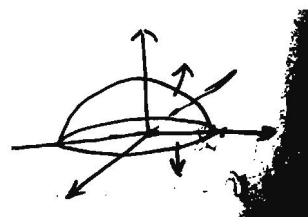
$$(5) \quad \iint_S xzdx dz + yzdz dx + x^2dxdy = -\frac{\pi}{2} a^4 - \frac{\pi}{4} a^4 = -\frac{3\pi}{4} a^4$$



3. 计算 $\iint_S \frac{xz^2 dydz + (x^2y - z^3) dzdx + (2xy + y^2z) dxdy}{x^2 + y^2 + z^2}$, 其中 S 表示上半

球面 $z = \sqrt{a^2 - x^2 - y^2}$ 的外侧.

【解析】(1) $\iint_S \frac{xz^2 dydz + (x^2y - z^3) dzdx + (2xy + y^2z) dxdy}{x^2 + y^2 + z^2}$



$$= \frac{1}{a^2} \iint_S xz^2 dydz + (x^2y - z^3) dzdx + (2xy + y^2z) dxdy ;$$

(2) 添加辅助曲面 $S_1: z=0$ ，方向向下， $D_{xy}: x^2 + y^2 \leq a^2$ ；

$$(3) \iint_S = \oiint_{S+S_1} - \iint_{S_1} ;$$

$$(4) \oiint_{S+S_1} = \iiint_{\Omega} (z^2 + x^2 + y^2) dV \stackrel{\text{球坐标}}{=} \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^a r^2 \cdot r^2 \sin \varphi dr = \frac{2}{5} \pi a^5 ;$$

$$(5) \iint_{S_1} \stackrel{\text{垂直性}}{=} \iint_{S_1} (2xy + y^2z) dxdy = -2 \iint_{D_{xy}} xy dxdy \stackrel{\text{对称性}}{=} 0 ;$$

$$(6) \iint_S \frac{xz^2 dydz + (x^2y - z^3) dzdx + (2xy + y^2z) dxdy}{x^2 + y^2 + z^2} = \frac{1}{a^2} \cdot \frac{2}{5} \pi a^5 = \frac{2}{5} \pi a^3$$

4. 已知流体的流速 $\vec{v}(x, y, z) = \{xy, yz, zx\}$ ，求由平面 $z=1, x=0, y=0$ 和锥面 $z = \sqrt{x^2 + y^2}$ 所围立体 Ω 向外流出的流量。（设流体密度为 1）

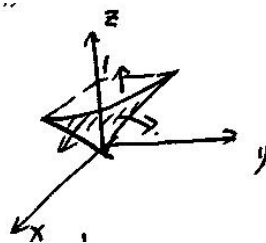
【解析】(1) $\Phi = \oiint_S xy dydz + yz dzdx + zx dxdy$ ；

$$(2) \Phi \stackrel{\text{高斯公式}}{=} \iiint_{\Omega} (x + y + z) dV \stackrel{\text{轮换对称性}}{=} 2 \iiint_{\Omega} x dV + \iiint_{\Omega} z dV \quad \left(\iiint_{\Omega} x dV = \iiint_{\Omega} y dV \right)$$

$$(3) \iiint_{\Omega} x dV \stackrel{\text{柱坐标}}{=} \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r \cos \theta \cdot r dr \int_r^1 z dz = \int_0^1 r^2 (1-r) dr = \frac{1}{12} ;$$

$$(4) \iiint_{\Omega} z dV \stackrel{\text{截面法}}{=} \int_0^1 z dz \iint_{D_z} dxdy = \int_0^1 z \cdot \frac{\pi}{4} z^2 dz = \frac{\pi}{16} ;$$

$$(5) \Phi = \frac{1}{6} + \frac{\pi}{16}$$

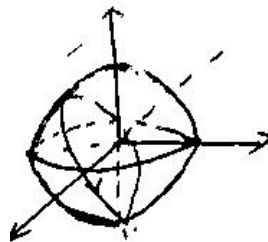


习题 12.7 Stokes 公式

1. 利用 Stokes 公式计算 $\oint_L ydx + zdy + xdz$, 其中 L 为圆周

$$L: x^2 + y^2 + z^2 = a^2, x + y + z = 0 \text{ 从 } z \text{ 轴正向看}$$

去沿逆时针方向.



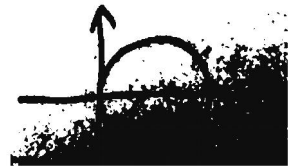
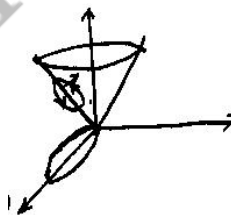
【解析】(1) $\sum: z = -x - y$, 方向向上; $\vec{n} = (1, 1, 1)$, $\vec{n}^0 = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$;

$$\begin{aligned} (2) \oint_L ydx + zdy + xdz &= \iint_{\sum} \begin{vmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix} dS = \frac{1}{\sqrt{3}} \iint_{\sum} [(0-1) - (1-0) + (0-1)] dS \\ &= \frac{1}{\sqrt{3}} \iint_{\sum} 1 dS = -\sqrt{3}\pi a^2 \end{aligned}$$

2. 计算 $I = \oint_L xydx + z^2dy + xzdz$, 其中 L 为锥面 $z = \sqrt{x^2 + y^2}$ 与柱面 $x^2 + y^2 = 2ax$ ($a > 0$) 的交线, 从 z 轴正向看去沿逆时针方向.

【解析】 $\sum: z = \sqrt{x^2 + y^2}$, 方向向上;

$$I = \iint_{\sum} \begin{vmatrix} dydz & dzdx & dxdy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & z^2 & zx \end{vmatrix} = \iint_{\sum} (-2z)dydz - zdzdx - xdx dy$$



$$= -\iint_{\sum} (2zdydz + zdzdx + xdx dy) \quad \vec{n} = \left(-\frac{x}{\sqrt{x^2 + y^2}}, -\frac{y}{\sqrt{x^2 + y^2}}, 1\right), \vec{n}^0 = \frac{1}{|\vec{n}|}\vec{n}$$

$$= -\iint_{\sum} (2z \cos \alpha + z \cos \beta + x \cos \gamma) dS = -\iint_{\sum} \left(2z \cdot \frac{\cos \alpha}{\cos \gamma} + z \cdot \frac{\cos \beta}{\cos \gamma} + x\right) \cos \gamma dS$$

$$= -\iint_{\sum} \left(2z \cdot \frac{-x}{\sqrt{x^2 + y^2}} + z \cdot \frac{-y}{\sqrt{x^2 + y^2}} + x\right) dxdy = \iint_{\sum} (2x + y - x) dxdy = \iint_{\sum} (x + y) dxdy$$

$$= \iint_{D_{xy}} (x + y) dxdy \quad D_{xy}: x^2 + y^2 \leq a^2 \quad (a > 0)$$

$$\text{对称性} \quad \iint_{D_{xy}} x dxdy = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{2a \cos \theta} r \cos \theta \cdot r dr = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \theta \cdot \left(\frac{1}{3} r^3 \Big|_0^{2a \cos \theta}\right) d\theta = \frac{8a^3}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4 \theta d\theta$$

$$= \frac{16a^3}{3} \int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta = \frac{16a^3}{3} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \pi a^3$$

习题 12.8 场论初步

1. 求函数 $u = \ln(x + \sqrt{y^2 + z^2})$ 在点 $A(1, 0, 1)$ 处沿 A 指向 $B(3, -2, 2)$ 方向的方向导数和梯度.

【解析】(1) $\overline{AB} = (2, -2, 1)$, $l = \overline{AB}^0 = \frac{1}{3}(2, -2, 1)$, 即 $\cos \alpha = \frac{2}{3}, \cos \beta = -\frac{2}{3}, \cos \gamma = \frac{1}{3}$;

$$(2) \left. \frac{\partial u}{\partial l} \right|_A = \left(\frac{\partial u}{\partial x} \cdot \cos \alpha + \frac{\partial u}{\partial y} \cdot \cos \beta + \frac{\partial u}{\partial z} \cdot \cos \gamma \right) \Big|_A$$

$$= \frac{1}{x + \sqrt{y^2 + z^2}} \left[1 \cdot \frac{2}{3} + \frac{y}{\sqrt{y^2 + z^2}} \cdot \left(-\frac{2}{3} \right) + \frac{z}{\sqrt{y^2 + z^2}} \cdot \frac{1}{3} \right] \Big|_A = \frac{1}{2};$$

$$(3) \text{grad} u|_A = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right) \Big|_A = \frac{1}{x + \sqrt{y^2 + z^2}} \left(1, \frac{y}{\sqrt{y^2 + z^2}}, \frac{z}{\sqrt{y^2 + z^2}} \right) \Big|_A = \left(\frac{1}{2}, 0, \frac{1}{2} \right)$$

2. 设 $u(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}$, 问 $u(x, y, z)$ 在点 (x, y, z) 处朝何方向的方向导数最大?

并求此时方向导数.

【解析】(1) 由方向导数与梯度关系可知 $u(x, y, z)$ 在点 (x, y, z) 处沿梯度方向的方向导数最大, 且方向导数为该梯度的模;

$$(2) \text{grad} u = \left(\frac{2x}{a^2}, \frac{2y}{b^2}, \frac{2z}{c^2} \right), \text{ 则 } \frac{\partial u}{\partial l} = 2 \sqrt{\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4}}$$

3. 设数量函数 $u = u(x, y, z)$ 具有二阶连续偏导数, 求

$$(1) \text{grad} u; \quad (2) \text{div}(\text{grad} u); \quad (3) \text{rot}(\text{grad} u).$$

【解析】(1) $\text{grad} u = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right)$;

$$(2) \text{div}(\text{grad} u) = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2};$$

$$(3) \text{rot}(\text{grad} u) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \end{vmatrix} = (0, 0, 0) = \vec{0}$$

自 测 题

一、填空题（每题 4 分，共 20 分）.

1. 设 L 为球面 $x^2 + y^2 + z^2 = a^2$ 与平面 $x + y + z = 0$ 的交线，则

$$\int_L (x^2 + y - z) ds = \underline{\hspace{2cm}}.$$

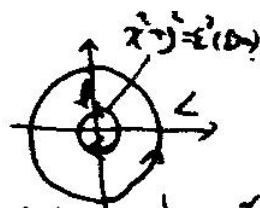
【解析】
$$\begin{aligned} \int_L (x^2 + y - z) ds &= \int_L x^2 ds + \int_L y ds - \int_L z ds \\ &= \frac{1}{3} \int_L (x^2 + y^2 + z^2) ds + \frac{1}{3} \int_L (x + y + z) ds - \frac{1}{3} \int_L (x + y + z) ds \\ &= \frac{1}{3} \int_L a^2 ds = \frac{a^2}{3} \cdot 2\pi a = \frac{2\pi a^3}{3} \end{aligned}$$

2. 设 L 为圆周 $x^2 + y^2 = a^2$ 按逆时针方向绕行，则

$$\oint_L \frac{(2xy - 3y) dx + (x^2 - 5x) dy}{x^2 + y^2} = \underline{\hspace{2cm}}.$$

【解析】 添加辅助曲线 $l: x^2 + y^2 \leq \varepsilon^2 (\varepsilon > 0)$ ，方向逆时针方向；

$$\begin{aligned} \oint_L &= \oint_{L+l} - \oint_l = 0 - \oint_l = \oint_l \frac{(2xy - 3y) dx + (x^2 - 5x) dy}{x^2 + y^2} \\ &= \frac{1}{\varepsilon^2} \oint_l (2xy - 3y) dx + (x^2 - 5x) dy = \frac{1}{\varepsilon^2} \iint_{D'} (-2) dx dy = -2 \cdot \frac{1}{\varepsilon^2} \cdot 2\pi \varepsilon^2 = -2\pi \end{aligned}$$



3. 设 S 是锥面 $z = \sqrt{x^2 + y^2}$ 被平面 $z = 2$ 所割下的有限部分，则

$$\iint_S (xy + yz + z^2) dS = \underline{\hspace{2cm}}.$$

【解析】 由对称性可知

$$\iint_S (xy + yz + z^2) dS = \iint_S z^2 dS = \sqrt{2} \iint_{D_{xy}} (x^2 + y^2) dx dy = \sqrt{2} \int_0^{2\pi} d\theta \int_0^2 r^2 \cdot r dr = 8\sqrt{2}\pi$$

4. 设 S 为球面 $x^2 + y^2 + z^2 = 1$ 的外侧，则 $\iint_S x^3 dy dz + y^3 dz dx + z^3 dx dy = \underline{\hspace{2cm}}.$

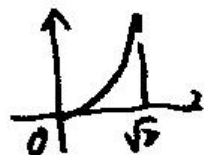
【解析】 利用高斯公式得 $\iint_S x^3 dy dz + y^3 dz dx + z^3 dx dy = 3 \iiint_{\Omega} (x^2 + y^2 + z^2) dV = \frac{12}{5} \pi$ (球坐标)

5. 设 $\vec{F}(x, y, z) = \{e^x \sin y, 2xy^2 + z, xzy^2\}$ ，则 $\operatorname{div} \vec{F}|_{(1,0,1)} = \underline{\hspace{2cm}},$

$$\operatorname{rot} \vec{F}|_{(1,0,1)} = \underline{\hspace{2cm}}.$$

【解析】 利用公式计算得 $\operatorname{div} \vec{F}|_{(1,0,1)} = 0$ ， $\operatorname{rot} \vec{F}|_{(1,0,1)} = (-1, 0, -e)$

二、选择题（每小题 4 分，共 20 分）.



6. 已知曲线 $L: y = x^2 (0 \leq x \leq \sqrt{2})$, 则 $\int_L x ds = ()$.

- (A) 2 (B) 0 (C) $\frac{13}{6}$ (D) $\frac{5}{6}$

【解析】 $\int_L x ds = \int_0^{\sqrt{2}} x \cdot \sqrt{4x^2 + 1} dx = \frac{13}{6}$

7. 设 L 是柱面方程 $x^2 + y^2 = 1$ 与平面 $z = x + y$ 的交线, 从 z 轴正向往 z 轴负向看去为逆时针方向, 则曲线

积分 $\oint_L xz dx + x dy + \frac{y^2}{2} dz = ()$.

- (A) π (B) 2π (C) 0 (D) 1

【解析】 $\sum: z = x + y; D_{xy}: x^2 + y^2 \leq 1$, 方向向上

$$\text{原式} = \iint_{\sum} \begin{vmatrix} -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & x & \frac{y^2}{2} \end{vmatrix} dS = \frac{1}{\sqrt{3}} \iint_{\sum} (-y - x + 1) dS = \frac{1}{\sqrt{3}} \iint_{D_{xy}} (-y - x + 1) \cdot \sqrt{3} dx dy = \pi$$

8. 设曲线积分 $\oint_L [f(x) - e^x] \sin y dx - f(x) \cos y dy$ 与路径无关, 其中 $f(x)$ 具有一阶连续导数, 且

$f(0) = 0$, 则 $f(x)$ 等于 ().

- (A) $\frac{1}{2}(e^{-x} - e^x)$ (B) $\frac{1}{2}(e^x - e^{-x})$ (C) $\frac{1}{2}(e^x + e^{-x}) - 1$ (D) $1 - \frac{1}{2}(e^x + e^{-x})$

【解析】 $P = (f(x) - e^x) \sin y$, $Q = -f(x) \cos y$, 因为积分与路径无关, 则

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow f'(x) + f(x) = e^x,$$

解一阶非齐次线性微分方程得: $f(x) = e^{-x} \left(\frac{1}{2} e^{2x} + C \right)$; 又 $f(0) = 0 \Rightarrow C = -\frac{1}{2}$, 则选 B

9. 设 S 为曲面 $x^2 + 2y^2 + z^2 = 1$, 在下面积分中, 积分值均为 0 的是 ().

- (A) $\iint_S z^2 dS$ 与 $\iint_S z^2 dx dy$ (B) $\iint_S z dS$ 与 $\iint_S z dx dy$
(C) $\iint_S z dS$ 与 $\iint_S z^2 dx dy$ (D) $\iint_S yz dS$ 与 $\iint_S x dy dz$

【解析】由对称性可知 $\iint_S z dS = \iint_S yz dS = 0$, $\iint_S z^2 dS \neq 0$, $\iint_S z dx dy \neq 0$, $\iint_S x dx dy \neq 0$, $\iint_S z^2 dx dy = 0$, 所以答案选 C

10. 设函数 $f(x, y, z) = x^2 + y^2 - z$ 在点 $P(1, 1, 1)$ 处沿单位向量 $\vec{v}$ 的方向增加最快, 则 $\vec{v} =$ ().

(A) $\frac{1}{3}(2, 2, -1)$ (B) $\frac{1}{3}(-2, -2, 1)$ (C) $\frac{1}{\sqrt{3}}(1, 1, 1)$ (D) $\frac{1}{\sqrt{3}}(1, 1, -1)$

【解析】(1) 该方向为梯度方向, 梯度 $\left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \Big|_P = (2, 2, -1)$;

(2) $\vec{v}$ 为梯度的单位向量 $\frac{1}{3}(2, 2, -1)$

三、解答题 (每小题 10 分, 共 60 分).

11. 已知一非均匀金属丝 L 的方程为 $L: x = a(t - \sin t), y = a(1 - \cos t), 0 \leq t \leq 2\pi$, 它在点 (x, y) 处的线密度

为 $\rho(x, y) = |y|$, 求该金属丝的质量.

【解析】(1) $M = \int_L \rho(x, y) ds = \int_L |y| ds$;

(2) $ds = \sqrt{x_t'^2 + y_t'^2} = \sqrt{[a(1 - \cos t)]^2 + [a \sin t]^2} dt = 2a \sin \frac{t}{2} dt$;

(3)

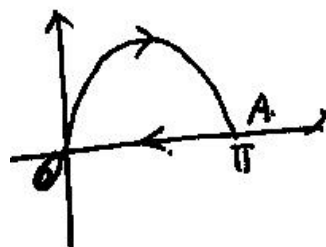
$$\begin{aligned} M &= \int_0^{2\pi} |a(1 - \cos t)| \cdot 2a \sin \frac{t}{2} dt = 2a^2 \int_0^{2\pi} (1 - \cos t) \cdot \sin \frac{t}{2} dt \\ &= 2a^2 \int_0^{2\pi} \left[1 - (1 - 2\sin^2 \frac{t}{2}) \right] \cdot \sin \frac{t}{2} dt \\ &= 4a^2 \int_0^{2\pi} \sin^3 \frac{t}{2} dt \stackrel{\frac{t}{2}=u}{=} 8a^2 \int_0^{\pi} \sin^3 u du = 8a^2 \int_0^{\pi} (1 - \cos^2 u) d(-\cos u) = \frac{32}{3} a^2 \end{aligned}$$

12. 计算曲线积分 $\int_L \sin 2x dx + 2(x^2 - 1)y dy$, 其中 L 是曲线 $y = \sin x$ 从 $(0, 0)$ 到 $(\pi, 0)$ 的一段.

【解析】(1) 添加 $\overline{AO}: y = 0$, 从 $A \rightarrow O$;

(2) $\int_L = \oint_{L+\overline{AO}} - \int_{\overline{AO}}$;

(3) $\oint_{L+\overline{AO}} \stackrel{\text{格林公式}}{=} - \iint_D 4xy dx dy = -4 \int_0^{\pi} x dx \int_0^{\sin x} y dy$



$$= -4 \int_0^{\pi} x \cdot \frac{1}{2} \sin^2 x dx = -\frac{1}{2} \pi^2;$$

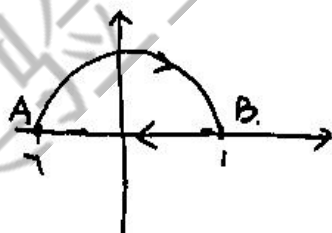
$$(4) \int_{AO} = \int_{\pi}^0 \sin 2x dx = 0;$$

$$(5) \int_L = \oint_{L+\overline{AO}} - \int_{\overline{AO}} = -\frac{1}{2} \pi^2$$

13. 计算曲线积分 $\int_L \frac{(xe^x + 5y^3x^2 + x - 4)dx - (3x^5 + \sin y)dy}{x^2 + y^2}$, 其中 L 为从点 $A(-1,0)$ 沿

$$\text{曲线 } y = \sqrt{1-x^2}$$

到点 $B(1,0)$ 一段弧.



【解析】(1) 原式 $= \int_L (xe^x + 5y^3x^2 + x - 4)dx - (3x^5 + \sin y)dy$;

(2) 添加 $\overline{BA}: y=0, x:1 \rightarrow -1$;

$$(3) \int_L = \oint_{L+\overline{BA}} - \int_{\overline{BA}};$$

$$(4) \oint_{L+\overline{BA}} = - \iint_D (-15x^4 - 15x^2y^2) dxdy = 15 \iint_D (x^4 + x^2y^2) dxdy$$

$$= 15 \int_0^{\pi} d\theta \int_0^1 (r^4 \cos^4 \theta + r^4 \cos^2 \theta \sin^2 \theta) \cdot r dr = 15 \int_0^{\pi} (\cos^4 \theta + \cos^2 \theta \sin^2 \theta) \cdot \frac{1}{6} d\theta$$

$$= \frac{5}{2} \int_0^{\pi} [\cos^4 \theta + \cos^2 \theta (1 - \cos^2 \theta)] d\theta = \frac{5}{2} \int_0^{\pi} \cos^2 \theta d\theta = 5 \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta = \frac{5\pi}{4};$$

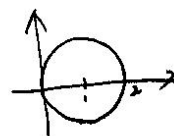
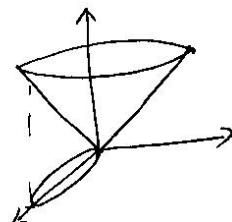
$$(5) \int_{\overline{BA}} = \int_1^{-1} (xe^2 + x - 4)dx = -2e^{-1} + 8;$$

$$(6) \text{原式} = \frac{5\pi}{4} - 8 + 2e^{-1}$$

14. 计算曲面积分 $\iint_S z dS$, 其中 S 为锥面 $z = \sqrt{x^2 + y^2}$ 在柱面 $x^2 + y^2 \leq 2x$ 内部分.

【解析】(1) $\iint_S z dS = \iint_{D_{xy}} \sqrt{x^2 + y^2} \cdot \sqrt{2} dxdy = \sqrt{2} \iint_{D_{xy}} \sqrt{x^2 + y^2} dxdy$

$$= \sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{2\cos\theta} r \cdot r dr = \frac{8\sqrt{2}}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^3 \theta d\theta = \frac{16\sqrt{2}}{3} \int_0^{\frac{\pi}{2}} \cos^3 \theta d\theta = \frac{32\sqrt{2}}{9}$$



15. 计算曲面积分 $\iint_S xdydz + ydzdx + zdx dy$, 其中 S 是柱面 $x^2 + y^2 = 1$ 被平面 $z = 0$ 及 $z = 3$ 所截得的在第一卦限内的部分前侧.

【解析】(1) 由垂直线可知 $\iint_S zdx dy = 0$, 则原式 $= \iint_S xdydz + ydzdx$;

$$(2) \iint_S xdydz = \iint_{D_{yz}} \sqrt{1-y^2} dydz = \int_0^1 dy \int_0^3 \sqrt{1-y^2} dz = 3 \int_0^1 \sqrt{1-y^2} dy = \frac{3\pi}{4} ;$$

$$(3) \iint_S ydzdx = \iint_{D_{yz}} \sqrt{1-x^2} dx dz = \int_0^1 dx \int_0^3 \sqrt{1-x^2} dz = 3 \int_0^1 \sqrt{1-x^2} dx = \frac{3\pi}{4} ;$$

$$(4) \text{原式} = \frac{3\pi}{2}$$

16. 计算曲面积分 $\iint_S (x^3 + az^2) dydz + (y^3 + ax^2) dzdx + (z^3 + ay^2) dx dy$, 其中 S 为上半球面

$z = \sqrt{a^2 - x^2 - y^2}$ 的上侧.

【解析】(1) 添加 $S_1: z = 0$, 方向向下;

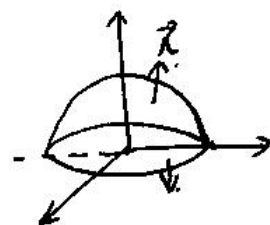
$$(2) \iint_S = \oiint_{S+S_1} - \iint_{S_1} ;$$

$$(3) \oiint_{S+S_1} = \iiint_{\Omega} (3x^2 + 3y^2 + 3z^2) dV = 3 \iiint_{\Omega} (x^2 + y^2 + z^2) dV \\ = 3 \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^a r^2 \cdot r^2 \sin \varphi dr = \frac{6\pi}{5} a^5 ;$$

$$(4) \iint_{S_1} \stackrel{\text{垂直性}}{=} \iint_{S_1} (z^3 + ay^2) dx dy = \iint_{S_1} ay^2 dx dy = - \iint_{D_{xy}} ay^2 dx dy \quad D_{xy}: x^2 + y^2 \leq a^2$$

$$= -a \iint_{D_{xy}} y^2 dx dy = -a \cdot \frac{1}{2} \iint_{D_{xy}} (x^2 + y^2) dx dy = -\frac{a}{2} \int_0^{2\pi} d\theta \int_0^a r \cdot r^2 dr = -\frac{\pi}{4} a^5 ;$$

$$(5) \iint_S = \frac{6\pi}{5} a^5 + \frac{\pi}{4} a^5 = \frac{29}{20} \pi a^5$$



第十三章 无穷级数

习题 13.1 数项级数的概念

1. 根据级数收敛与发散的定定义判别下列级数敛散性.

$$(1) \sum_{n=1}^{\infty} \frac{1}{(3n-2)(3n+1)};$$

【解析】(1) $\frac{1}{(3n-2)(3n+1)} = \frac{1}{3} \left(\frac{1}{3n-2} - \frac{1}{3n+1} \right);$

$$(2) S_n = \frac{1}{3} \left[\left(1 - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{7} \right) + \left(\frac{1}{7} - \frac{1}{10} \right) + \cdots + \left(\frac{1}{3n-2} - \frac{1}{3n+1} \right) \right] = \frac{1}{3} \left(1 - \frac{1}{3n+1} \right);$$

$$(3) \lim_{n \rightarrow \infty} S_n = \frac{1}{3}, \text{ 则级数收敛}$$

$$(2) \sum_{n=1}^{\infty} \frac{1}{\sqrt{n+1} + \sqrt{n}};$$

【解析】(1) $\frac{1}{\sqrt{n+1} + \sqrt{n}} = \sqrt{n+1} - \sqrt{n};$

$$(2) S_n = (\sqrt{2} - 1) + (\sqrt{3} - \sqrt{2}) + \cdots + (\sqrt{n+1} - \sqrt{n}) = \sqrt{n+1} - 1;$$

$$(3) \lim_{n \rightarrow \infty} S_n = \infty, \text{ 则级数发散}$$

$$(3) \sum_{n=1}^{\infty} (-1)^{n-1} \left(\frac{9}{10} \right)^n;$$

【解析】 $\sum_{n=1}^{\infty} (-1)^{n-1} \left(\frac{9}{10} \right)^n = \sum_{n=1}^{\infty} (-1)^{n-1} \left(\frac{9}{10} \right)^{n-1} \cdot \frac{9}{10} = \frac{9}{10} \sum_{n=1}^{\infty} \left(-\frac{9}{10} \right)^{n-1},$

由于公比 $q = -\frac{9}{10}$, $|q| < 1$, 所以原级数收敛

$$(4) \sum_{n=1}^{\infty} \sin \frac{n}{6} \pi$$

【解析】 $u_{12n+k} = \sin \left(\frac{12n+k}{6} \pi \right) = \sin \left(2n\pi + \frac{k}{6} \pi \right) = \sin \frac{k}{6} \pi = u_k$, 所以

$$u_{12n} = u_{12} = 0, \text{ 而 } u_{12n+1} = u_1 = \sin \frac{\pi}{6} = \frac{1}{2},$$

则 $\lim_{n \rightarrow \infty} u_{12n} = 0 \neq \lim_{n \rightarrow \infty} u_{12n+1} = \frac{1}{2}$, 极限不存在, 所以原级数发散.

2. 判别下列级数敛散性.

$$(1) \sum_{n=1}^{\infty} \frac{n^2 + 3^n}{n^2 \cdot 3^n};$$

【解析】(1) $\frac{n^2 + 3^n}{n^2 \cdot 3^n} = \frac{1}{3^n} + \frac{1}{n^2};$

(2) $\sum_{n=1}^{\infty} \frac{1}{3^n}, q = \frac{1}{3} < 1$, 该级数收敛; 又 $\sum_{n=1}^{\infty} \frac{1}{n^2}, p = 2 > 1$, 该级数收敛;

(3) 由性质可知原级数收敛

$$(2) \sum_{n=2}^{\infty} n \tan \frac{\pi}{n}.$$

【解析】 $\lim_{n \rightarrow \infty} n \tan \frac{\pi}{n} = \lim_{n \rightarrow \infty} n \cdot \frac{\pi}{n} = \pi \neq 0$

3. 设级数 $\sum_{n=1}^{\infty} a_n$ 收敛, 且 $\lim_{n \rightarrow \infty} na_n = 0$, 证明 $\sum_{n=1}^{\infty} (n+1)(a_{n+1} - a_n)$ 收敛.

【解析】 $\sum_{n=1}^{\infty} a_n$ 部分和数列为 S_n , $\sum_{n=1}^{\infty} (n+1)(a_{n+1} - a_n)$ 部分和数列为 σ_n ;

因为 $\sum_{n=1}^{\infty} a_n$ 收敛, 且 $\lim_{n \rightarrow \infty} na_n = 0$, 则 $\lim_{n \rightarrow \infty} S_n = S$;

又因为 $\sigma_n = 2(a_2 - a_1) + 3(a_3 - a_2) + \cdots + (n+1)(a_{n+1} - a_n) = -a_1 - S_n + (n+1)a_{n+1}$, 则

$\lim_{n \rightarrow \infty} \sigma_n = -a_1 - S$, 所以 $\sum_{n=1}^{\infty} (n+1)(a_{n+1} - a_n)$ 收敛

习题 13.2 数项级数的收敛判别法

1. 用比较判别法收敛或其极限形式判别下列级数敛散性.

$$(1) \sum_{n=1}^{\infty} \left(1 - \cos \frac{2}{n}\right); \quad (2) \sum_{n=1}^{\infty} \frac{1}{\ln(n+1)};$$

$$(3) \sum_{n=1}^{\infty} \frac{1}{n^n \sqrt[n]{n}}; \quad (4) \sum_{n=1}^{\infty} \frac{1}{1+a^n}.$$

【解析】(1) $n \rightarrow \infty, 1 - \cos \frac{2}{n} \sim \frac{2}{n^2}$, 又 $\sum_{n=1}^{\infty} \frac{1}{n^2}$ 收敛, 所以原级数收敛;

$$(2) \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{\frac{1}{\ln(n+1)}} = \lim_{n \rightarrow \infty} \frac{\ln(n+1)}{n} = 0, \text{ 而 } \sum_{n=1}^{\infty} \frac{1}{n} \text{ 发散, 则原级数发散;}$$

$$(3) \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{\frac{1}{n^n \sqrt[n]{n}}} = \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1, \text{ 而 } \sum_{n=1}^{\infty} \frac{1}{n} \text{ 发散, 则原级数发散;}$$

$$(4) \textcircled{1} \quad 0 < a < 1 \text{ 时, } \lim_{n \rightarrow \infty} \frac{1}{1+a^n} = 1 \neq 0, \text{ 发散;}$$

$$\textcircled{2} \quad a = 1 \text{ 时, } \frac{1}{1+a^n} = \frac{1}{2}, \text{ 通项极限不区域零, 则发散;}$$

$$\textcircled{3} \quad a > 1 \text{ 时, } \lim_{n \rightarrow \infty} \frac{\frac{1}{a^n}}{\frac{1}{1+a^n}} = \lim_{n \rightarrow \infty} \left(\frac{1}{a^n} + 1\right) = 1, \text{ 而 } \sum_{n=1}^{\infty} \frac{1}{a^n} \text{ 收敛, 所以原级数收敛.}$$

2. 用比值判别法或根值判别法判别下列级数的敛散性.

$$(1) \sum_{n=1}^{\infty} \frac{n^n}{(2n)!}; \quad (2) \sum_{n=1}^{\infty} n \tan \frac{\pi}{2^{n+1}};$$

$$(3) \sum_{n=1}^{\infty} \frac{3+(-1)^n}{2^n}; \quad (4) \sum_{n=1}^{\infty} \frac{2^n}{\left(1+\frac{1}{n}\right)^{2n}}.$$

【解析】(1) $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1}}{n^n} \cdot \frac{(2n)!}{[2(n+1)]!} = \lim_{n \rightarrow \infty} \frac{1}{2(2n+1)} \left(1 + \frac{1}{n}\right)^n = 0 < 1$, 原级数收敛;

$$(2) \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1) \tan \frac{\pi}{2^{n+2}}}{n \tan \frac{\pi}{2^{n+1}}} = \lim_{n \rightarrow \infty} \frac{n+1}{n} \cdot \frac{2^{\frac{n+2}{2}}}{\frac{\pi}{2^{n+1}}} = \frac{1}{2} < 1, \text{ 原级数收敛;}$$

$$(3) \lim_{n \rightarrow \infty} \sqrt[n]{\frac{3+(-1)^n}{2^n}} = \frac{1}{2} \lim_{n \rightarrow \infty} \sqrt[n]{3+(-1)^n} = \frac{1}{2} < 1, \text{ 原级数收敛};$$

$$(4) \lim_{n \rightarrow \infty} \sqrt[n]{\frac{2^n}{\left(1+\frac{1}{n}\right)^{2n}}} = \lim_{n \rightarrow \infty} \frac{2}{\left(1+\frac{1}{n}\right)^2} = 2 > 1, \text{ 原级数发散}$$

3. 判别下列级数敛散性, 若收敛, 说明是条件收敛还是绝对收敛.

$$(1) \sum_{n=2}^{\infty} \frac{(-1)^n}{\sqrt{n^2-n}};$$

$$(2) \sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{3^{n-1}};$$

$$(3) \sum_{n=1}^{\infty} (-1)^{n-1} \frac{2^{n^2}}{n!};$$

$$(4) \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{\ln(n+1)}.$$

【解析】(1) $\sum_{n=2}^{\infty} \left| \frac{(-1)^n}{\sqrt{n^2-n}} \right| = \sum_{n=2}^{\infty} \frac{1}{\sqrt{n^2-n}},$ 而 $\frac{1}{\sqrt{n^2-n}} > \frac{1}{n},$ 又 $\sum_{n=1}^{\infty} \frac{1}{n}$ 发散, 所以由比较判

别法可知 $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n^2-n}}$ 发散; 而 $\sum_{n=2}^{\infty} \frac{(-1)^n}{\sqrt{n^2-n}}$ 中 $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^2-n}} = 0,$ 且 $u_n = \frac{1}{\sqrt{n^2-n}}$ 随 n 的

增大而减小, 由莱布尼兹判别法得 $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2-n}}$ 收敛; 综上所述原级数条件收敛.

$$(2) \sum_{n=1}^{\infty} \left| (-1)^{n-1} \frac{n}{3^{n-1}} \right| = \sum_{n=1}^{\infty} \frac{n}{3^{n-1}}, \quad \lim_{n \rightarrow \infty} \frac{n+1}{3} \cdot \frac{3^{n-1}}{n} = \lim_{n \rightarrow \infty} \frac{n+1}{3n} = \frac{1}{3} < 1, \text{ 原级数绝对收敛}.$$

$$(3) \sum_{n=1}^{\infty} \left| (-1)^{n-1} \frac{2^{n^2}}{n!} \right| = \sum_{n=1}^{\infty} \frac{2^{n^2}}{n!}, \quad \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{2^{(n+1)^2}}{(n+1)!} \cdot \frac{n!}{2^{n^2}} = \lim_{n \rightarrow \infty} \frac{2^{2n+1}}{n+1} \rightarrow \infty, \text{ 原级数发}$$

散.

$$(4) \sum_{n=1}^{\infty} \left| (-1)^{n-1} \frac{1}{\ln(n+1)} \right| = \sum_{n=1}^{\infty} \frac{1}{\ln(n+1)}, \quad \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{\frac{1}{\ln(n+1)}} = 0, \text{ 则 } \sum_{n=1}^{\infty} \frac{1}{\ln(n+1)} \text{ 发散};$$

而 $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{\ln(n+1)}$ 中 $\lim_{n \rightarrow \infty} \frac{1}{\ln(n+1)} = 0,$ 且 $u_n = \frac{1}{\ln(n+1)}$ 随 n 的增大而减小, 由莱布尼兹判别法得

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{\ln(n+1)} \text{ 收敛; 综上所述原级数条件收敛}.$$

4. 若级数 $\sum_{n=1}^{\infty} a_n$ 绝对收敛, 且 $a_n \neq -1 (n=1, 2, \dots)$, 试证 $\sum_{n=1}^{\infty} \frac{a_n}{1+a_n}$ 和 $\sum_{n=1}^{\infty} \frac{a_n^2}{1+a_n^2}$ 都收敛,

而级数 $\sum_{n=1}^{\infty} \frac{1}{1+a_n}$ 发

散.

【证明】(1) $\sum_{n=1}^{\infty} a_n$ 绝对收敛 $\Rightarrow \sum_{n=1}^{\infty} a_n, \sum_{n=1}^{\infty} |a_n|$ 都收敛且 $\lim_{n \rightarrow \infty} a_n = 0$

(2) $\lim_{n \rightarrow \infty} (1+a_n) = 1, \exists \varepsilon = \frac{1}{2}, \exists N \in \mathbb{N}, \forall n > N, 1+a_n > \frac{1}{2}, \left| \frac{1}{1+a_n} \right| \leq 2, \left| \frac{a_n}{1+a_n} \right| \leq 2|a_n|,$

而 $\sum_{n=1}^{\infty} |a_n|$ 收敛, 由比较判别法可知 $\sum_{n=1}^{\infty} \left| \frac{a_n}{1+a_n} \right|$ 收敛, 则 $\sum_{n=1}^{\infty} \frac{a_n}{1+a_n}$ 绝对收敛, 则 $\sum_{n=1}^{\infty} \frac{a_n}{1+a_n}$

必收敛;

(3) $\left| \frac{a_n^2}{1+a_n^2} \right| \leq |a_n|^2$, 而 $\lim_{n \rightarrow \infty} a_n = 0 \Rightarrow \exists M > 0, |a_n| \leq M \Rightarrow |a_n|^2 \leq M|a_n|$, 由比较判别

法可知 $\sum_{n=1}^{\infty} \frac{a_n^2}{1+a_n^2}$ 绝对收敛, 则原级数收敛;

(4) (反证法) 假设 $\sum_{n=1}^{\infty} \frac{1}{1+a_n}$ 收敛, 则 $\lim_{n \rightarrow \infty} \frac{1}{1+a_n} = 0$, 进而 $\lim_{n \rightarrow \infty} a_n = \infty$ 与已知 $\lim_{n \rightarrow \infty} a_n = 0$ 矛

盾, 所以 $\sum_{n=1}^{\infty} \frac{1}{1+a_n}$

发散.

习题 13.3 幂级数

1. 求下列幂级数的收敛半径和收敛域.

$$(1) \sum_{n=1}^{\infty} \frac{(x-2)^n}{n5^n};$$

【解析】(1)

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}(x)}{u_n(x)} \right| = \lim_{n \rightarrow \infty} \left| \frac{\frac{(x-2)^{n+1}}{(n+1) \cdot 5^{n+1}}}{\frac{(x-2)^n}{n \cdot 5^n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{n}{n+1} \cdot \frac{1}{5} (x-2) \right| = \frac{1}{5} |x-2| < 1 \Rightarrow |x-2| < 5,$$

$$R = 5;$$

(2) 解得收敛区间为 $(-3, 7)$;

(3) 当 $x = -3$ 时, $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ 收敛; 当 $x = 7$ 时, $\sum_{n=1}^{\infty} \frac{1}{n}$ 发散;

(4) 收敛域为 $[-3, 7)$.

$$(2) \sum_{n=1}^{\infty} (\sqrt{n+1} - \sqrt{n}) \cdot 2^n \cdot x^{2n}.$$

$$\text{【解析】(1) } \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}(x)}{u_n(x)} \right| = \lim_{n \rightarrow \infty} \left| \frac{(\sqrt{n+2} - \sqrt{n+1}) \cdot 2^{n+1} \cdot x^{2n+2}}{(\sqrt{n+1} - \sqrt{n}) \cdot 2^n \cdot x^{2n}} \right| = 2|x|^2 < 1 \Rightarrow |x| < \frac{1}{\sqrt{2}},$$

$$R = \frac{1}{\sqrt{2}};$$

(2) 收敛区间为 $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$;

(3) $x = \pm \frac{1}{\sqrt{2}}$ 时, $\sum_{n=1}^{\infty} (\sqrt{n+1} - \sqrt{n}) \cdot 2^n \cdot \frac{1}{2^n} = \sum_{n=1}^{\infty} (\sqrt{n+1} - \sqrt{n})$ 发散 (因为部分和极限不存在);

(4) 收敛域为 $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.

2. 求下列幂级数的和函数.

$$(1) \sum_{n=1}^{\infty} n(n+1)x^n \quad (|x| < 1);$$

【解析】(1) 计算得收敛域为 $(-1, 1)$;

$$(2) \quad \forall x \in (-1, 1), \quad S(x) = \sum_{n=1}^{\infty} n(n+1)x^n = 1 \cdot 2x + 2 \cdot 3x^2 + 3 \cdot 4x^3 + \cdots,$$

$$S(x) = x(1 \cdot 2 + 2 \cdot 3x + 3 \cdot 4x^2 + \cdots) = x(x^2 + x^3 + x^4 + \cdots)' = x \cdot \left(\frac{x^2}{1-x} \right)' = \frac{2x}{(1-x)^3}.$$

$$(2) \quad \sum_{n=1}^{\infty} (-1)^{n-1} \cdot \frac{x^{2n-1}}{2n-1} \quad (|x| < 1), \text{ 并求级数 } \sum_{n=1}^{\infty} \frac{(-1)^n}{2n-1} \left(\frac{3}{4} \right)^n \text{ 的和.}$$

【解析】(1) 计算得收敛域为 $[-1, 1]$;

$$(2) \quad \forall x \in (-1, 1), \quad \text{令 } S(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \cdot \frac{x^{2n-1}}{2n-1} = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \cdots, \quad S(0) = 0;$$

$$S'(x) = 1 - x^2 + x^4 - x^6 + \cdots = \frac{1}{1+x^2};$$

$$(3) \quad \int_0^x S'(t) dt = \int_0^x \frac{1}{1+t^2} dt = \arctan x, \quad S(x) - S(0) = \arctan x \Rightarrow S(x) = \arctan x,$$

$$x \in [-1, 1];$$

$$(4) \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{2n-1} \left(\frac{3}{4} \right)^n = - \sum_{n=1}^{\infty} \frac{(-1)^n}{2n-1} \left(\frac{\sqrt{3}}{2} \right)^{2n-1} \cdot \frac{\sqrt{3}}{2} = - \frac{\sqrt{3}}{2} S \left(\frac{\sqrt{3}}{2} \right) = - \frac{\sqrt{3}}{2} \arctan \frac{\sqrt{3}}{2}$$

$$(3) \quad \sum_{n=0}^{\infty} \frac{(2n+1)x^{2n}}{n!}, \text{ 并求级数 } \sum_{n=0}^{\infty} \frac{2n+1}{n! 2^n} \text{ 的和.}$$

【解析】(1) 计算得收敛域为 R

$$(2) \quad \forall x \in R, \quad S(x) = \sum_{n=0}^{\infty} \frac{(2n+1)x^{2n}}{n!}, \text{ 则 } \int_0^x S(t) dt = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{n!} = x \cdot \sum_{n=0}^{\infty} \frac{(x^2)^n}{n!} = xe^{x^2};$$

两边求导, 得

$$S(x) = (1 + 2x^2)e^{x^2}, \quad \forall x \in R$$

$$(3) \quad \sum_{n=0}^{\infty} \frac{2n+1}{n! 2^n} = \sum_{n=0}^{\infty} \frac{2n+1}{n!} \left(\frac{1}{\sqrt{2}} \right)^{2n} = S'_1 \left(\frac{1}{\sqrt{2}} \right) = \left[1 + 2 \left(\frac{1}{\sqrt{2}} \right)^2 \right] e^{\left(\frac{1}{\sqrt{2}} \right)^2} = 2e^{\frac{1}{2}}.$$

3. 将函数 $f(x) = (1+x)\ln(1+x)$ 展开成 x 的幂级数.

$$\text{【解析】(1) } f'(x) = 1 + \ln(1+x), f''(x) = \frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n, \quad x \in (-1, 1);$$

(2) 逐项积分得:

$$f'(x) - f'(0) = \int_0^x f''(t) dt = \int_0^x \sum_{n=0}^{\infty} (-1)^n t^n dt = \sum_{n=0}^{\infty} \int_0^x (-1)^n t^n dt = \sum_{n=0}^{\infty} (-1)^n \frac{1}{n+1} x^{n+1}$$

又 $f'(0)=1$, 则 $f'(x) = 1 + \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$;

(3) 逐项积分得:

$$\begin{aligned} f(x) - f(0) &= \int_0^x f'(t) dt = \int_0^x \left[1 + \sum_{n=0}^{\infty} (-1)^n \frac{t^{n+1}}{n+1} \right] dt = x + \sum_{n=0}^{\infty} \int_0^x (-1)^n \frac{t^{n+1}}{n+1} dt \\ &= x + \sum_{n=0}^{\infty} (-1)^n \frac{1}{(n+1)(n+2)} x^{n+2}, \end{aligned}$$

又 $f(0)=0$, 则 $f(x) = x + \sum_{n=0}^{\infty} (-1)^n \frac{1}{(n+1)(n+2)} x^{n+2}$, $x \in (-1, 1)$

$$= x + \sum_{n=2}^{\infty} (-1)^n \frac{1}{n(n-1)} x^n, \quad x \in (-1, 1);$$

(4) $x=1$ 时, 收敛; $x=-1$ 时, 收敛, 所以 $f(x) = x + \sum_{n=2}^{\infty} (-1)^n \frac{1}{n(n-1)} x^n$, $x \in [-1, 1]$.

4. 将函数 $f(x) = \frac{1}{x^2 + 4x + 3}$ 展开成 $x-1$ 的幂级数.

【解析】(1)
$$\begin{aligned} f(x) &= \frac{1}{x^2 + 4x + 3} = \frac{1}{(x+1)(x+3)} = \frac{1}{2} \left(\frac{1}{x+1} - \frac{1}{x+3} \right) \\ &= \frac{1}{2} \left[\frac{1}{2+(x-1)} - \frac{1}{4+(x-1)} \right] = \frac{1}{2} \left[\frac{1}{2} \cdot \frac{1}{1+\frac{x-1}{2}} - \frac{1}{4} \cdot \frac{1}{1+\frac{x-1}{4}} \right] \\ &= \frac{1}{2} \left[\frac{1}{2} \cdot \sum_{n=0}^{\infty} \left(-\frac{x-1}{2} \right)^n - \frac{1}{4} \cdot \sum_{n=0}^{\infty} \left(-\frac{x-1}{4} \right)^n \right] \\ &= \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{2^{n+2}} - \frac{1}{2^{2n+3}} \right) \cdot (x-1)^n; \end{aligned}$$

(2)
$$\begin{cases} \left| -\frac{x-1}{2} \right| < 1 \Rightarrow |x-1| < 2 \\ \left| -\frac{x-1}{4} \right| < 1 \Rightarrow |x-1| < 4 \end{cases} \Rightarrow |x-1| < 2;$$

$$(3) \quad f(x) = \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{2^{n+1}} - \frac{1}{2^{2n+1}} \right) \cdot (x-1)^n \quad (|x-1| < 2).$$

安徽大学 ACM/ICPC 实验班

习题 13.4 Fourier 级数

1. 设 $f(x)$ 是以 2π 为周期的周期函数, 且

$$f(x) = \begin{cases} x & -\pi < x < 0; \\ 0 & 0 \leq x \leq \pi. \end{cases}$$

将 $f(x)$ 展开成以 2π 为周期的 Fourier 级数.

【解析】(1) 间断点 $x = k\pi (k = \pm 1, \pm 3, \pm 5, \dots)$ 处收敛于 $-\frac{\pi}{2}$;

(2) $x \neq k\pi (k = \pm 1, \pm 3, \pm 5, \dots)$ 收敛于 $f(x)$;

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^0 x dx = -\frac{\pi}{2},$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^0 x \cos nx dx = \frac{1}{\pi n} \int_{-\pi}^0 x d(\sin nx) = \frac{1 - \cos n\pi}{\pi n^2} = \frac{1 - (-1)^n}{\pi n^2} = \begin{cases} \frac{2}{\pi n^2}, & n = 1, 3, 5, \dots \\ 0, & n = 2, 4, 6, \dots \end{cases}$$

,

$$b_n = \frac{1}{\pi} \int_{-\pi}^0 x \sin nx dx = -\frac{1}{\pi n} \int_{-\pi}^0 x d(\cos nx) = -\frac{1}{n} \cos n\pi = \frac{(-1)^{n+1}}{n},$$

$$f(x) = -\frac{\pi}{4} + \sum_{n=1}^{\infty} \left[\frac{1 - (-1)^n}{\pi n^2} \cos nx + \frac{(-1)^{n+1}}{n} \sin nx \right].$$

2. 将函数 $f(x) = x - 1 (0 \leq x \leq 2)$ 展开成以 4 为周期的余弦级数.

【解析】所给函数定义在半周期上, 因此作偶延拓及周期延拓展成余弦函数.

$$a_0 = \frac{2}{l} \int_0^l f(x) dx = \int_0^2 (x-1) dx = 0;$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx = \int_0^2 (x-1) \cos \frac{n\pi x}{2} dx = \frac{4}{n^2 \pi^2} [(-1)^n - 1] = \begin{cases} -\frac{8}{\pi^2 n^2}, & n = 1, 3, 5, \dots \\ 0, & n = 2, 4, 6, \dots \end{cases}$$

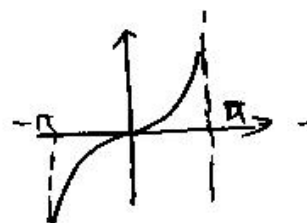
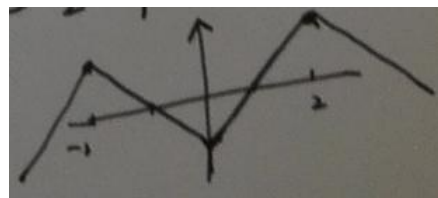
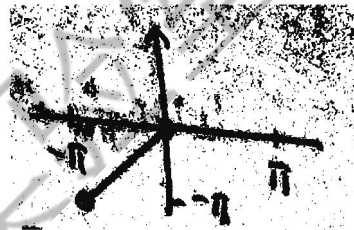
;

$$\text{则 } f(x) = -\frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos \frac{(2n-1)\pi x}{2} \quad x \in [0, 2]$$

3. 将函数 $f(x) = x^2 (0 \leq x \leq \pi)$ 分别展开成正弦级数和余弦级数.

【解析】(1) 正弦级数

$$a_n = 0 \quad (n = 0, 1, 2, \dots);$$



$$b_n = \frac{2}{\pi} \int_0^{\pi} x^2 \sin nx \, dx = \frac{2\pi}{n} (-1)^{n+1} + \frac{4}{n^3 \pi} [(-1)^n - 1];$$

$$\text{所以 } f(x) = \sum_{n=1}^{\infty} \left\{ \frac{2\pi}{n} (-1)^{n+1} + \frac{4}{n^3 \pi} [(-1)^n - 1] \right\} \cdot \sin nx \quad x \in (0, \pi);$$

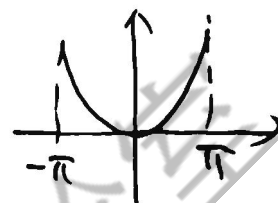
(2) 余弦级数

$$b_n = 0 \quad (n = 0, 1, 2, \dots);$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 \, dx = \frac{2\pi^2}{3}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx \, dx = (-1)^n \frac{4}{n^2};$$

$$\text{所以 } f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} (-1)^n \frac{4}{n^2} \cdot \cos nx \quad x \in [0, \pi]$$



自 测 题

一、填空题（每题 4 分，共 20 分）.

1. 幂级数 $\sum_{n=1}^{\infty} \frac{n}{(-3)^n + 2^n} x^{2n-1}$ 的收敛半径 $R = \underline{\sqrt{3}}$.

【解析】 $\lim_{n \rightarrow \infty} \left| \frac{\frac{n+1}{(-3)^{n+1} + 2^{n+1}} x^{2n+1}}{\frac{n}{(-3)^n + 2^n} x^{2n-1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{n+1}{n} \cdot \frac{(-3)^n + 2^n}{(-3)^{n+1} + 2^{n+1}} \cdot x^2 \right| = \frac{1}{3} |x| < 1 \Rightarrow |x| < \sqrt{3}$

2. 设幂级数 $\sum_{n=0}^{\infty} a_n x^n$ 的收敛半径为 3, 则幂级数 $\sum_{n=0}^{\infty} n a_n (x-1)^{n+1}$ 的收敛区间为 $\underline{(-2, 4)}$.

【解析】 逐项求导和逐项求积分不改变级数的收敛半径和收敛区间, 由 $\sum_{n=0}^{\infty} a_n x^n$ 的收敛半径为 3, 可得数

$\sum_{n=0}^{\infty} n a_n x^{n-1}$ 的收敛半径也为 3, 即数 $\sum_{n=0}^{\infty} n a_n (x-1)^{n+1}$ 的收敛半径也为 3, 则 $-3 < x-1 < 3$, 解得 $(-2, 4)$.

3. $\sum_{n=1}^{\infty} n \left(\frac{1}{2} \right)^{n-1} = \underline{4}$.

【解析】 考查幂级数 $\sum_{n=1}^{\infty} n x^{n-1}$, 在收敛区间内, 记 $S(x) = \sum_{n=1}^{\infty} n x^{n-1} = \left(\sum_{n=1}^{\infty} x^n \right)' = \left(\frac{x}{1-x} \right)' = \frac{1}{(1-x)^2}$,

则 $\sum_{n=1}^{\infty} n \left(\frac{1}{2} \right)^{n-1} = S\left(\frac{1}{2}\right) = 4$.

4. 函数 $f(x) = \begin{cases} -1 & -\pi \leq x \leq 0 \\ 1+x^2 & 0 < x \leq \pi \end{cases}$, 以 2π 为周期的傅里叶级数在点 $x = \pi$ 处收敛于 $\underline{\frac{\pi^2}{2}}$.

【解析】 函数在点 $x = \pi$ 处是间断点, 由狄里克莱收敛定理可知在点 $x = \pi$ 处收敛于

$$\frac{f(\pi^-) + f(\pi^+)}{2} = \frac{1 + \pi^2 + (-1)}{2} = \frac{\pi^2}{2}.$$

5. $f(x) = \pi x + x^2$ ($-\pi < x < \pi$) 的傅里叶级数展开式中系数 $b_3 = \underline{\frac{2}{3}\pi}$.

【解析】 $b_3 = \frac{1}{\pi} \int_{-\pi}^{\pi} (\pi x + x^2) \sin 3x dx = \frac{2}{3}\pi$.

二、选择题（每小题 4 分，共 20 分）.

6. 下列选项正确的是 (A).

(A) 若 $\sum_{n=1}^{\infty} a_n^2$ 和 $\sum_{n=1}^{\infty} b_n^2$ 都收敛, 则 $\sum_{n=1}^{\infty} (a_n + b_n)^2$ 收敛; (B) 若 $\sum_{n=1}^{\infty} |a_n b_n|$ 收敛, 则 $\sum_{n=1}^{\infty} a_n^2$ 与 $\sum_{n=1}^{\infty} b_n^2$ 都收敛;

(C) 若正项级数 $\sum_{n=1}^{\infty} a_n$ 发散, 则 $a_n \geq \frac{1}{n}$;

(D) 若级数 $\sum_{n=1}^{\infty} a_n$ 发散, 则 $a_n \geq \frac{1}{n}$.

【解析】A 正确, $(a_n + b_n)^2 = a_n^2 + b_n^2 + 2a_nb_n$, 又 $a_nb_n \leq \frac{1}{2}(a_n^2 + b_n^2)$, 由比较判别法可得级数收敛;

B 错误, 反例取 $a_n = \frac{1}{n^{\frac{3}{2}}}, b_n = \frac{1}{n^{\frac{1}{2}}}$;

C 错误, 正项级数的比较判别法只是充分而非充要条件;

D 错误, 反例 $a_n = \frac{0.5}{n}$

7. 级数 $\sum_{n=1}^{\infty} \left(\frac{\sin na}{n^2} - \frac{1}{\sqrt{n}} \right)$ (a 为常数) (C).

(A) 绝对收敛

(B) 条件收敛

(C) 发散

(D) 收敛性与 a 有关

【解析】 $\sum_{n=1}^{\infty} \frac{\sin na}{n^2}$: $\left| \frac{\sin na}{n^2} \right| \leq \frac{1}{n^2}$, 由比较判别法, 该级数绝对收敛;

$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$: 由 p -级数可知该级数发散; 再由级数的性质, 可知原级数发散.

8. 设级数 $\sum_{n=1}^{\infty} a_n^2$ 收敛, 则级数 $\sum_{n=1}^{\infty} (-1)^n \cdot \frac{|a_n|}{\sqrt{n^2 + \alpha}}$ ($\alpha > 0$) (A).

(A) 绝对收敛

(B) 条件收敛

(C) 发散

(D) 收敛性与 α 有关

【解析】 $\sum_{n=1}^{\infty} \left| (-1)^n \cdot \frac{|a_n|}{\sqrt{n^2 + \alpha}} \right| = \sum_{n=1}^{\infty} \frac{|a_n|}{\sqrt{n^2 + \alpha}}$, 而 $\frac{|a_n|}{\sqrt{n^2 + \alpha}} \leq \frac{1}{2} \left(a_n^2 + \frac{1}{n^2 + \alpha} \right)$, 由级数的性质、 p -级数以及比较判别法可知原级数绝对收敛.

9. 设幂级数 $\sum_{n=0}^{\infty} a_n (x-1)^n$ 在 $x=-1$ 处收敛, 该幂级数在 $x=2$ 处 (B).

(A) 条件收敛

(B) 绝对收敛

(C) 发散

(D) 敛散性不定

【解析】由于 $\sum_{n=0}^{\infty} a_n (x-1)^n$ 在 $x=-1$ 处收敛, 可知 $\sum_{n=0}^{\infty} a_n t^n$ 在 $t=-2$ 处收敛, 则 $\sum_{n=0}^{\infty} a_n t^n$ 在 $(-2, 2)$ 的开区间内

绝对收敛, 则当 $x=2$ 处 $\sum_{n=0}^{\infty} a_n (x-1)^n$ 绝对收敛.

10. 幂级数 $\sum_{n=1}^{\infty} (-1)^n \frac{(x-3)^n}{2n \cdot 2^n}$ 的收敛域为 (C).

(A) $(1, 5)$

(B) $[1, 5)$

(C) $(1, 5]$

(D) $[1, 5]$

【解析】 $\lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} \frac{(x-3)^{n+1}}{2(n+1) \cdot 2^{n+1}}}{(-1)^n \frac{(x-3)^n}{2n \cdot 2^n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{2n}{2(n+1)} \cdot \frac{2^n}{2^{n+1}} \cdot (x-3) \right| = \frac{1}{2} |x-3| < 1 \Rightarrow |x-3| < 2$

则收敛半径为 2，收敛区间为 $(1, 5)$ ；验证可知在 $x=1$ 点处 $\sum_{n=1}^{\infty} (-1)^n \frac{(-2)^n}{2n \cdot 2^n} = \sum_{n=1}^{\infty} \frac{1}{2n}$ 发散，而在 $x=5$ 点处

$\sum_{n=1}^{\infty} (-1)^n \frac{(2)^n}{2n \cdot 2^n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{2n}$ ，由交错级数的判别法可知收敛，所以收敛域为 $(1, 5]$

三、解答题（每小题 10 分，共 60 分）.

11. 判别下列级数敛散性：

(1) $\sum_{n=1}^{\infty} \frac{n^2}{\left(n + \frac{1}{n}\right)^n}$ ；

【解析】该级数是正项级数，利用根值判别法进行判别： $\lim_{n \rightarrow \infty} \sqrt[n]{\frac{n^2}{\left(n + \frac{1}{n}\right)^n}} = \lim_{n \rightarrow \infty} \frac{\sqrt[n]{n^2}}{n + \frac{1}{n}} = 0 < 1$ ，级数收敛.

(2) $\sum_{n=1}^{\infty} 2^n \sin \frac{\pi}{3^n}$.

【解析】该级数是正项级数，利用比较判别法的极限形式

$n \rightarrow \infty, 2^n \sin \frac{\pi}{3^n} \sim 2^n \cdot \frac{\pi}{3^n} = \pi \left(\frac{2}{3}\right)^n$ ，而 $\sum_{n=1}^{\infty} \pi \left(\frac{2}{3}\right)^n$ 收敛，所以原级数收敛.

12. 讨论下列级数是绝对收敛，还是条件收敛，或发散.

(1) $\sum_{n=1}^{\infty} \sin \left(n\pi + \frac{\pi}{n} \right)$ ；

【解析】 $\sin \left(n\pi + \frac{\pi}{n} \right) = (-1)^n \cdot \sin \left(\frac{\pi}{n} \right)$ ，则 $\sum_{n=1}^{\infty} \sin \left(n\pi + \frac{\pi}{n} \right) = \sum_{n=1}^{\infty} (-1)^n \cdot \sin \left(\frac{\pi}{n} \right)$ ；

$\sum_{n=1}^{\infty} \left| (-1)^n \cdot \sin \left(\frac{\pi}{n} \right) \right| = \sum_{n=1}^{\infty} \left| \sin \left(\frac{\pi}{n} \right) \right| \sim \sum_{n=1}^{\infty} \frac{\pi}{n}$ ，发散；而 $\sum_{n=1}^{\infty} (-1)^n \cdot \frac{\pi}{n}$ 为交错级数，收敛，所以该级数条件收敛.

(2) $\sum_{n=1}^{\infty} (-1)^{n-1} (\sqrt{n+1} - \sqrt{n}) \ln \left(1 + \frac{1}{n} \right)$.

【解析】 $\sum_{n=1}^{\infty} \left| (-1)^{n-1} (\sqrt{n+1} - \sqrt{n}) \ln \left(1 + \frac{1}{n} \right) \right| = \sum_{n=1}^{\infty} (\sqrt{n+1} - \sqrt{n}) \ln \left(1 + \frac{1}{n} \right) \sim \sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n}}{n}$ ，

而 $\frac{\sqrt{n+1}-\sqrt{n}}{n} = \frac{1}{(\sqrt{n+1}+\sqrt{n})n} \leq \frac{1}{2n^{\frac{3}{2}}}$, 由比较判别法可知 $\sum_{n=1}^{\infty} \frac{\sqrt{n+1}-\sqrt{n}}{n}$ 收敛, 则原级数绝对收敛.

13. 求幂级数 $\sum_{n=1}^{\infty} \frac{2n-1}{2^n} x^{2n-2}$ 在收敛区间内和函数 $S(x)$, 并计算 $\sum_{n=1}^{\infty} \frac{2n-1}{2^n}$ 的值.

【解析】(1) $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}(x)}{u_n(x)} \right| = \frac{1}{2} |x|^2 < 1 \Rightarrow |x| < \sqrt{2}$, 则 $R=2$, 收敛区间为 $(-\sqrt{2}, \sqrt{2})$;

当 $x = \pm\sqrt{2}$ 时, $\frac{1}{2} \sum_{n=1}^{\infty} (2n-1)$ 发散, 则收敛域为 $(-\sqrt{2}, \sqrt{2})$;

$$(2) \forall x \in (-\sqrt{2}, \sqrt{2}), S(x) = \sum_{n=1}^{\infty} \frac{2n-1}{2^n} x^{2n-2} = \frac{1}{x^2} \sum_{n=1}^{\infty} (2n-1) \left(\frac{x}{\sqrt{2}} \right)^{2n} = \frac{1}{x^2} f\left(\frac{x}{\sqrt{2}} \right),$$

$$\begin{aligned} \text{而 } f(x) &= \sum_{n=1}^{\infty} (2n-1)x^{2n} = \sum_{n=0}^{\infty} (2n+1)x^{2n+2} = x^2 \sum_{n=0}^{\infty} (2n+1)x^{2n} = x^2 \left(\sum_{n=0}^{\infty} x^{2n+1} \right)' \\ &= x^2 \left(\frac{x}{1-x^2} \right)' = x^2 \cdot \frac{x^2+1}{(1-x^2)^2} \end{aligned}$$

$$\text{所以 } S(x) = \frac{1}{x^2} f\left(\frac{x}{\sqrt{2}} \right) = \frac{1}{x^2} \cdot \left(\frac{x}{\sqrt{2}} \right)^2 \cdot \frac{\left(\frac{x}{\sqrt{2}} \right)^2 + 1}{\left[1 - \left(\frac{x}{\sqrt{2}} \right)^2 \right]^2} = \frac{x^2+2}{(2-x^2)^2}$$

$$(3) \sum_{n=1}^{\infty} \frac{2n-1}{2^n} = S(1) = 3$$

14. 将函数 $f(x) = \ln(4-3x-x^2)$ 展开成 x 的幂级数.

【解析】(1) $f(x) = \ln(4-3x-x^2) = \ln[(4+x)(1-x)] = \ln(4+x) + \ln(1-x)$;

$$\begin{aligned} (2) \ln(4+x) &= \ln 4 + \ln\left(1 + \frac{x}{4}\right) = 2\ln 2 + \sum_{n=1}^{\infty} (-1)^n \frac{1}{n} \left(\frac{x}{4}\right)^n \\ &= 2\ln 2 + \sum_{n=1}^{\infty} \frac{(-1)^n}{n \cdot 4^n} x^n \quad \left(-1 < \frac{x}{4} \leq 1 \Rightarrow -4 < x \leq 4 \right) \end{aligned}$$

$$(3) \ln(1-x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n} (-x)^n = -\sum_{n=1}^{\infty} \frac{1}{n} x^n \quad (-1 < -x \leq 1 \Rightarrow -1 \leq x < 1)$$

$$(4) f(x) = \ln(4-3x-x^2) = 2\ln 2 + \sum_{n=1}^{\infty} \frac{(-1)^n}{n \cdot 4^n} x^n - \sum_{n=1}^{\infty} \frac{1}{n} x^n = 2\ln 2 + \sum_{n=1}^{\infty} \frac{1}{n} \left[\frac{(-1)^n}{4^n} - 1 \right] x^n$$

$$(5) \begin{cases} -4 < x \leq 4 \\ -1 \leq x < 1 \end{cases} \Rightarrow -1 \leq x < 1$$

$$(6) f(x) = 2 \ln 2 + \sum_{n=1}^{\infty} \frac{1}{n} \left[\frac{(-1)^n}{4^n} - 1 \right] x^n \quad (-1 \leq x < 1).$$

15. 将函数 $f(x) = \frac{2x+1}{x^2+x-2}$ 展开成 $x-2$ 的幂级数.

$$\begin{aligned} \text{【解析】 } f(x) &= \frac{2x+1}{x^2+x-2} = \frac{(x+2)+(x-1)}{(x+2)(x-1)} = \frac{1}{x-1} + \frac{1}{x+2} \\ &= \frac{1}{1+(x-2)} + \frac{1}{4+(x-2)} = \frac{1}{1+(x-2)} + \frac{1}{4} \cdot \frac{1}{1+\frac{x-2}{4}} \\ &= \sum_{n=0}^{\infty} [-(x-2)]^n + \frac{1}{4} \cdot \sum_{n=0}^{\infty} \left[-\frac{x-2}{4} \right]^n = \sum_{n=0}^{\infty} (-1)^n (x-2)^n + \sum_{n=0}^{\infty} \frac{(-1)^n}{4^{n+1}} (x-2)^n \\ &= \sum_{n=0}^{\infty} (-1)^n \left[1 + \frac{(-1)^n}{4^{n+1}} \right] (x-2)^n; \end{aligned}$$

$$(2) \begin{cases} |-(x-2)| < 1 \\ \left| -\frac{x-2}{4} \right| < 1 \end{cases} \Rightarrow \begin{cases} |x-2| < 1 \\ |x-2| < 4 \end{cases} \Rightarrow |x-2| < 1;$$

$$(3) f(x) = \sum_{n=0}^{\infty} (-1)^n \left[1 + \frac{(-1)^n}{4^{n+1}} \right] (x-2)^n \quad (|x-2| < 1)$$

16. 将函数 $f(x) = x+2$ 在区间 $[0, 4]$ 上展开成正弦级数.

【解析】 $a_n = 0$;

$$b_n = \frac{2}{4} \int_0^4 (x+2) \sin \frac{n\pi}{4} x dx = \frac{4}{n\pi} [1 - 3(-1)^n];$$

$$\text{则 } f(x) = \sum_{n=0}^{\infty} \frac{4}{n\pi} [1 - 3(-1)^n] \sin \frac{n\pi x}{4} \quad (0 < x < 4)$$

